GLOBAL CLASSICAL STABLE SOLUTIONS OF THE ONE-PHASE BERNOULLI PROBLEM IN DIMENSION FIVE

Abstract. We prove that every global classical stable solution of the one-phase Bernoulli problem in $\mathbb{R}^5$ has zero Hessian in its positive phase. An exact rational tensor certificate gives a curvature Caccioppoli inequality at power 39/10. We combine this estimate with the selected free-boundary tree of Fernández-Real and Serra. Summing over all active tree pieces gives an $L^{6/5}$ Hessian estimate with the full power of the selected excess. A weighted cutoff transfers that power to the higher curvature moment. The resulting neck estimate lowers the critical selection exponent to 11/18 and bounds each relevant neck radius by $C\varepsilon^{12/11}R$. These gains allow the signed Neumann iteration to reach the amplitude scale. A five-dimensional Gaussian calibration then contradicts the logarithmic excess lower bound. We give the dimension-dependent calculations, the multiplier in ordinary tensor notation, and the complete rational factor data in an electronic supplement. The scalar-weight and boundary inequalities are proved analytically. The full interior inequality is established by exact rational reconstruction, so this part of the proof is computer-assisted.

Draft status. This candidate proof was obtained by our team on November 07, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

1.1. **The problem and the main result.** The one-phase Bernoulli functional is

$$\mathcal{J}(v;D)=\int_D\left(|\nabla v|^2+\mathbf{1}_{\{v>0\}}\right)dx,\qquad v\geq 0,$$

where D is a bounded open set. Its classical critical points satisfy

$$\Delta u=0\quad\text{in }\Omega:=\{u>0\},\qquad u=0,\quad |\nabla u|=1\quad\text{on }\Gamma:=\partial\Omega. \tag{1.1}$$

We call a solution classical if u is continuous, the free boundary is smooth and locally one-sided, and u is smooth up to the boundary from the positive side. One may start with the usual classical boundary regularity; the Bernoulli equation then gives the smoothness used here. Stability means that the second inner variation of $\mathcal{J}$ is nonnegative for every compactly supported deformation. Thus, if $\Psi_t(x)=x+t\xi(x)$ with $\xi\in C_c^\infty(D;\mathbb{R}^5)$, then

$$\left.\frac{d^2}{dt^2}\right|_{t=0}\mathcal{J}(u\circ\Psi_t;D)\geq 0.$$

The use of a bounded D only localizes the deformation. The solution itself is defined on all of $\mathbb{R}^5$.

The theorem below is the principal result.

Theorem 1.1 (Global rigidity)**.** *Let $u:\mathbb{R}^5\to[0,\infty)$ be a global classical stable solution of (1.1). Then*

$$D^2u\equiv 0\quad\text{in }\{u>0\}.$$

If $\Gamma\neq\varnothing$, there is a unit vector $e\in S^4$ such that either

$$u(x)=(e\cdot x-a)_+$$

for some $a\in\mathbb{R}$, or

$$u(x)=(e\cdot x-a)_++(b-e\cdot x)_+,\qquad b<a.$$

If $\Gamma=\varnothing$, then u is constant.

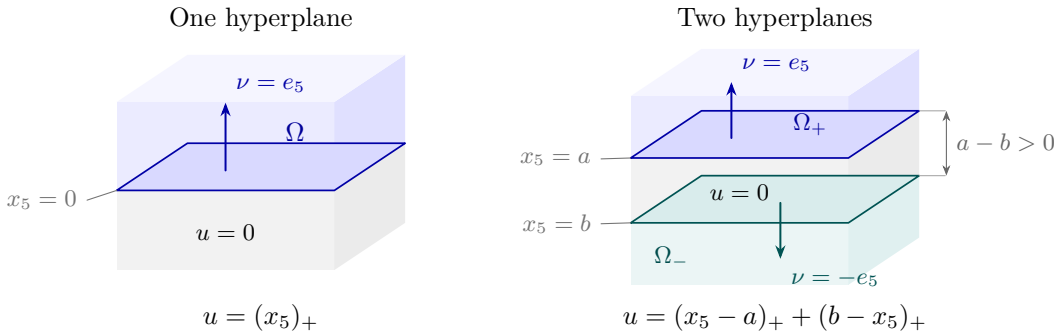


FIGURE 1. The flat solutions with nonempty free boundary. These are cropped three-dimensional sections of configurations in $\mathbb{R}^5$; two tangential coordinates are suppressed. The displayed planar pieces extend to whole hyperplanes. The arrows point into the positive phase. On the right, $b < a$, so the zero strip has positive width.

The strict inequality $b < a$ expresses the one-sided classical condition. The function $|e \cdot x - a|$ is a limiting vee, but its positive phase is two-sided at the ridge. It is used as a blow-down model in the proof.

1.2. **Research status.** Alt and Caffarelli established the basic existence and regularity theory for its minimizers in [1]. A global minimizer compares its energy with every admissible competitor in every ball. Stability is weaker: it requires a nonnegative second variation under compactly supported deformations of the independent variable. The classification of global classical stable solutions is a free-boundary version of the Bernstein problem. It asks whether stability forces the free boundary to be flat in low dimensions.

Two related classification questions must be distinguished. The first concerns one-homogeneous solutions, whose positive sets are cones. The second concerns arbitrary global solutions, without a homogeneity assumption. For minimizers, monotonicity and improvement of flatness connect these questions. For stable solutions, a further difficulty appears. Two portions of the positive phase can approach opposite sides of a hyperplane at large scales. Their limit can be the vee $V(x) = |x \cdot e|$. This limit has two positive sides, whereas a classical free boundary is locally one-sided. Excluding nonflat smooth cones therefore leaves a separate problem of controlling necks near a vee.

The homogeneous problem has a substantial history. Caffarelli, Jerison, and Kenig proved the three-dimensional result in [2]. Jerison and Savin proved that stable one-homogeneous solutions with smooth cross-section are flat through dimension four [7]. Their proof constructs suitable functions of the harmonic Hessian and studies the linearized free-boundary problem. In the opposite direction, De Silva and Jerison constructed a nonflat homogeneous global minimizer in dimension seven [4]. These results placed the first dimension admitting a singular minimizing cone between five and seven.

The preprint of OpenAI dated 24 September 2026 [9] reports that this critical dimension is exactly seven. Its theorem states that every nonzero one-homogeneous global minimizer in dimensions at most six is a half-space solution. The proof combines a scalar Hessian test with a separate tensor vector field. Exact rational inequalities control the interior and boundary contributions simultaneously. We use that construction with its stated preprint status. The extension from its minimizing-cone formulation to the smooth stable cones needed here is explained separately; it does not follow merely by replacing the word “minimizing” by “stable” in the theorem statement.

For arbitrary global classical stable solutions, dimension two follows from a logarithmic cutoff in the stability inequality; see [8]. Chan, Fernández-Real, Figalli, and Serra established the classification in dimension three in [3]. That work also treats the free-boundary Allen–Cahn equation

and develops compactness and neck estimates for the Bernoulli problem. Fernández-Real and Serra proved the four-dimensional classification in [6]. Their argument uses a combined Weiss–vee excess, a selection of exponent, center, and scale, and a signed decomposition of the positive phase. The dimension-general parts of that framework are used in the present proof.

The upper dimensional obstruction is also known for the classical stable problem. The singular cone of [4] does not itself belong to the globally classical class. However, De Silva, Jerison, and Shahgholian [5, Theorem 1.1 and Lemma 3.1] construct smooth nonhomogeneous global minimizers asymptotic to a given homogeneous minimizing cone whose only singular point is its vertex. Applied to the seven-dimensional cone, their theorem gives nonflat global classical stable solutions in dimension seven.[1] Extending these functions independently of extra coordinates gives such examples in every higher dimension. Indeed, write the extension as $U(x,y) = u(x)$ with $x \in \mathbb{R}^7$ and $y \in \mathbb{R}^m$. For an admissible competitor on a ball in $\mathbb{R}^{7+m}$, apply the minimizing property of u to almost every x-slice and integrate in y; the additional term $\int |\nabla_y v|^2$ in the competitor's energy is nonnegative. Thus U is still a global minimizer. Its free boundary is the smooth product $\partial\{u > 0\} \times \mathbb{R}^m$, and minimality implies stability. These results place dimensions five and six between the existing global classification and the higher-dimensional examples. We address the five-dimensional case in the present draft.

1.3. **The estimates that close the proof.** Put $w = |D^2 u|$. The main analytic estimate obtained from the Euclidean multiplier is

$$\int_\Omega \eta^2 w^{39/10} \le C \int_\Omega w^{19/10} |\nabla \eta|^2, \qquad \eta \in C_c^\infty(\mathbb{R}^5). \tag{1.2}$$

Proposition 3.1 also controls a weighted third derivative and a boundary curvature integral. The explicit polynomial formulas are in Appendix B. Appendix A proves the scalar-weight and boundary estimates analytically and identifies the exact interior certificate used to derive (1.2).

The global growth consequence of (1.2) is only $\int_{B_R \cap \Omega} w^{39/10} \le C R^{11/10}$. To obtain rigidity, we use the extra smallness at the selected scales. If ε is the selected excess at scale R, a summation over the active tree gives

$$\int_{B_{R/8}(z_0) \cap \Omega} w^{6/5} \le C \varepsilon^{6/5} R^{19/5}.$$

[1] For a minimizer, a smooth free-boundary patch is locally one-sided. If the positive phase occupied both sides of such a patch in a small ball B, then $u > 0$ almost everywhere in B. Its harmonic replacement h, with the same boundary values, is strictly positive in B, so the phase terms of their energies agree. Orthogonality of the harmonic replacement gives

$$\int_B |\nabla u|^2 - \int_B |\nabla h|^2 = \int_B |\nabla (u-h)|^2 > 0.$$

The inequality is strict because u vanishes at an interior free-boundary point while h is positive there, and both functions are continuous. This contradicts minimality. For $\phi \in C_c^\infty(\{u > 0\})$, the competitors $u + t\phi$ preserve the positive phase for sufficiently small positive and negative t. Differentiation gives $\int \nabla u \cdot \nabla \phi = 0$, hence $\Delta u = 0$. Smooth boundary regularity for the zero Dirichlet problem gives smoothness from the positive side. Write $\Omega = \{u > 0\}$ and let n be its outward normal. For $\Phi_t = \mathrm{id} + t\xi$, with ξ smooth and compactly supported in a small ambient neighborhood of the patch, minimality and change of variables for $u \circ \Phi_t^{-1}$ give

$$0 = \int_\Omega \left[(|\nabla u|^2 + 1) \operatorname{div} \xi - 2 u_i u_j \partial_j \xi_i \right] = \int_{\partial\Omega} (1 - |\nabla u|^2) \xi \cdot n.$$

Here repeated indices are summed. The second equality follows by integration by parts: the stress tensor $T_{ij} = (|\nabla u|^2 + 1)\delta_{ij} - 2u_i u_j$ satisfies $\partial_j T_{ij} = -2u_i \Delta u = 0$, while the zero boundary value makes ∇u normal to $\partial\Omega$. Arbitrary smooth normal traces of ξ therefore imply $|\nabla u| = 1$ on the patch. This proves the classical Bernoulli conditions. Minimality also gives nonnegative second inner variation and thus stability.

A weighted cutoff keeps the same power of ε in the higher moment. For the relevant threshold neck radii r_i, this gives

$$\sum_i (r_i/R)^{11/10} \le C\varepsilon^{6/5}. \tag{1.3}$$

It follows that the neck charge and the largest nearby neck satisfy

$$\rho(cR) \le C\varepsilon^{18/11}, \qquad r_{\max} \le C\varepsilon^{12/11}R.$$

The first estimate lowers the critical selection exponent to $\alpha_* \le 11/18 < 2/3$. The second places every relevant neck below the amplitude scale. Both gains are needed in the final argument.

The proof uses $\alpha = 5/8$ and a boundary-flux exponent $p_2 = 41/20 > 2$. We give explicit choices for all remaining small parameters. The final comparison is

$$\frac{c\varepsilon}{1 + C|\log\varepsilon|} \le \mathcal{E}_{z_0}(\rho) \le C\varepsilon^{1+\sigma/2}, \qquad \sigma > 0,$$

which is impossible as $\varepsilon \downarrow 0$.

1.4. **Organization.** Section 2 fixes the normal convention and explains why the smooth stable-cone exclusion required by the geometric framework is available. Section 3 proves the curvature estimate, including the cutoff at zeros of the Hessian. Section 4 introduces the neck centers, the combined excess, and the selected tree. It then proves the low-power Hessian estimate. Section 5 derives the higher moment, the charge gain, and the individual-radius bound. Section 6 chooses the parameters and proves the oscillation estimate down to the amplitude scale. Section 7 gives the Gaussian calibration and the contradiction. Appendix A gives the elementary scalar and boundary proofs and the exact interior certificate. Appendix B gives the multiplier polynomials without monomial codes.

The proof is partly computer-assisted. The Euclidean interior inequality uses the rational factors supplied with the paper and a reconstruction in exact integer and rational arithmetic. Appendix A.3 explains their tensor interpretation and the coercivity deduction. The scalar-weight and Euclidean boundary estimates are proved analytically. The cone lemma uses the spherical tensor inequalities of [9], with their hypotheses checked in its proof. Those spherical inequalities have different coefficients and exponent from the Euclidean certificate used here. A purely analytic replacement for the full Euclidean interior calculation is not asserted.

2. Preliminaries and smooth stable cones

2.1. **Notation, normal convention, and stability.** We work in $\mathbb{R}^5$ unless a different dimension is stated. A ball is $B_r(x)$, and cB denotes the concentric dilation of a ball B by a factor $c > 0$. We write

$$\fint_E f = \frac{1}{|E|}\int_E f$$

when $0 < |E| < \infty$. Tensor norms are Euclidean Frobenius norms. Repeated indices are summed. The space $\mathrm{STF}^k(\mathbb{R}^5)$ consists of fully symmetric rank-k tensors whose trace in any pair of slots is zero. Constants may depend on the fixed coefficient polynomials and on parameters chosen before a selected sequence is taken. They never depend on its index.

We use the elementary level-set fact that, for a locally Lipschitz function v, one has $\nabla v = 0$ almost everywhere on $\{v = c\}$. Indeed, almost every point of this set is both a density point of the set and a differentiability point of v. A nonzero derivative there would force $v \ne c$ in a fixed cone of nearby directions, contradicting density one. The classical derivative agrees almost everywhere with the weak gradient, which proves the assertion.

On the physical free boundary put

$$\nu = \nabla u, \qquad H = -u_{\nu\nu}.$$

The vector ν is the inward unit normal. The symbol H denotes the sum of the principal curvatures with this convention, without division by four. The scalar consequence of inner stability is

$$Q(\phi) := \int_\Omega |\nabla\phi|^2\,dx - \int_\Gamma H\phi^2\,d\mathcal{H}^4 \geq 0 \tag{2.1}$$

for compactly supported smooth tests up to the positive side of the boundary; see [2, Lemma 1] and [6, Definition 6.1]. We explain the test class and the scaling, since the latter source states its local inequality for nonnegative ambient tests in B_1. Choose $B_L(x_c)$ containing the support of a nonnegative ambient test ϕ compactly, and set

$$v(y) = L^{-1}u(x_c + Ly), \qquad \psi(y) = \phi(x_c + Ly).$$

The rescaled solution is classically stable in B_1: the Bernoulli energy, including its compact inner variations, is multiplied by L^{-5}. Its inward normal is unchanged and its curvature is $H_v(y) = LH(x_c + Ly)$. Writing $\Omega_v = \{v > 0\}$ and $\Gamma_v = \partial\Omega_v$, the change of variables gives

$$\int_{\Omega_v} |\nabla_y\psi|^2 = L^{-3}\int_\Omega |\nabla_x\phi|^2, \qquad \int_{\Gamma_v} H_v\psi^2\,d\mathcal{H}^4 = L^{-3}\int_\Gamma H\phi^2\,d\mathcal{H}^4.$$

The common positive factor cancels, proving (2.1) for such nonnegative tests at every scale.

For a signed ambient test, use

$$\phi_\varepsilon = \sqrt{\phi^2 + \varepsilon^2} - \varepsilon \geq 0, \qquad \nabla\phi_\varepsilon = \frac{\phi}{\sqrt{\phi^2 + \varepsilon^2}}\nabla\phi.$$

These are smooth, have support contained in $\operatorname{supp}\phi$, and satisfy $|\nabla\phi_\varepsilon| \leq |\nabla\phi|$. The level-set fact above gives $\nabla\phi = 0$ almost everywhere on $\{\phi = 0\}$, so dominated convergence yields $\int |\nabla\phi_\varepsilon|^2 \to \int |\nabla\phi|^2$. Also $\phi_\varepsilon^2 \to \phi^2$ uniformly. On their common compact boundary support, H is bounded and the smooth boundary has finite area, so the boundary integrals converge as well. Passing to the limit proves (2.1) for signed ambient tests. This argument uses convergence of the squared gradients; ϕ_ε itself converges to $|\phi|$.

To obtain such tests from the ambient formulation, flatten the smooth boundary in finitely many charts over their compact support, reflect across the flattened boundary, and use a partition of unity. This gives a compactly supported ambient H^1 extension. Mollifying it gives smooth ambient tests converging in H^1 and, by the local trace theorem, in L^2 on the boundary. Since H is bounded on the support, the boundary quadratic terms converge as well. More generally, the same passage admits H^1 limits whenever their boundary terms converge.

The global gradient bound and perimeter estimate are

$$|\nabla u| \leq 1, \qquad \mathcal{H}^4(\Gamma \cap B_r(z)) \leq Cr^4 \quad (z \in \Gamma, r > 0); \tag{2.2}$$

see [3, Lemmas 3.2 and 3.3]. For the perimeter estimate in the form of [6, Lemma 3.6], given $z \in \Gamma$ and $r > 0$, choose an auxiliary radius $R_0 > \max\{|z|, 2r\}$. Continuity gives $u(z) = 0$, hence $z \in \{u = 0\} \cap B_{R_0}$, and $r < R_0/2$. The cited estimate then gives the displayed bound with a constant independent of R_0. Since $|\nabla u|^2 = 1$ on Γ and is at most one in Ω, its inward derivative is nonpositive. Consequently $2u_{\nu\nu} = \partial_\nu|\nabla u|^2 \leq 0$, so $H \geq 0$. Later, $\nu_\pm$ will denote the *outward* normals to the signed domains $U_\pm$. Thus $\nu_\pm = -\nu$ on their regular physical boundaries.

2.2. **The cone threshold used in the geometric argument.** Let n_{cone} be the least dimension $n \geq 2$ admitting a global one-homogeneous stable Bernoulli solution $v : \mathbb{R}^n \to [0, \infty)$ with $0 \in \partial\{v > 0\}$, classical away from the origin, and not equal to $(e \cdot x)_+$ for any $e \in S^{n-1}$; this is [6, Definition 4.1]. The vertex condition excludes the zero solution. Indeed, for a continuous nonnegative one-homogeneous function it is equivalent to being nonzero: homogeneity gives $v(0) = 0$, and any point with $v(x) > 0$ gives positive points tx approaching the origin. The cone results through dimension four give $n_{\text{cone}} \geq 5$; see [2, 7].

Lemma 2.1 (Smooth stable cones in dimension five)**.** *A nonzero one-homogeneous stable Bernoulli solution in $\mathbb{R}^5$ that is classical away from the origin is a half-space solution. In particular, $n_{\mathrm{cone}} > 5$.*

Proof. Suppose that the cone is nonflat. All derivatives on a sphere in this proof are covariant derivatives, and all tensor norms use its metric. We use the previously established scalar stability and tensor inequalities, checking explicitly the geometry and the cutoffs needed to apply them to a smooth stable cone.

Step 1: the original link and its boundary. Choose a connected component $D \subset S^4$ of the positive link and write

$$g = u|_D, \qquad T_o = \nabla^2_{S^4} g + gI, \qquad w_o = |T_o|, \qquad B_o = \nabla T_o.$$

In the radial and tangential orthonormal frame, homogeneity gives

$$\nabla_{\mathbb{R}^5} u(ry) = g(y)y + \nabla_{S^4} g(y), \qquad D^2 u(ry) = r^{-1} \begin{pmatrix} 0 & 0 \\ 0 & T_o(y) \end{pmatrix}.$$

Indeed, Euler's identity makes the radial row and column of the Hessian zero; tangential differentiation of the gradient gives the block $\nabla^2 g + gI$. In particular, this tensor vanishes exactly when u is linear on the corresponding positive component. If it vanished on every component, homogeneity would remove the constant term of each affine map, and continuity would make that component the whole half-space on which the map is positive. Two disjoint half-spaces through the origin are opposite, and their union is locally two-sided along the common hyperplane. The classical hypothesis excludes this possibility. Thus we may choose D with $T_o \not\equiv 0$.

The closure of D is compact and its boundary is smooth. To see the local statement, intersect a regular free-boundary chart of the cone with S^4. The radial vector is tangent to the cone boundary, so the intersection is transverse. The locally one-sided condition gives one positive side of a single smooth hypersurface in each such chart. A finite cover of the spherical free boundary then gives a smooth compact closure and separates D from other components near its boundary. This boundary is nonempty: otherwise $D = S^4$, and integrating the equation below would give $0 = -4\int_{S^4} g < 0$.

Taking the trace of the polar Hessian formula and using the Bernoulli data yields

$$\Delta_{S^4} g = -4g \quad \text{in } D, \qquad g = 0, \quad |\nabla_{S^4} g| = 1 \quad \text{on } \partial D.$$

Thus T_o is trace-free. The unit-sphere tensor identities [9, Proposition 3.1], applied with spherical dimension four, give the Codazzi identity $\nabla_i (T_o)_{jk} = \nabla_j (T_o)_{ik}$. Together with the symmetry of T_o, it makes B_o fully symmetric. Differentiating $\operatorname{tr} T_o = 0$ gives $(B_o)_{ijj} = 0$; symmetry gives all other traces, and hence $\operatorname{div} T_o = 0$.

Set $s_o^2 = g^2 + |\nabla g|^2$. The two scalar identities we need can also be read directly from these facts:

$$\nabla_l s_o^2 = 2g\nabla_l g + 2(\nabla_i g)\big((T_o)_{li} - g\delta_{li}\big) = 2(T_o)_{li}\nabla_i g,$$
$$\Delta s_o^2 = 2(\nabla_l (T_o)_{li})\nabla_i g + 2(T_o)_{li}\big((T_o)_{li} - g\delta_{li}\big) = 2w_o^2.$$

Since $s_o^2 = 1$ on ∂D, the maximum principle gives $s_o^2 \le 1$. Equality at an interior point would force this subharmonic function to be constant, and then $T_o \equiv 0$. Consequently $s_o^2 < 1$ in D.

Let N_o be the inward unit normal to ∂D. Positivity of g inside D and the boundary data give $\nabla g = N_o$ there. Hopf's lemma, with the inward normal, therefore gives

$$0 > \partial_{N_o} s_o^2 = 2T_o(N_o, N_o), \qquad H_o := -T_o(N_o, N_o) > 0.$$

This H_o is the sum of the three principal curvatures of $\partial D \subset S^4$ with our convention. Indeed, for tangent vectors τ, σ one has $\langle \nabla_\tau N_o, \sigma \rangle = \nabla^2 g(\tau, \sigma) = T_o(\tau, \sigma)$, and the trace-free condition gives their sum as $-T_o(N_o, N_o)$. Tangential differentiation of $s_o^2 = 1$ also gives $T_o(\tau, N_o) = 0$. Hence

$$T_o N_o = -H_o N_o, \qquad w_o \ge H_o > 0 \quad \text{on } \partial D.$$

The Euclidean cone boundary adds a zero radial principal curvature; its other principal curvatures at ry are r^{-1} times those of ∂D. By compactness, w_o has a positive lower bound in a neighborhood of the boundary, so its zero set is a compact subset of D.

Step 2: suspension and scalar stability. Put $C_o = \{ry : r > 0,\ y \in D\}$. A compact test support avoiding the origin lies in an annulus $\{c_1 < |x| < c_2\}$ with $c_1 > 0$. The dimension-independent annular stability inequality [7, Section 2.3, equation (2.4)] therefore applies. It uses inner stability for deformations supported in that annulus, which is part of our hypothesis; it does not use the classification of stable cones. For a test on the regular closure of C_o, its compact support has a collar disjoint from the other positive components. The test-extension argument in Section 2 therefore allows the test to be zero on those components before applying stability to the original cone.

Extend the cone independently of a sixth coordinate, obtaining $\widetilde{C} = C_o \times \mathbb{R}$. For a compact smooth test ϕ whose support avoids the axis $\{0\} \times \mathbb{R}$, slicing gives

$$Q_{\widetilde{C}}(\phi) = \int_{\mathbb{R}} Q_{C_o}(\phi(\cdot, x_6))\, dx_6 + \int_{C_o \times \mathbb{R}} |\partial_{x_6}\phi|^2 \geq 0.$$

The boundary curvature is unchanged under this product, because its new principal curvature is zero. All the sliced tests avoid the vertex, so the preceding local stability statement applies to each of them.

The suspended link $\widetilde{D} = \widetilde{C} \cap S^5$ is parametrized by

$$(t, y) \longmapsto (\sin t\, y, \cos t), \qquad 0 < t < \pi, \quad y \in D.$$

Write $s = \sin t$ and $c = \cos t$. The differential maps ∂_t to $(cy, -s)$ and $v \in T_yD$ to $(sv, 0)$. They are orthogonal with lengths 1 and $s|v|$, respectively. Thus

$$ds^2_{\widetilde{D}} = dt^2 + s^2 ds^2_D, \qquad dV = s^4\, dt\, dV_o, \qquad dS = s^3\, dt\, dS_o,$$

where dS_o is the boundary measure on ∂D. The boundary measure formula concerns only the regular side boundary; the two omitted tips lie on the product axis.

For clarity, we compute the tensors in this metric. Let $e_1, \ldots, e_4$ be a local orthonormal frame on D, lifted independently of t, and put $E_0 = \partial_t$ and $E_\alpha = s^{-1}e_\alpha$. At a chosen point take $\nabla^D_{e_\alpha} e_\beta = 0$. The warped metric has coordinate Christoffel symbols $\widetilde{\Gamma}^t_{ab} = -sch_{ab}$ and $\widetilde{\Gamma}^a_{tb} = (c/s)\delta^a_b$, where h_{ab} is the original metric. Its other tangential symbols are those of h. In the orthonormal frame this gives, at the chosen point,

$$\widetilde{\nabla}_{E_0} E_0 = 0, \qquad \widetilde{\nabla}_{E_0} E_\alpha = 0, \qquad \widetilde{\nabla}_{E_\alpha} E_0 = \frac{c}{s} E_\alpha,$$
$$\widetilde{\nabla}_{E_\alpha} E_\beta = -\frac{c}{s}\delta_{\alpha\beta} E_0.$$

In $\widetilde{\nabla}_{E_0} E_\alpha = 0$, the derivative of s^{-1} cancels the corresponding Christoffel term. For $\widetilde{g} = sg$, one has $E_0\widetilde{g} = cg$ and $E_\alpha\widetilde{g} = e_\alpha g$. The formula $\nabla^2 f(V, W) = V(Wf) - (\nabla_V W)f$ therefore yields

$$(\widetilde{\nabla}^2\widetilde{g})_{00} = -sg, \qquad (\widetilde{\nabla}^2\widetilde{g})_{0\alpha} = 0,$$
$$(\widetilde{\nabla}^2\widetilde{g})_{\alpha\beta} = s^{-1}(\nabla^2 g)_{\alpha\beta} + \frac{c^2}{s} g\delta_{\alpha\beta}.$$

Since $s^2 + c^2 = 1$, the tensor $\widetilde{T} = \widetilde{\nabla}^2\widetilde{g} + \widetilde{g}I$ satisfies

$$\widetilde{T}_{00} = \widetilde{T}_{0\alpha} = 0, \qquad \widetilde{T}_{\alpha\beta} = s^{-1}(T_o)_{\alpha\beta}.$$

Differentiating once more, with $\widetilde{B}_{ijk} = (\widetilde{\nabla}_{E_i}\widetilde{T})(E_j, E_k)$, gives all its possibly nonzero components:

$$\widetilde{B}_{\alpha\beta\gamma} = s^{-2}(B_o)_{\alpha\beta\gamma},$$
$$\widetilde{B}_{0\alpha\beta} = \widetilde{B}_{\alpha 0\beta} = \widetilde{B}_{\alpha\beta 0} = -cs^{-2}(T_o)_{\alpha\beta}.$$

For example, the first mixed component is $\partial_t(s^{-1}(T_o)_{\alpha\beta})$, while the second is $-\widetilde{T}(\widetilde{\nabla}_{E_\alpha}E_0, E_\beta)$. Components with two or three zero indices vanish. Squaring and summing over ordered indices counts the single zero index in three positions. Consequently, with $\widetilde{w} = |\widetilde{T}|$,

$$\widetilde{w} = s^{-1}w_o, \qquad |\widetilde{B}|^2 = s^{-4}(|B_o|^2 + 3c^2w_o^2), \qquad \widetilde{H} = s^{-1}H_o. \tag{2.3}$$

For the curvature formula, the inward unit normal is $N = \sum_\alpha (N_o)_\alpha E_\alpha$. The three tangential principal curvatures from ∂D are multiplied by s^{-1}, and the E_0 direction has curvature zero. This also follows from $-\widetilde{T}(N,N) = s^{-1}H_o$. These formulas agree with [9, Proposition 3.4]. The same Hessian and gradient computations give

$$\Delta_{S^5}\widetilde{g} = -sg + s^{-1}(\Delta_{S^4}g + 4c^2 g) = -5sg,$$
$$\widetilde{g}^2 + |\nabla\widetilde{g}|^2 = s^2g^2 + c^2g^2 + |\nabla g|^2 = s_o^2 < 1.$$

We next pass from stability on $\widetilde{C}$ to its spherical form. Let ψ be smooth up to the regular boundary of $\widetilde{D}$, supported away from the tips, and let $0 \neq \zeta \in C_c^\infty((0,\infty))$. In polar coordinates of $\mathbb{R}^6$, the volume measure is $r^5\,dr\,dV$, the side-boundary measure is $r^4\,dr\,dS$, and the boundary curvature is $r^{-1}\widetilde{H}$. For $\phi(r,\theta) = \zeta(r)\psi(\theta)$, stability becomes

$$\left(\int_0^\infty r^3\zeta^2\,dr\right)\int_{\partial\widetilde{D}}\widetilde{H}\psi^2 \le \left(\int_0^\infty r^3\zeta^2\,dr\right)\int_{\widetilde{D}}|\nabla\psi|^2 + \left(\int_0^\infty r^5|\zeta'|^2\,dr\right)\int_{\widetilde{D}}\psi^2.$$

The test is admissible because its radial support is an annulus and its angular support avoids the tips, so its support avoids the product axis. Fix $0 \neq \eta \in C_c^\infty(\mathbb{R})$ and take $\zeta_k(r) = r^{-2}\eta((\log r)/k)$. With $q = (\log r)/k$,

$$\zeta_k'(r) = r^{-3}(-2\eta(q) + k^{-1}\eta'(q)), \qquad dr/r = k\,dq.$$

Thus

$$\int_0^\infty r^3\zeta_k^2\,dr = k\int_{\mathbb{R}}\eta^2,$$
$$\int_0^\infty r^5|\zeta_k'|^2\,dr = 4k\int_{\mathbb{R}}\eta^2 - 4\int_{\mathbb{R}}\eta\eta' + k^{-1}\int_{\mathbb{R}}|\eta'|^2 = 4k\int_{\mathbb{R}}\eta^2 + k^{-1}\int_{\mathbb{R}}|\eta'|^2.$$

The cross term vanishes because $\int\eta\eta' = \frac{1}{2}\int(\eta^2)' = 0$. Dividing by $k\int\eta^2$ and letting $k\to\infty$ proves

$$\int_{\partial\widetilde{D}}\widetilde{H}\psi^2 \le \int_{\widetilde{D}}(|\nabla\psi|^2 + 4\psi^2). \tag{2.4}$$

Step 3: application of the spherical multiplier. Use the universal tensor inequalities of [9, Proposition 4.1 and Section 5], with exponent $a = 29/50$ and homogenizing variable $z = 1$. On $\{\widetilde{w} > 0\}$ put

$$A = \widetilde{T}/\widetilde{w}, \qquad X = \widetilde{B}/\widetilde{w}, \qquad p = \nabla\widetilde{g}.$$

The component formulas show that A is symmetric and trace-free with $|A| = 1$, that X is fully symmetric and trace-free, and that $\widetilde{g}^2 + |p|^2 \le 1$. On the regular boundary $p = N$. The suspension of $T_oN_o = -H_oN_o$ gives $Ap = \lambda p$, where $\lambda = -\widetilde{H}/\widetilde{w} < 0$. These are exactly the interior and boundary hypotheses of the cited inequalities. Define

$$h = \widetilde{w}^a\sqrt{f(A,p,\widetilde{g})},$$
$$Z_l = \widetilde{w}^{2a}(G_{lijk}X_{ijk} + \widetilde{w}M_l + P_l), \qquad G_{lijk} = \delta_{li}U_{jk} + K_{lijk} - K_{iljk}.$$

Here f, U, K, M, P are the coefficient polynomials of that spherical certificate, and f has a positive lower bound. We use the established local conclusions [9, Propositions 5.2 and 5.4, Lemma 5.5]:

$$\begin{aligned} \operatorname{div} Z &\geq |\nabla h|^2 + 4h^2 + \epsilon_* \widetilde{w}^{2a}, \qquad N \cdot Z + \widetilde{H} h^2 \geq 0, \\ |h| &\leq C\widetilde{w}^a, \qquad |\nabla h| \leq C\widetilde{w}^a(1 + \widetilde{w} + |X|), \qquad |Z| \leq C\widetilde{w}^{2a}(1 + \widetilde{w} + |X|), \end{aligned} \tag{2.5}$$

where $\epsilon_* > 0$ and N is inward. The constants depend only on the fixed polynomials. These inequalities concern local normalized tensor data and require no minimizing comparison. Smoothness and scalar stability, which are the geometric inputs needed to integrate them, have been supplied independently in Steps 1 and 2.

Step 4: the integrable weight. Apply the weighted Bochner estimate [9, Lemma 7.1] to the original component, with $m_o = 4$ and $a = 29/50$. Its inputs are a smooth compact link, the tensor identities of Proposition 3.1 there, and positivity of w_o at the boundary; all hold by Step 1. The proof of that estimate uses these inputs and $1/2 < a < 1$, without a minimizing hypothesis. In particular, its Bochner identity here is

$$\Delta w_o^2 = 2|B_o|^2 + 8w_o^2.$$

On $\{w_o > 0\}$, differentiation and Cauchy–Schwarz also give $\nabla_l w_o = w_o^{-1}(T_o)_{ij}(B_o)_{lij}$ and $|\nabla w_o| \leq |B_o|$, as used below. The weighted estimate gives

$$\int_{D \cap \{w_o > 0\}} w_o^{2a-2} |B_o|^2 \, dV_o < \infty.$$

Define

$$D_o(y) = w_o^{2a}(1 + |B_o|/w_o)^2 \quad (w_o > 0), \qquad D_o(y) = 0 \quad (w_o = 0).$$

The inequality $(1 + b)^2 \leq 2 + 2b^2$ shows directly that

$$\int_D D_o \, dV_o \leq 2 \int_D w_o^{2a} \, dV_o + 2 \int_{D \cap \{w_o > 0\}} w_o^{2a-2} |B_o|^2 \, dV_o < \infty.$$

The first integral is finite because w_o is bounded on the compact closure. No size or regularity assumption on its zero set is needed.

Step 5: the two cutoffs and the contradiction. Fix $\vartheta \in C^\infty([0, \infty))$, with values in $[0, 1]$, equal to zero on $[0, 1]$ and one on $[2, \infty)$. For $j \geq 3$ set

$$\alpha_j(y) = \vartheta(j w_o(y)), \qquad \beta_j(t) = \vartheta(j \sin t), \qquad \xi_j = \alpha_j \beta_j.$$

The normalized fields need only be evaluated where $w_o > 0$. Extend the products $\xi_j h$ and $\xi_j^2 Z$ by zero where $w_o = 0$. They are smooth because they vanish on a whole neighborhood of that set; they also vanish near both tips. We may integrate on a truncated suspended link whose end slices lie in these zero regions. There are no end-slice fluxes, and its remaining outward normal is $-N$.

Here is the cancellation before taking any limit. By (2.5) and integration by parts,

$$\begin{aligned} \int_{\widetilde{D}} \xi_j^2 (|\nabla h|^2 + 4h^2 + \epsilon_* \widetilde{w}^{2a}) &\leq - \int_{\partial \widetilde{D}} \xi_j^2 N \cdot Z - 2 \int_{\widetilde{D}} \xi_j Z \cdot \nabla \xi_j \\ &\leq \int_{\partial \widetilde{D}} \widetilde{H} (\xi_j h)^2 - 2 \int_{\widetilde{D}} \xi_j Z \cdot \nabla \xi_j. \end{aligned}$$

Apply (2.4) to $\psi = \xi_j h$ and expand

$$|\nabla(\xi_j h)|^2 = \xi_j^2 |\nabla h|^2 + 2\xi_j h \nabla h \cdot \nabla \xi_j + h^2 |\nabla \xi_j|^2.$$

The terms $\xi_j^2 |\nabla h|^2$ and $4\xi_j^2 h^2$ cancel, leaving

$$\epsilon_* \int_{\widetilde{D}} \xi_j^2 \widetilde{w}^{2a} \leq \int_{\widetilde{D}} \mathcal{E}_j, \qquad \mathcal{E}_j = h^2 |\nabla \xi_j|^2 + 2\xi_j (h \nabla h - Z) \cdot \nabla \xi_j.$$

Thus the boundary terms have been eliminated at each fixed j; only the following bulk domination is needed for the limit.

Put $L = \|\vartheta'\|_\infty$, $\kappa = 2L$, and $b_o = |B_o|/w_o$ on $\{w_o > 0\}$. On the transition band of α_j, one has $1 < jw_o < 2$, so

$$|\nabla_D \alpha_j| \le jL|\nabla_D w_o| \le \kappa b_o.$$

Likewise $1 < js < 2$ on the transition band of β_j, and $|\beta_j'| = |j\vartheta'(js)c| \le \kappa/s$. Both derivatives vanish off their respective bands. The warped metric therefore gives

$$|\nabla \xi_j|^2 = \alpha_j^2|\beta_j'|^2 + s^{-2}\beta_j^2|\nabla_D \alpha_j|^2 \le \kappa^2 s^{-2}(1 + b_o^2).$$

Moreover, (2.3) gives the exact expression

$$|X| = s^{-1}\sqrt{b_o^2 + 3c^2}.$$

Since w_o is bounded and $s \le 1$, it follows that

$$1 + \widetilde{w} + |X| \le C_D s^{-1}(1 + b_o), \qquad |\nabla \xi_j| \le \kappa s^{-1}(1 + b_o).$$

Using the field bounds, $0 \le \xi_j \le 1$, and $\widetilde{w} = s^{-1}w_o$, we consequently obtain

$$\begin{aligned} |\mathcal{E}_j| &\le C\widetilde{w}^{2a}(|\nabla \xi_j|^2 + (1 + \widetilde{w} + |X|)|\nabla \xi_j|) \\ &\le C_D s^{-2a-2} w_o^{2a}(1 + b_o)^2 = C_D s^{-2a-2} D_o(y). \end{aligned}$$

After multiplication by the volume factor s^4, the common dominator is $C_D s^{2-2a} D_o(y)$. It is integrable on $(0, \pi) \times D$ because $2 - 2a = 21/25 > -1$ and $D_o \in L^1(D)$; near the endpoints, $\sin t$ is comparable to t or $\pi - t$. All constants are independent of j.

At each point with $0 < t < \pi$ and $w_o(y) > 0$, the cutoff is eventually one and its derivative is zero, so $\mathcal{E}_j \to 0$. At $w_o = 0$ every error is zero by the smooth zero extension. Dominated convergence therefore sends the right side to zero. For the left side, the product-coordinate density is bounded by $s^{4-2a} w_o^{2a}$, an integrable function because $4 - 2a = 71/25 > -1$. Its limit is

$$\epsilon_* \left(\int_0^\pi (\sin t)^{4-2a}\, dt \right) \left(\int_D w_o^{2a}\, dV_o \right) > 0.$$

The second factor is positive since the smooth tensor T_o is not identically zero. This contradicts the preceding inequality and proves the lemma. □

The geometric results in Sections 4–11 of [6] are formulated or proved for dimensions $n < n_{\mathrm{cone}}$, with the numerical restrictions used here valid for $n \le 6$. Lemma 2.1 permits their use at $n = 5$. We state each required consequence before applying it. The final dimension-four argument of that paper is replaced below by the five-dimensional parameter calculation and calibration.

3. The Euclidean multiplier and the curvature estimate

3.1. **Statement.** Write $w = |D^2 u|$. The interior tensor certificate and the analytic scalar-weight and boundary inequalities give the following estimate.

Proposition 3.1. *For a global classical stable solution in $\mathbb{R}^5$, every compact smooth ambient cutoff η satisfies*

$$\begin{aligned} \int_\Omega \eta^2 w^{39/10}\, dx + \int_\Omega \eta^2 w^{-1/10}|D^3 u|^2\, dx + \int_\Gamma \eta^2 w^{29/10}\, d\mathcal{H}^4 \\ \le C \int_\Omega w^{19/10}|\nabla \eta|^2\, dx. \end{aligned} \tag{3.1}$$

Consequently, for every $x_0 \in \mathbb{R}^5$ and $R > 0$,

$$\int_{B_R(x_0)\cap\Omega} w^{39/10} + \int_{B_R(x_0)\cap\Omega} w^{-1/10}|D^3 u|^2 + \int_{B_R(x_0)\cap\Gamma} w^{29/10} \le CR^{11/10}. \tag{3.2}$$

Measures are the appropriate volume or surface measures. The constant depends only on the fixed multiplier. The weighted derivative integrand has its almost-everywhere interpretation at $w = 0$ explained below.

The power $39/10$ is below the ambient dimension. Thus the growth bound (3.2) must be combined with the small selected excess. This is done in Sections 4 and 5.

3.2. **The finite algebraic statement.** Fix

$$b = \frac{19}{20}, \qquad A \in \mathrm{STF}^2(\mathbb{R}^5), \quad |A| = 1, \quad |p| \le 1, \quad X \in \mathrm{STF}^3(\mathbb{R}^5).$$

The tensor X is fully symmetric and trace-free in every pair. Temporarily let $w \in \mathbb{R}$ and $\ell \in \mathbb{R}^5$ be independent algebraic variables. Set

$$q_l = A_{ij}X_{lij}, \qquad D_l A_{ij} = X_{lij} - q_l A_{ij}, \qquad D_l p_i = wA_{li}.$$

The operator D_l acts on polynomials in (A, p) by the chain rule. Define the polynomial fields f, U, K, M by the ordinary tensor formulas in Appendix B, and put

$$G_{lijk} = \delta_{li}U_{jk} + K_{lijk} - K_{iljk}.$$

Define

$$D_{\mathrm{div}} = (D_l G_{lijk})X_{ijk} + (2b-1)q_l G_{lijk}X_{ijk} + wD_l M_l + (2b+1)wq_l M_l, \tag{3.3}$$

$$F_l = D_l f + 2bfq_l, \qquad I_b = D_{\mathrm{div}} - F \cdot \ell + f|\ell|^2. \tag{3.4}$$

The full uniform interior estimate is the following finite algebraic result.

Proposition 3.2 (Uniform interior coercivity)**.** *For every $A \in \mathrm{STF}^2(\mathbb{R}^5)$ with $|A| = 1$, every $p \in \mathbb{R}^5$ with $|p| \le 1$, every $X \in \mathrm{STF}^3(\mathbb{R}^5)$, every $w \in \mathbb{R}$, and every $\ell \in \mathbb{R}^5$, the fields of Appendix B satisfy*

$$I_b \ge \mu(w^2 + |X|^2 + |\ell|^2), \qquad \mu = \frac{1}{1250000} > 0. \tag{3.5}$$

Proof. This is the uniform tensor inequality proved in Appendix A.3 by exact rational reconstruction and a uniform residual estimate. That proof treats (w, X, ℓ) as independent variables and imposes no boundary eigenvector condition on (A, p). □

For the boundary, assume additionally $|p| = 1$, $Ap = \lambda p$, and $\lambda \le 0$. Put

$$\begin{aligned} S_0 &= \lambda A - A^2, \\ V_{ijk} &= p_i(S_0)_{jk} + p_j(S_0)_{ki} + p_k(S_0)_{ij} + p_i p_j p_k, \\ S(A, p) &= -\lambda f + p_l G_{lijk} V_{ijk} + p \cdot M. \end{aligned}$$

For these same fields, the elementary argument in Appendix A.4 proves

$$S \ge \beta, \qquad \beta = \frac{1}{10^6} > 0. \tag{3.6}$$

The lower bound for f also has an elementary proof: Lemma A.1 gives

$$\frac{1}{6} \le f \le 5.$$

These estimates are the algebraic inputs to the following PDE argument.

3.3. **Substitution into the PDE.** On $\{w > 0\}$ set

$$A = \frac{D^2u}{w}, \qquad p = \nabla u, \qquad X = \frac{D^3u}{w}, \qquad h = w^b\sqrt{f}, \qquad Z_l = w^{2b}(G_{lijk}X_{ijk} + wM_l).$$

Here and below the coefficient fields are evaluated at (A, p). Harmonicity gives $\operatorname{tr} A = \Delta u/w = 0$ and $X_{lii} = \partial_l \Delta u/w = 0$. Symmetry of the derivatives gives the remaining symmetries and traces of X; also $|A| = 1$. The gradient bound (2.2) gives $|p| \le 1$. Thus all the hypotheses of Proposition 3.2 hold.

We first relate the formal derivation D_l to spatial derivatives. Differentiating $w^2 = u_{ij}u_{ij}$ gives

$$2w\partial_l w = 2u_{ij}u_{lij} = 2w^2 A_{ij}X_{lij} = 2w^2 q_l.$$

The quotient rule therefore gives

$$\partial_l w = wq_l, \qquad \partial_l A_{ij} = X_{lij} - q_l A_{ij}, \qquad \partial_l p_i = wA_{li}, \qquad \partial_l X_{ijk} = \frac{u_{lijk}}{w} - q_l X_{ijk}.$$

Consequently $\partial_l f = D_l f$, and the same equality holds componentwise for U, K, M, G. This chain rule applies to the coefficient fields, which depend on (A, p); the displayed quotient rules are used separately for w and X.

Expanding the two parts of the divergence separately, one obtains

$$\begin{aligned}\partial_l(w^{2b}G_{lijk}X_{ijk}) &= w^{2b}((D_lG_{lijk})X_{ijk} + (2b-1)q_lG_{lijk}X_{ijk}) + w^{2b-1}G_{lijk}u_{lijk},\\ \partial_l(w^{2b+1}M_l) &= w^{2b+1}(D_lM_l + (2b+1)q_lM_l).\end{aligned}$$

The coefficient $2b-1$ combines the derivative of w^{2b} with the term $-q_lX_{ijk}$ in $\partial_l X_{ijk}$. The only fourth derivatives cancel:

$$G_{lijk}u_{lijk} = U_{jk}\partial_{jk}\Delta u + (K_{lijk} - K_{iljk})u_{lijk} = 0.$$

The first term is zero by harmonicity. In the second, interchanging the summation indices l, i changes the coefficient's sign and leaves u_{lijk} unchanged. Adding the two divergence expansions now gives

$$w^{-2b}\operatorname{div} Z = D_{\mathrm{div}}.$$

For the scalar test function, differentiation gives

$$\partial_l h = bw^{b-1}(\partial_l w)\sqrt{f} + \frac{w^b}{2\sqrt{f}}D_l f = \frac{w^b}{2\sqrt{f}}F_l, \qquad |\nabla h|^2 = \frac{w^{2b}}{4f}|F|^2.$$

Since $f \ge 1/6$, completing the square in the independent vector ℓ gives the exact identity

$$I_b = D_{\mathrm{div}} - \frac{|F|^2}{4f} + f\left|\ell - \frac{F}{2f}\right|^2.$$

Proposition 3.2 holds for every ℓ, so we may choose $\ell_* = F/(2f)$ at each point. It follows that

$$\operatorname{div} Z - |\nabla h|^2 = w^{2b}\left(D_{\mathrm{div}} - \frac{|F|^2}{4f}\right) \ge \mu w^{2b}(w^2 + |X|^2 + |\ell_*|^2).$$

Discarding the last nonnegative term and using $w^{2b}|X|^2 = w^{2b-2}|D^3u|^2$ yields

$$\operatorname{div} Z - |\nabla h|^2 \ge \mu\left(w^{2b+2} + w^{2b-2}|D^3u|^2\right). \tag{3.7}$$

We next compute the physical boundary flux at a point with $w > 0$. Choose an orthonormal frame $e_1, \dots, e_4, \nu$ there, with $e_1, \dots, e_4$ tangent to Γ. Subscripts in the following computations mean Euclidean derivatives evaluated in this frame. Since $u = 0$ and $\nabla u = \nu$ on Γ, tangential differentiation of $|\nabla u|^2 = 1$ gives

$$0 = \partial_\alpha|\nabla u|^2 = 2u_{\nu\alpha}.$$

Write the second fundamental form as $L_{\alpha\beta} = \langle D_{e_\alpha}\nu, e_\beta\rangle = u_{\alpha\beta}$. Harmonicity and the definition of H give

$$H = \operatorname{tr} L = -u_{\nu\nu}, \qquad A = \begin{pmatrix} L/w & 0 \\ 0 & \lambda \end{pmatrix}, \qquad p = \nu, \qquad \lambda = -H/w \le 0.$$

Thus the hypotheses of the boundary inequality hold.

To compute the third derivatives, extend the tangent frame so that its intrinsic covariant derivatives vanish at the chosen point. Its ambient derivatives still have the normal components

$$D_{e_\beta}\nu = \sum_\gamma L_{\beta\gamma} e_\gamma, \qquad D_{e_\beta} e_\alpha = -L_{\beta\alpha}\nu.$$

Differentiating the identity $D^2u(\nu, e_\alpha) = 0$ along e_β therefore gives

$$\begin{aligned} 0 &= D^3u(e_\beta, \nu, e_\alpha) + D^2u(D_{e_\beta}\nu, e_\alpha) + D^2u(\nu, D_{e_\beta}e_\alpha) \\ &= u_{\nu\alpha\beta} + (L^2)_{\alpha\beta} - L_{\alpha\beta}u_{\nu\nu}. \end{aligned}$$

Consequently

$$u_{\nu\alpha\beta} = -HL_{\alpha\beta} - (L^2)_{\alpha\beta}.$$

Taking the trace and differentiating $\Delta u = 0$ in the normal direction gives

$$u_{\nu\nu\nu} = -\sum_\alpha u_{\nu\alpha\alpha} = H^2 + |L|^2 = |D^2u|^2 = w^2.$$

The penultimate equality uses the block form of D^2u, whose mixed normal–tangential entries vanish.

In this frame $p = e_5$, and direct substitution into S_0, V gives

$$\begin{gathered} (S_0)_{\alpha\beta} = \frac{-HL_{\alpha\beta} - (L^2)_{\alpha\beta}}{w^2}, \qquad (S_0)_{55} = (S_0)_{5\alpha} = 0, \\ V_{5\alpha\beta} = (S_0)_{\alpha\beta}, \qquad V_{555} = 1, \qquad V_{55\alpha} = V_{\alpha\beta\gamma} = 0, \end{gathered}$$

with the corresponding permutations. Thus V agrees with D^3u/w^2 in every component having an odd number of normal slots. Only these components are determined by the preceding calculation.

To see why they suffice for the flux, let $\mathcal{R} = I - 2p\otimes p$, the reflection reversing the normal. The block form of A gives $\mathcal{R}A\mathcal{R}^{\mathsf{T}} = A$, while $\mathcal{R}p = -p$. The tensor formulas in Appendix B transform orthogonally in their free indices, and each monomial of U, K contains an even number of p factors. Hence

$$\mathcal{R}^{\otimes 4} G(A, p) = G(\mathcal{R}A\mathcal{R}^{\mathsf{T}}, \mathcal{R}p) = G(A, -p) = G(A, p).$$

For $C_{ijk} = p_l G_{lijk}$, contracting the first slot and using $\mathcal{R}p = -p$ yields $\mathcal{R}^{\otimes 3}C = -C$. A component with m normal slots is multiplied by $(-1)^m$ under this reflection. For even m it must therefore vanish. This proves the contraction identity

$$C_{ijk}\frac{u_{ijk}}{w^2} = C_{ijk}V_{ijk}.$$

No symmetry of the four-tensor G is required for this argument.

Finally, $X = D^3u/w = w(D^3u/w^2)$ and $H = -\lambda w$, so

$$\begin{aligned} \nu\cdot Z + Hh^2 &= w^{2b+1}(C_{ijk}V_{ijk} + p\cdot M - \lambda f) \\ &= w^{2b+1}S(A, p) \ge \beta w^{2b+1}. \end{aligned} \tag{3.8}$$

We record also the bounds needed for the mixed cutoff terms. The fields and their first derivatives with respect to (A, p) are bounded on the compact normalized coefficient domain. In the actual spatial chain rule,

$$\begin{aligned} |DA|^2 &= \sum_l |X_l - q_l A|^2 = |X|^2 - |q|^2 \le |X|^2, \\ |Dp|^2 &= w^2|A|^2 = w^2, \qquad |q| \le |X|. \end{aligned}$$

Here $X_l = (X_{lij})_{ij}$ and $A : X_l = q_l$. It follows that $|Df| \le C(|X| + w)$ and hence $|F| \le C(|X| + w)$. Together with the exact identity $h\nabla h = \frac{1}{2}w^{2b}F$, the definition of Z, and $f \le 5$, this gives

$$h^2 \le 5w^{2b}, \qquad |h\nabla h| + |Z| \le Cw^{2b}(|X| + w). \tag{3.9}$$

All constants here depend only on the fixed Euclidean multiplier. The exponent $b = 19/20$ and the fields in this section are different from the spherical exponent and fields used in Lemma 2.1.

3.4. **The Hessian zero set.** We now justify integration across $\{D^2u = 0\}$, including its intersection with the physical boundary. The weighted estimate on the spherical link in Section 2 assumed positivity of the Hessian norm at that boundary. No such assertion is available here, so we include the boundary argument.

Put $T = D^2u$, $W_\delta = w^2 + \delta^2$ for $\delta > 0$, and $J_l = T : \partial_l T$. Each component of T is harmonic. Thus

$$\Delta W_\delta = 2|DT|^2, \qquad |\nabla W_\delta|^2 = 4\sum_l J_l^2 \le 4w^2|DT|^2 \le 4W_\delta|DT|^2.$$

Unlike w itself, W_δ is smooth even where $T = 0$. The chain rule gives the positive-term identity

$$\begin{aligned}
\Delta W_\delta^b &= 2bW_\delta^{b-1}|DT|^2 - 4b(1-b)W_\delta^{b-2}\sum_l J_l^2 \\
&= 2b(2b-1)W_\delta^{b-1}|DT|^2 \\
&\quad + 4b(1-b)W_\delta^{b-2}\left(W_\delta|DT|^2 - \sum_l J_l^2\right) \\
&\ge 2b(2b-1)W_\delta^{b-1}|DT|^2.
\end{aligned}$$

The last line uses $1/2 < b < 1$ and Cauchy–Schwarz. For the chosen $b = 19/20$, the two coefficients in the second expression are $171/100$ and $19/100$, respectively.

Choose $0 \le \chi \in C_c^\infty(\mathbb{R}^5)$ equal to one on a neighborhood of a given compact set. Green's formula with outward normal $n = -\nu$ gives

$$\int_\Omega \chi\Delta W_\delta^b = \int_\Omega W_\delta^b\Delta\chi + \int_\Gamma \left(\chi\partial_n W_\delta^b - W_\delta^b\partial_n\chi\right).$$

We bound every term on the right uniformly for $0 < \delta \le 1$. The classical solution has bounded T, DT on the compact support, and $W_\delta^b \le (w^2+1)^b$ there. For $w > 0$, since $b < 1$,

$$|\nabla W_\delta^b| \le 2bW_\delta^{b-1}w|DT| \le 2bw^{2b-1}|DT|.$$

At $w = 0$ the first gradient is zero, and the same bound holds with right side zero because $2b - 1 = 9/10 > 0$. Hence the volume term is bounded by a fixed supremum times $\|\Delta\chi\|_{L^1(\Omega)}$. The two boundary terms are bounded by the corresponding local suprema times $\|\chi\|_{L^1(\Gamma)}$ and $\|\partial_n\chi\|_{L^1(\Gamma)}$, which are finite by (2.2). The differential inequality therefore implies

$$\int_\Omega \chi W_\delta^{b-1}|DT|^2 \le C_{u,\chi} \quad (0 < \delta \le 1).$$

As $\delta \downarrow 0$, the integrand increases on $\{w > 0\}$, because $b - 1 < 0$. Monotone convergence gives

$$w^{2b-2}|D^3u|^2 \in L^1_{\mathrm{loc}}(\overline{\Omega}).$$

The level-set fact from Section 2, applied componentwise to T, gives $DT = 0$ almost everywhere on $\{T = 0\}$. We define the weighted integrand to be zero there. The preliminary constant $C_{u,\chi}$ only establishes this local integrability; it will not enter the constant in (3.1).

Use the fixed cutoff profile ϑ from Lemma 2.1, and set $\vartheta_\delta = \vartheta(w/\delta)$. On its transition band, $\delta < w < 2\delta$, so

$$|\nabla\vartheta_\delta| \le \frac{\|\vartheta'\|_\infty}{\delta}|\nabla w| \le \frac{2\|\vartheta'\|_\infty}{w}|DT|.$$

Off this band the derivative is zero. Consequently

$$\int_\Omega \eta^2 w^{2b}|\nabla\vartheta_\delta|^2 \le C\int_{\{\delta<w<2\delta\}} \eta^2 w^{2b-2}|D^3u|^2 \longrightarrow 0. \tag{3.10}$$

Indeed, the integrable function on the right, extended by zero at $w=0$, dominates it, while the transition-band indicator tends pointwise to zero. For each fixed δ, the cutoff vanishes on an open neighborhood of the zero set. All products of the normalized fields with this cutoff extend smoothly by zero, so their integrations over Ω have only the physical boundary Γ.

3.5. **Stability, absorption, and growth.** Fix the ambient cutoff η in Proposition 3.1 and put $\eta_\delta = \eta\vartheta_\delta$. The test $\eta_\delta h$ is smooth up to the positive side of the boundary and has compact support. It is admissible for stability by Section 2. Multiplying (3.7) by η_δ^2 and integrating the smoothly extended products gives

$$\mu\int_\Omega \eta_\delta^2(w^{2b+2}+w^{2b-2}|D^3u|^2) \le -\int_\Gamma \eta_\delta^2 \nu\cdot Z - 2\int_\Omega \eta_\delta Z\cdot\nabla\eta_\delta - \int_\Omega \eta_\delta^2|\nabla h|^2.$$

The boundary inequality (3.8) gives $-\nu\cdot Z \le Hh^2 - \beta w^{2b+1}$. Stability gives

$$\int_\Gamma H(\eta_\delta h)^2 \le \int_\Omega \left(\eta_\delta^2|\nabla h|^2 + 2\eta_\delta h\nabla h\cdot\nabla\eta_\delta + h^2|\nabla\eta_\delta|^2\right).$$

Substitution cancels the terms $\int \eta_\delta^2|\nabla h|^2$, leaving

$$\mu\int_\Omega \eta_\delta^2(w^{2b+2}+w^{2b-2}|D^3u|^2) + \beta\int_\Gamma \eta_\delta^2 w^{2b+1} \le \int_\Omega h^2|\nabla\eta_\delta|^2 + 2\int_\Omega \eta_\delta(h\nabla h - Z)\cdot\nabla\eta_\delta.$$

To make the absorption explicit, put, on $\{w>0\}$,

$$A_\delta = |\eta_\delta| w^b\sqrt{|X|^2+w^2}, \qquad B_\delta = w^b|\nabla\eta_\delta|.$$

The mixed integrand is at most $C_0A_\delta B_\delta$ by (3.9) and $|X|+w \le \sqrt{2}\sqrt{|X|^2+w^2}$. Young's inequality gives

$$C_0A_\delta B_\delta \le \frac{\mu}{2}A_\delta^2 + \frac{C_0^2}{2\mu}B_\delta^2.$$

Since $A_\delta^2 = \eta_\delta^2(w^{2b+2}+w^{2b-2}|D^3u|^2)$ and $h^2|\nabla\eta_\delta|^2 \le 5B_\delta^2$, we obtain

$$\frac{\mu}{2}\int_\Omega \eta_\delta^2(w^{2b+2}+w^{2b-2}|D^3u|^2) + \beta\int_\Gamma \eta_\delta^2 w^{2b+1} \le C\int_\Omega w^{2b}|\nabla\eta_\delta|^2.$$

This constant depends only on the fixed multiplier and its positive constants μ,β, and is independent of u,δ.

For the limit, use

$$|\nabla\eta_\delta|^2 \le 2\vartheta_\delta^2|\nabla\eta|^2 + 2\eta^2|\nabla\vartheta_\delta|^2.$$

The second weighted integral tends to zero by (3.10). Dominated convergence applies to the first because w is bounded on the compact support. On $\{w>0\}$ one has $\eta_\delta\to\eta$; on $\{w=0\}$ the volume integrands have the zero interpretation fixed above. The boundary integrand also has the correct pointwise limit at every zero of w, since $2b+1>0$. Fatou's lemma on the two volume integrals and the boundary integral therefore gives the same estimate with η in place of η_δ. Finally,

$$2b+2=\frac{39}{10}, \qquad 2b-2=-\frac{1}{10}, \qquad 2b+1=\frac{29}{10}, \qquad 2b=\frac{19}{10}.$$

Absorbing the fixed factors $\mu/2$ and β into the constant proves (3.1).

For the growth estimate put $P = 39/10$ and choose $\psi \in C_c^\infty(B_{2R}(x_0))$ with $0 \le \psi \le 1$, equal to one on $B_R(x_0)$ and satisfying $|\nabla\psi| \le C/R$. The cutoff $\eta = \psi^{P/2}$ is admissible as a limit of smooth cutoffs. Explicitly, for $\varepsilon > 0$ the functions

$$\eta_\varepsilon = (\psi+\varepsilon)^{P/2} - \varepsilon^{P/2}, \qquad \nabla\eta_\varepsilon = \frac{P}{2}(\psi+\varepsilon)^{P/2-1}\nabla\psi$$

are smooth with the same compact support and converge uniformly with their first derivatives to $\psi^{P/2}$, since $P/2 = 39/20 > 1$. The local integrability already proved, and finite local boundary area, permit passage to the limit in (3.1).

Write

$$A_0 = \int_\Omega w^P \psi^P, \qquad B_0 = \int_\Omega |\nabla\psi|^P.$$

The full right side of (3.1) is bounded by a constant times $\int w^{P-2}\psi^{P-2}|\nabla\psi|^2$. Hölder's inequality, with conjugate exponents $39/19$ and $39/20$, gives

$$\begin{aligned}\int_\Omega w^{P-2}\psi^{P-2}|\nabla\psi|^2 &= \int_\Omega (w^P\psi^P)^{19/39}(|\nabla\psi|^P)^{20/39} \\ &\le A_0^{19/39} B_0^{20/39}.\end{aligned}$$

In particular, $A_0 \le C A_0^{19/39} B_0^{20/39}$. If $A_0 > 0$, division by $A_0^{19/39}$ and taking the power $39/20$ give $A_0 \le C B_0$. The full right side is then at most CB_0 as well. If $A_0 = 0$, the same Hölder bound makes that right side zero, so all three nonnegative left-side integrals vanish. Finally,

$$B_0 \le C R^{-P}|B_{2R}(x_0)| \le C R^{5-P} = C R^{11/10}.$$

Because $\psi = 1$ on $B_R(x_0)$, the full left side controls each of the three integrals in (3.2). This proves that estimate and completes the proof of Proposition 3.1.

4. Selected excess and the free-boundary tree

4.1. **Normalization and threshold neck radii.** Throughout the contradiction argument, a global solution v is called nonflat when $D^2 v \not\equiv 0$ in $\{v > 0\}$. Suppose that such a global classical stable solution exists in $\mathbb{R}^5$. This is precisely the nonflatness hypothesis of [6, Lemma 5.2], whose contradiction reduction produces a normalized global classical stable solution u. The reduction may involve point-picking and passage to a limit. With $\Omega = \{u > 0\}$, $\Gamma = \partial\Omega$ and $w = |D^2 u|$, its conclusions and the global gradient bound are

$$0 \in \Gamma, \qquad w(0) = 1, \qquad w \le 1, \qquad |\nabla u| \le 1.$$

Lemma 2.1 permits the dimension-general framework of [6]. Proposition 3.1 supplies

$$\int_\Omega w^P \eta^2 \le C \int_\Omega w^{P-2}|\nabla\eta|^2, \qquad P = \frac{39}{10}, \tag{4.1}$$

for every compact cutoff η. Its proof includes the Hessian zero set. We next state the precise geometric and selection estimates that enter the argument.

Let $\eta_*(5) > 0$ be the Hessian epsilon-regularity threshold in [6, Lemma 3.4], and fix

$$\eta_0 = \min\{\eta_*(5), |B_1|^{1/5}\} > 0,$$

as in [6, Assumption 5.3]. Thus the threshold is in the admissible range of that regularity estimate, and its value agrees with the standing choice in the geometric results used below. For $y \in \Gamma$, define

$$r_*(y) = \inf\left\{ r > 0 : \int_{B_r(y)\cap\Omega} w^5 \ge \eta_0^5 \right\}.$$

There is at least one finite threshold radius. Indeed, if $r_*(0) = \infty$, the scale-invariant energy is below η_0^5 in every $B_r(0)$. The rescaled Hessian epsilon-regularity estimate [6, Lemma 3.4] would

give $w(0) \le C\eta_0/r$ for every $r > 0$, contradicting $w(0) = 1$. Thus the first nonempty radius band below produces a neck center. Put $r_{\min} := \eta_0 |B_1|^{-1/5}$; the lower bound below shows that every finite threshold radius is at least $r_{\min}$. Select the neck centers $\mathcal{Z}$ as in [6, Definition 5.4]. More explicitly, split the finite radii into bands $[2^j r_{\min}, 2^{j+1} r_{\min})$. In the first band, take a disjoint Vitali family of threshold balls. In each later band, first discard a candidate y if $B_{4r_*(y)}(y)$ contains a center selected earlier. Then take a disjoint Vitali family in the remaining band. More precisely, if $\mathcal{Y}_j$ is the retained candidate set in band j and $\mathcal{V}_j \subset \mathcal{Y}_j$ is its chosen set of centers, require

$$\bigcup_{y \in \mathcal{Y}_j} B_{r_*(y)}(y) \subset \bigcup_{z \in \mathcal{V}_j} B_{4r_*(z)}(z),$$

with the balls $B_{r_*(z)}(z)$, $z \in \mathcal{V}_j$, pairwise disjoint. To obtain the factor four explicitly, write the band as $[L, 2L)$ and first choose a maximal disjoint family among its retained balls of radii in $[3L/2, 2L)$. Among the smaller retained balls disjoint from that family, choose a maximal disjoint family with radii in $[L, 3L/2)$. Every unchosen ball $B_r(y)$ meets a chosen ball $B_s(z)$ with $r < 3s/2$. Consequently

$$B_r(y) \subset B_{2r+s}(z) \subset B_{4s}(z).$$

This proves the displayed covering; an arbitrary maximal disjoint family without radius ordering would not by itself give this factor. The families are countable, since disjoint open balls contain distinct points of $\mathbb{Q}^5$. The union of the chosen centers is $\mathcal{Z} = \bigcup_{j \ge 0} \mathcal{V}_j$. At each selected center,

$$\int_{\Omega \cap B_{r_*(z)}(z)} w^5 = \eta_0^5, \qquad \sup_{\Omega \cap B_{r_*(z)}(z)} w \le C/r_*(z). \tag{4.2}$$

The threshold is attained because the integral is continuous in its radius. Also, $w \le 1$ implies

$$\eta_0^5 \le |B_1| r_*(z)^5, \qquad r_*(z) \ge r_{\min} > 0.$$

The selected threshold balls are pairwise disjoint. Indeed, disjointness within each band is part of the construction. If z is in a later band than z', then $r_*(z') < r_*(z)$ and $|z - z'| \ge 4r_*(z) > r_*(z) + r_*(z')$. In particular, every bounded region contains only finitely many of these balls with a fixed upper radius bound.

We shall use the following form of local neck comparability: for each fixed $M \ge 1$,

$$\sup_{\Omega \cap B_{Mr_*(z)}(z)} w \le \frac{C_M}{r_*(z)} \qquad (z \in \mathcal{Z}). \tag{4.3}$$

This is [6, Lemma 5.7]. Its case $M = 1$ gives the second statement in (4.2). The constants C_M are independent of the chosen center.

4.2. **The combined excess and the selection.** Use the unnormalized Gaussian $G_t(x) = t^{-5} e^{-|x|^2/t^2}$ and define

$$W_{t,y} = \frac{1}{2} \int_{\mathbb{R}^5} (|\nabla u|^2 + \mathbf{1}_\Omega) G_t(x - y)\, dx - t^{-2} \int_{\mathbb{R}^5} u^2 G_t(x - y)\, dx.$$

For the normalized nonflat solution, [6, Proposition 4.12 and Lemma 7.3] give the vee limit $W_\infty = \pi^{5/2}/2$. This is the value of the Gaussian functional on a vee: for $u(x) = |x_5|$, one has $|\nabla u| = 1$ and $\mathbf{1}_\Omega = 1$ almost everywhere, while

$$\int_{\mathbb{R}^5} G_t = \pi^{5/2}, \qquad \int_{\mathbb{R}^5} x_5^2 G_t = \frac{t^2}{2} \pi^{5/2}.$$

Substitution gives $W_{t,0} = \pi^{5/2}/2$ at every scale. With $V_{z,e}(x) = |e \cdot (x - z)|$, put

$$\Theta^e_{t,z} = \sup_{y \in (z + e^\perp) \cap B_t(z)} (W_\infty - W_{t,y}),$$

$$M^e_{t,z} = t^{-7} \int_{B_t(z)} (u - V_{z,e})^2\, dx,$$

$$\mathcal{E}_z(t;e)^2 = \Theta^e_{t,z} + M^e_{t,z}, \qquad \mathcal{E}_z(t) = \min_{e\in S^4} \mathcal{E}_z(t;e).$$

These are the combined excess quantities of [6, Section 7]. We justify their nonnegativity and the minimizing direction. The Gaussian Weiss monotonicity formula [6, Lemma 7.2] is

$$\partial_t W_{t,y} = \frac{2}{t^3}\int_{\mathbb{R}^5} (u-(x-y)\cdot\nabla u)^2 G_t(x-y)\,dx \geq 0.$$

Together with $W_{T,y} \to W_\infty$ for every fixed y, it gives $W_{t,y} \leq W_\infty$. Thus both terms defining $\mathcal{E}_z(t;e)^2$ are nonnegative. For fixed t, z, the map $y \mapsto W_{t,y}$ is continuous: the energy density is bounded, $u(x) \leq u(z) + |x-z|$, and translated Gaussians over a compact set of centers have a common integrable bound given by a Gaussian times a quadratic polynomial. Indeed,

$$e^{-|x-y|^2/t^2} \leq e^{|y-z|^2/t^2} e^{-|x-z|^2/(2t^2)},$$

and the first factor is uniformly bounded on that compact set. Dominated convergence applies to both integrals.

Put $K_e = \{z+v : v\cdot e = 0,\ |v| \leq t\}$. The supremum defining $\Theta^e_{t,z}$ equals the maximum over K_e, since a boundary point can be approached radially within the open ridge ball. If $e_j \to e$, maximizers in K_{e_j} have convergent subsequences in $\overline{B_t(z)}$ whose limits belong to K_e; this gives the upper limit inequality for Θ. Conversely, if $z+v$ is a maximizer in K_e, then $v_j = v - (v\cdot e_j)e_j$ satisfies $v_j \cdot e_j = 0$, $|v_j| \leq t$, and $v_j \to v$, giving the lower limit inequality. Hence $\Theta^e_{t,z}$ is continuous in e. The same is true of $M^e_{t,z}$, by dominated convergence and $|V_{z,e} - V_{z,e'}| \leq t|e-e'|$ on $B_t(z)$. The nonnegative square root $\mathcal{E}_z(t;e)$ is therefore continuous, and its minimum exists on the compact sphere S^4. The same continuity argument gives the minimizing vee in (4.13) below. Define the neck charge by

$$\rho_z(t)^2 = \sum_{z_i \in \mathcal{Z}_{t/2}\cap B_{t/2}(z)} \left(\frac{r_*(z_i)}{t}\right)^3, \qquad \mathcal{Z}_s = \{z\in\mathcal{Z} : r_*(z) \leq s\}. \tag{4.4}$$

Define

$$F^a_u(t) = \sup_{z\in\mathcal{Z}_t} \frac{\mathcal{E}_z(8t)}{\rho_z(2t)^a}, \qquad \alpha_* = \inf\left\{a>0 : \limsup_{t\to\infty} F^a_u(t) = \infty\right\}.$$

We set $F^a_u(t) = 0$ when $\mathcal{Z}_t = \varnothing$. For large t, the supremum is over a nonempty set. Each denominator is positive because the selected center itself contributes to the charge. The compactness statement [6, Lemma 7.8] also gives a universal nondecreasing modulus ω_*, with $\omega_*(s) \to 0$ as $s \downarrow 0$, such that

$$\mathcal{E}_z(t) \leq \omega_*\left(\frac{r_*(z)}{t}\right) \qquad (z\in\mathcal{Z},\ t>0). \tag{4.5}$$

We make the source notation explicit; the relevant definitions and comparisons are [6, Definitions 6.6 and 6.12, equations (6.6)–(6.8), Proposition 6.13, and Lemma 7.7]. In dimension n, the source charge is

$$\varrho_x(S)^2 = \sum_{\zeta\in\mathcal{Z}_{S/2}\cap B_{S/2}(x)} \left(\frac{r_*(\zeta)}{S}\right)^{n-2}, \qquad \mathcal{Z}_s = \{\zeta\in\mathcal{Z} : r_*(\zeta) \leq s\}.$$

The cutoff, center ball, and denominator agree with (4.4). Here $n-2 = 3$ and both charges are nonnegative square roots, so $\varrho_x(S) = \rho_x(S)$ exactly. Write the source's symmetric gradient excess as

$$E^{\mathrm{sym}}_x(S) = \min_{e\in S^4}\left(S^{-5}\int_{B_S(x)} |\nabla u - (\nabla u\cdot e)e|^2\,dx\right)^{1/2}.$$

Its Proposition 6.13 gives $\varrho_x(S) \le CE_x^{\mathrm{sym}}(4S)$, and its Lemma 7.7 gives $E_x^{\mathrm{sym}}(S) \le C\mathcal{E}_x(S)$. Both hold for every center and every positive scale in the normalized framework already established above. Applying the first at $S = 2t$ and the second at $S = 8t$ yields

$$\rho_z(2t) = \varrho_z(2t) \le CE_z^{\mathrm{sym}}(8t) \le C\mathcal{E}_z(8t).$$

Together with the uniform bounds proved next, this gives

$$\rho_z(2t) \le C\mathcal{E}_z(8t), \qquad \mathcal{E}_z(8t) + \rho_z(2t) \le C \quad (z \in \mathcal{Z}_t). \tag{4.6}$$

For completeness, the uniform upper bound also uses the global Lipschitz bound. If $z \in \Gamma$ and $y \in B_t(z)$, then $u(y + t\xi) \le t(1 + |\xi|)$, so

$$t^{-2} \int_{\mathbb{R}^5} u(x)^2 G_t(x - y)\, dx \le \int_{\mathbb{R}^5} (1 + |\xi|)^2 e^{-|\xi|^2}\, d\xi =: C_G < \infty.$$

The other term of $W_{t,y}$ is nonnegative, hence $W_{t,y} \ge -C_G$. Also $u, V_{z,e} \le t$ on $B_t(z)$ gives $M_{t,z}^e \le 4|B_1|$. Consequently $\mathcal{E}_z(t; e)^2 \le W_\infty + C_G + 4|B_1|$, uniformly in t, z, e. The first comparison in (4.6) then bounds the charge as well. For $z \in \mathcal{Z}_t$, the center itself contributes to the charge, so $\rho_z(2t) \ge (r_{\min}/(2t))^{3/2}$. In particular, for each $T > 0$, the uniform upper bound for the excess gives

$$\sup_{0<t\le T} F_u^a(t) \le C \left(\frac{2T}{r_{\min}}\right)^{3a/2} < \infty.$$

Thus both F_u^a and its running maximum are finite on bounded intervals, including intervals where some of the sets $\mathcal{Z}_t$ are empty. To prove $\alpha_* \le 1$, fix one center $z \in \mathcal{Z}$. For every $a > 1$, (4.6) gives

$$F_u^a(t) \ge c_a \mathcal{E}_z(8t)^{1-a} \longrightarrow \infty.$$

Here $t \to \infty$ and (4.5) gives $\mathcal{E}_z(8t) \to 0$. The excess is positive because the charge is positive.

We now give the selection statement with its precise range of exponents. This will also justify its later use with $a = 5/8$. For $0 < \zeta < 1$, set

$$\mathfrak{N}(\zeta; R, z) := (\zeta R)^{-5} \left| \bigcup_{y \in \mathcal{Z}_{\zeta R} \cap B_R(z)} B_{\zeta R}(y) \right|.$$

Lemma 4.1 (Selection package)**.** *Fix $\beta_\circ = 1/4$ and an exponent $\alpha_* < a \le 101/100$. There are sequences $R_k \to \infty$, $z_k \in \mathcal{Z}_{R_k}$, and $\varepsilon_k \downarrow 0$ with the following properties. Suppressing the index k, one has*

$$\mathcal{E}_{z_0}(8R) \le C_a \varepsilon, \qquad \mathcal{E}_{z_0}(t) \ge \frac{c_a \varepsilon}{1 + \log(R/t)} \quad (0 < t \le R). \tag{4.7}$$

Moreover,

$$\mathfrak{N}(\zeta; R, z_0) \le C\zeta^{-3-2\beta_\circ} \qquad (0 < \zeta < 1), \tag{4.8}$$

and, whenever $0 < \zeta < 1/2$ and $\mathcal{F} \subset \mathcal{Z}_{\zeta R} \cap B_R(z_0)$ is such that the balls $\{B_{\zeta R}(z) : z \in \mathcal{F}\}$ are pairwise disjoint,

$$\sum_{z \in \mathcal{F}} \mathcal{E}_z(8\zeta R)^{2/a} \le C_a \zeta^{-3} \varepsilon^{2/a}. \tag{4.9}$$

All constants are independent of k.

Proof. We explain the selection from [6, Section 8], including the point that its proofs permit every positive $a > \alpha_*$ in the stated range.

First, the divergence set in the definition of α_* is upward closed. If $a > b > 0$, the uniform upper bound for the charge gives $F_u^a(t) \ge C^{b-a} F_u^b(t)$. This set is nonempty because it contains every $a > 1$. If $a > \alpha_*$, its definition as an infimum therefore supplies an exponent $b < a$ in that set, and upward closure puts a in the same set. No assertion at the endpoint $a = \alpha_*$ is needed. It follows that

$$\widetilde{F}_u^a(T) := \sup_{0<t\le T} F_u^a(t) \longrightarrow \infty.$$

To choose scales at which this running maximum is almost attained, first take $S_k \to \infty$ with $\widetilde{F}_u^a(S_k) \to \infty$. Choose $T_k \le S_k$ and $y_k \in \mathcal{Z}_{T_k}$ such that

$$\frac{\mathcal{E}_{y_k}(8T_k)}{\rho_{y_k}(2T_k)^a} \ge \frac{1}{2}\widetilde{F}_u^a(S_k) \ge \frac{1}{2}\widetilde{F}_u^a(T_k).$$

The boundedness of the running maximum on bounded intervals forces $T_k \to \infty$. Thus these scales almost realize their own running maximum. Put $\epsilon_k^0 = \mathcal{E}_{y_k}(8T_k)$. Then

$$\mathcal{E}_z(8t) \le 2\epsilon_k^0 \left(\frac{\rho_z(2t)}{\rho_{y_k}(2T_k)}\right)^a \quad (z \in \mathcal{Z}_t,\ t \le T_k). \tag{4.10}$$

The numerator in the quotient defining F_u^a is bounded. Consequently $Q_k := \rho_{y_k}(2T_k) \to 0$. The contribution of y_k to this charge gives

$$\left(\frac{r_*(y_k)}{2T_k}\right)^3 \le Q_k^2, \qquad \frac{r_*(y_k)}{T_k} \le 2Q_k^{2/3} \longrightarrow 0.$$

Equation (4.5) then gives $\epsilon_k^0 \to 0$.

The refined selection in [6, Lemma 8.4] chooses $s_k \in (0, 1/4]$ and a center z_k so that

$$R_k = s_k T_k, \qquad \varepsilon_k = s_k^{a\beta_\circ}\epsilon_k^0, \qquad B_{4R_k}(z_k) \subset B_{T_k}(y_k),$$

and

$$\rho_{z_k}(4R_k) \le 8 s_k^{\beta_\circ}\rho_{y_k}(2T_k). \tag{4.11}$$

Here is the precise infimum choice. Call $s \in (0, 1/4]$ admissible if some $z \in \mathcal{Z}_{sT_k}$ satisfies

$$B_{4sT_k}(z) \subset B_{T_k}(y_k), \qquad \rho_z(4sT_k) \le 8s^{\beta_\circ}Q_k,$$

and denote the infimum of these scales by σ_k. The admissible class is nonempty: $s = 1/4$ and $z = y_k$ work for large k. Indeed, $r_*(y_k)/T_k \to 0$ gives $y_k \in \mathcal{Z}_{T_k/4}$ eventually, and the ball inclusion is then immediate. The index set of $\rho_{y_k}(T_k)$ is contained in that of $\rho_{y_k}(2T_k)$, so the dimension-five normalization of the charge gives

$$\rho_{y_k}(T_k) \le 2^{3/2}\rho_{y_k}(2T_k) < 8\,4^{-1/4}\rho_{y_k}(2T_k).$$

This is precisely the admissibility inequality with $s = 1/4$. Moreover, $z \in \mathcal{Z}_{sT_k}$ implies $r_{\min} \le r_*(z) \le sT_k$, so $\sigma_k \ge r_{\min}/T_k > 0$. Choose an admissible s_k with

$$\sigma_k \le s_k \le \min\{2\sigma_k, 1/4\},$$

and a center z_k witnessing its admissibility. Such a choice exists by the definition of the infimum; when $\sigma_k = 1/4$, use the already admissible scale $1/4$. In particular, no attainment of the infimum is being assumed. The contribution of z_k to its own charge gives

$$\left(\frac{r_{\min}}{4R_k}\right)^{3/2} \le \rho_{z_k}(4R_k) \le 8s_k^{\beta_\circ}\rho_{y_k}(2T_k) \longrightarrow 0.$$

It follows that $R_k \to \infty$.

We recall why this selection yields the two family estimates. Minimality of the refined scale gives

$$\rho_z(4tR_k) > t^{\beta_\circ}\rho_{z_k}(4R_k)$$

whenever $0 < t < 1/4$, $z \in \mathcal{Z}_{tR_k}$, and $B_{4tR_k}(z) \subset B_{4R_k}(z_k)$. Indeed, otherwise ts_k would be admissible in the original selection: the ball inclusion is inherited and

$$\rho_z(4tR_k) \le t^{\beta_\circ}\rho_{z_k}(4R_k) \le 8(ts_k)^{\beta_\circ}\rho_{y_k}(2T_k).$$

But $ts_k < s_k/4 \le \sigma_k/2 < \sigma_k$, contradicting the definition of σ_k. For a disjoint family as in (4.9), the definition of the charge gives

$$\zeta^3 \sum_{z\in\mathcal{F}} \rho_z(2\zeta R_k)^2 \le 8\rho_{z_k}(4R_k)^2. \tag{4.12}$$

Indeed, put $I_z = \mathcal{Z}_{\zeta R_k} \cap B_{\zeta R_k}(z)$. These index sets are disjoint. Since $z \in B_{R_k}(z_k)$ and $\zeta < 1/2$, their union is contained in $\mathcal{Z}_{2R_k} \cap B_{2R_k}(z_k)$. Thus

$$\zeta^3 \sum_{z\in\mathcal{F}} \rho_z(2\zeta R_k)^2 = \sum_{z\in\mathcal{F}} \sum_{z_i\in I_z} \frac{r_*(z_i)^3}{8R_k^3} = 8 \sum_{z_i\in\bigcup I_z} \left(\frac{r_*(z_i)}{4R_k}\right)^3,$$

which proves (4.12). Apply (4.10) at $t = \zeta R_k$, raise it to power $2/a$, and sum over $\mathcal{F}$. Equations (4.11) and (4.12) give

$$\begin{aligned}\sum_{z\in\mathcal{F}} \mathcal{E}_z(8\zeta R_k)^{2/a} &\le 2^{2/a}(\epsilon_k^0)^{2/a} Q_k^{-2} \sum_{z\in\mathcal{F}} \rho_z(2\zeta R_k)^2 \\ &\le 2^{2/a+9}\zeta^{-3}(\epsilon_k^0)^{2/a} s_k^{2\beta_\circ} = 2^{2/a+9}\zeta^{-3}\varepsilon_k^{2/a}.\end{aligned}$$

The last equality uses $(s_k^{a\beta_\circ})^{2/a} = s_k^{2\beta_\circ}$. This is (4.9). Applying the initial maximality at $t = R_k$, and using $\rho_{z_k}(2R_k) \le 2^{3/2}\rho_{z_k}(4R_k)$, also gives $\mathcal{E}_{z_k}(8R_k) \le C_a\varepsilon_k$.

Here are the details of the covering count. For $\zeta \ge 1/2$, the union in its definition lies in $B_{2R_k}(z_k)$, so the desired bound is immediate. Suppose $\zeta < 1/2$ and split its centers into those with $r_*(z) \le \zeta R_k/2$ and those with $\zeta R_k/2 < r_*(z) \le \zeta R_k$. Choose a maximal disjoint family of radius-ζR_k balls from the first group. Their triples cover all the balls in that group. Minimality above, with $t = \zeta/2$, applies at each chosen center, since $B_{2\zeta R_k}(z) \subset B_{4R_k}(z_k)$, and yields

$$\rho_z(2\zeta R_k) > (\zeta/2)^{\beta_\circ}\rho_{z_k}(4R_k).$$

The latter charge is positive, so it can be cancelled after combining this inequality with (4.12). The chosen family $\mathcal{F}$ therefore satisfies

$$\#\mathcal{F}\,(\zeta/2)^{2\beta_\circ} \le 8\zeta^{-3}, \qquad \#\mathcal{F} \le 8\,2^{2\beta_\circ}\zeta^{-3-2\beta_\circ}.$$

The volume of its triples, divided by $(\zeta R_k)^5$, is at most $3^5|B_1|\#\mathcal{F}$. Every center in the second group contributes more than $(\zeta/8)^3$ to $\rho_{z_k}(4R_k)^2$. Thus its cardinality is at most

$$512\zeta^{-3}\rho_{z_k}(4R_k)^2 \le C\zeta^{-3},$$

where the charge is uniformly bounded by (4.11) and (4.6). Adding the volumes of its radius-ζR_k balls to the first group's covered volume proves (4.8). This is the two-family argument in Step 3 of [6, Lemma 8.4], with the dimension-five charge power.

It remains to obtain the lower bound in (4.7). We spell out the application of the scale propagation and recentering estimates used in [6, Proposition 8.5]. For $0 < t \le 8T_k$, choose a direction e_t minimizing $\mathcal{E}_{z_k}(t; e)$. Apply [6, Proposition 7.9] with outer radius $32T_k$ and ratio $t/(32T_k) \le 1/4$. Also, $|z_k - y_k| < T_k$ implies $B_{8T_k}(y_k) \subset B_{32T_k}(z_k)$, which is the geometric hypothesis of [6, Lemma 7.10]. These two results give

$$\epsilon_k^0 = \mathcal{E}_{y_k}(8T_k) \le C\mathcal{E}_{z_k}(32T_k; e_t) \le C(1 + \log(8T_k/t))\mathcal{E}_{z_k}(t).$$

At $t = 8R_k$, comparison with the upper bound just obtained yields

$$s_k^{a\beta_\circ}(1 + \log(1/s_k)) \ge c_a.$$

Since $a\beta_\circ > 0$, the expression on the left tends to zero as $s_k \downarrow 0$. Thus $s_k \ge c_a > 0$. It follows that R_k and T_k are comparable, and so are ε_k and ϵ_k^0. More explicitly, for $0 < t \le R_k$,

$$1 + \log(8T_k/t) = 1 + \log(8/s_k) + \log(R_k/t) \le C_a(1 + \log(R_k/t)).$$

Together with the preceding lower bound, this proves (4.7).

The argument uses $a > \alpha_*$, $a > 0$, and $0 < \beta_\circ < 1/2$; it never uses $a \ge 3/4$. The dimension-five admissibility check was the explicit inequality $2^{3/2} < 8\,4^{-1/4}$ above. Finally, pass to a subsequence so that ε_k decreases to zero; all the estimates are preserved. □

For the first application we may take $a = 101/100$, since $\alpha_* \le 1$. We fix such a selected sequence and suppress its index. Constants may depend on this fixed exponent. At any fixed positive fraction of R, the lower bound in (4.7) is simply $c_a\varepsilon$.

4.3. **The dyadic tree and its signed domains.** For $B = B_{r_B}(x_B)$ with $x_B \in \Gamma$, define

$$h_B^2 = \min_{e\in S^4} r_B^{-7} \int_{16B} \big|u - |e\cdot(x-x_B)|\big|^2. \tag{4.13}$$

Choose a minimizing direction and write $\ell_B(x) = e_B\cdot(x-x_B)$ and $V_B = |\ell_B|$. The choice of V_B determines e_B only up to sign. We fix these signs along the tree below. This is the centered excess of [6, Definition 9.1].

We now specify the geometric notation used in the rest of the proof. The ball tree is the one constructed in [6, Section 9], rooted at $B_R(z_0)$. Its generation-j radii are $r_j = 2^{-j}R$. For each $j \ge 1$, choose a maximal $r_j/4$-separated family of free-boundary centers in the root ball and the associated radius-r_j balls. The active family $\mathcal{D}^0$ consists of the root. An active ball B is branching when $4B \cap \mathcal{Z} \ne \varnothing$ and $h_B < \delta$, where $\delta > 0$ is fixed sufficiently small. Its children are the balls at the next generation whose centers lie in $\frac{5}{4}B$. Recursively, $\mathcal{D}^{j+1}$ consists of all children of the branching balls in $\mathcal{D}^j$, with each ball counted once even if it has several parents. A predecessor is chosen for each child to propagate the sign of its affine map.

An active ball which does not branch is terminal. It is regular when $4B \cap \mathcal{Z} = \varnothing$, and is a neck terminal otherwise. Write $\mathcal{B}^j, \mathcal{R}^j, \mathcal{N}^j$ for these three subfamilies and $\mathcal{B}, \mathcal{R}, \mathcal{N}$ for their unions over the generations. Thus a neck terminal satisfies $h_B \ge \delta$. The construction has no artificial last generation; its stopping rules are exactly these geometric rules.

For the root, the vee term of the combined excess and its fixed-scale propagation give

$$h_{B_R(z_0)} \le 16^{7/2}\mathcal{E}_{z_0}(16R) \le C\mathcal{E}_{z_0}(8R) \le C_a\varepsilon.$$

The first factor is the change from the normalization R^{-7} in (4.13) to $(16R)^{-7}$ in the vee excess. The middle inequality is [6, Proposition 7.9] with outer radius $16R$ and scale ratio $1/2$. Its constant is dimensional; only the last constant depends on the fixed selection exponent. Thus the root satisfies $h_{B_R(z_0)} < \delta$ for all sufficiently large k, including after the later reselection with $a = 5/8$. Since $z_0 \in \bar{\mathcal{Z}}$, it also meets the neck-center condition and hence is branching.

The geometric arguments below use the selection only through root flatness and separation from the root threshold radius. Both follow for every fixed $a > \alpha_*$ admitted by the selection package. Indeed, the center $z_0 \in \mathcal{Z}_R$ contributes to its charge, so the charge–excess comparison gives

$$\left(\frac{r_*(z_0)}{2R}\right)^{3/2} \le \rho_{z_0}(2R) \le C\mathcal{E}_{z_0}(8R) \le C_a\varepsilon.$$

Consequently

$$\frac{r_*(z_0)}{R} \le C_a\varepsilon^{2/3} \longrightarrow 0, \qquad h_{B_R(z_0)} \le C_a\varepsilon \longrightarrow 0.$$

In particular, this separation is available before the active-radius estimate is proved. The other inputs are the normalized classical stable solution in dimension $5 < n_{\mathrm{cone}}$, the fixed threshold and neck properties of Section 5 of [6], and the tree's geometric stopping rules. The following verification of the active-radius, terminal-comparison, signed-covering and boundary estimates uses exactly these inputs. The exponent a does not enter any of their local arguments; it enters only the selection upper bound and the later generation summations. Constants may depend on the fixed a through C_a and the index from which the root estimates hold.

For every branching ball B, write $\mathrm{Ch}(B)$ for its local child family. The maximal next-generation net gives

$$\Gamma \cap (9/8)B \cap B_R(z_0) \subset \bigcup_{B'\in \mathrm{Ch}(B)} \tfrac{1}{4}B'.$$

Indeed, for any point p on the left, maximality supplies a next-generation center $x_{B'}$ with $|p-x_{B'}| < r_B/8$. Since $|p-x_B| < 9r_B/8$, one has $|x_{B'}-x_B| < 5r_B/4$, so B' is a local child. Its radius is $r_B/2$, and $|p-x_{B'}| < r_B/8 = r_{B'}/4$, as required.

We record how the centered L^2 error enters the lower-slab estimate. Put $c_{\mathrm{av}} = (16^5|B_1|)^{-1/2}$. Cauchy–Schwarz gives

$$\begin{aligned}\frac{1}{r_B}\fint_{16B}|u-V_B| &\le \frac{1}{r_B}\left(\frac{1}{|16B|}\int_{16B}|u-V_B|^2\right)^{1/2}\\ &= \frac{1}{r_B}\left(\frac{h_B^2 r_B^7}{16^5|B_1|r_B^5}\right)^{1/2} = c_{\mathrm{av}}h_B.\end{aligned}$$

For $0 < c_{\mathrm{av}}h_B < 1$, apply [6, Corollary 9.2] with its parameter $\eta = c_{\mathrm{av}}h_B$. Its conclusions are

$$u \ge V_B - Ch_B r_B \quad \text{in } 12B, \qquad \Gamma \cap 12B \subset \{V_B \le Ch_B r_B\},$$

and the corresponding positive-cap inclusion. If $h_B = 0$, apply the same result with every $\eta \in (0,1)$ and let $\eta \downarrow 0$; its dimensional constant is independent of η. For branching parents, the required upper bound on η follows by first fixing $c_{\mathrm{av}}\delta < 1$. The inherited-flatness calculation below then gives a dimensional constant C_{inh}. Decrease the fixed δ further so that $c_{\mathrm{av}}C_{\mathrm{inh}}\delta < 1$. The conversion therefore applies to every active ball, including terminal balls. This order of choices uses no inherited-flatness conclusion to justify its own proof.

Every active ball satisfies the inherited flatness estimate

$$h_B \le C_{\mathrm{inh}}\delta. \tag{4.14}$$

To verify it, let A be a branching parent of B. The centered lower-slab bound gives $|\ell_A(x_B)| \le Ch_A r_A$. Put $\widehat{V}_B(x) = |e_A \cdot (x - x_B)|$. The elementary inequality $||s|-|t|| \le |s-t|$ gives

$$|V_A(x) - \widehat{V}_B(x)| \le |\ell_A(x_B)| \le Ch_A r_A.$$

Also, $r_A = 2r_B$ and $|x_B - x_A| \le 5r_A/4$, so $16B \subset B_{(5/4+8)r_A}(x_A) \subset 16A$. Using $\widehat{V}_B$ as the competitor in (4.13), we obtain

$$\begin{aligned}h_B^2 &\le 2r_B^{-7}\int_{16B}|u-V_A|^2 + 2r_B^{-7}|16B|\,C^2h_A^2r_A^2\\ &\le 2(r_A/r_B)^7h_A^2 + C(r_A/r_B)^2h_A^2 \le Ch_A^2 \le C\delta^2.\end{aligned}$$

First we need the following consequence of the local splitting estimate of [6, Lemma 9.7]. If $z \in \mathcal{Z}$ and

$$u \ge |e\cdot(x-z)| - h \quad \text{in } B_{4s}(z), \qquad s = r_*(z), \qquad h \ge 0,$$

then $s \le C_{\mathrm{trap}}h$. Indeed, the rescaling $\bar{u}(y) = s^{-1}u(z+sy)$ satisfies $0 \in \partial\{\bar{u} > 0\}$, $|\nabla\bar{u}| \le 1$, and

$$\|D^2\bar{u}\|_{L^\infty(B_3\cap\{\bar{u}>0\})} \le C_3, \qquad \int_{B_1\cap\{\bar{u}>0\}}|D^2\bar{u}|^5 = \eta_0^5.$$

The first inequality is [6, Lemma 5.7], with $M = 3$; the second is the threshold identity. If $s > Ah$, rotate e to e_5 to obtain $\bar{u} \ge |y_5| - A^{-1}$ on B_4. Choose A so large that $A^{-1} \le h_{\mathrm{split}}(5, C_3)$ and $|B_1|^{1/5}C_{\mathrm{split}}(5,C_3)A^{-1} < \eta_0$. The explicitly local statement of [6, Lemma 9.7] then gives $|D^2\bar{u}| \le C_{\mathrm{split}}A^{-1}$ on B_1 in its positive phase, contradicting the threshold identity. For $h = 0$ the same contradiction follows by applying the result with every $h' > 0$ and letting $h' \downarrow 0$. This proves the asserted trapping consequence, including its zero-error case, using no selected-scale assertion.

Let $\rho \in (0, 1/16]$ be fixed. Suppose that B is an active ball and $z \in \mathcal{Z} \cap 8B$ has $s = r_*(z) \le \rho r_B$. The normalized L^1 consequence of h_B and [6, Corollary 9.2] give

$$u \ge V_B - Ch_B r_B \quad \text{in } 12B, \qquad |\ell_B(z)| \le Ch_B r_B,$$

where the second assertion follows by setting $x = z$, since $u(z) = 0$. We have $B_{4s}(z) \subset 12B$, because $8 + 4\rho < 12$, and

$$\big|V_B(x) - |e_B \cdot (x - z)|\big| \le |\ell_B(z)|.$$

The trapping consequence just proved therefore gives the conditional bound

$$r_*(z) \le C_{\mathrm{nr}} h_B r_B.$$

Here C_{nr} is fixed before δ is chosen. Choose δ small enough that $2C_{\mathrm{nr}}\delta \le \rho$, and also so that the inherited bound $h_B \le C_{\mathrm{inh}}\delta$ puts every active ball in the small normalized L^1 range used above.

We check the initialization rather than assume the conditional hypothesis at the root. Let C_M be the constant in [6, Lemma 5.7], with $M = 32/\rho$. It suffices to assume $r_*(z_0) < \rho R/C_M$, which holds here for large selected indices by the root separation established above. If $z \in \mathcal{Z} \cap B_{32R}(z_0)$ had $r_*(z) \ge \rho R$, then

$$|z - z_0| < 32R \le (32/\rho) r_*(z).$$

Applying that lemma with center z and nearby center z_0 would imply $r_*(z_0) \ge r_*(z)/C_M \ge \rho R/C_M$, a contradiction. All neck centers in $8B_{\mathrm{root}}$ therefore satisfy the conditional hypothesis. The root is branching and satisfies $h_{B_{\mathrm{root}}} < \delta$ by the root calculation above, so this completes the initial step.

Now suppose the radius bound has been proved at some generation and B' is a child of the branching predecessor B. Since $r_B = 2r_{B'}$ and $|x_{B'} - x_B| \le 5r_B/4$,

$$8B' \subset (21/4)B \subset 8B.$$

For $z \in \mathcal{Z} \cap 8B'$ the preceding-generation bound yields

$$r_*(z) \le C_{\mathrm{nr}} h_B r_B < 2C_{\mathrm{nr}}\delta r_{B'} \le \rho r_{B'}.$$

Applying the conditional bound to B' proves it at the next generation. We have proved, without a restriction on the selected exponent,

$$r_*(z) \le C h_B r_B \quad (z \in \mathcal{Z} \cap 8B), \qquad r_*(z) \le C_a \varepsilon R \quad (z \in \mathcal{Z} \cap B_{8R}(z_0)). \tag{4.15}$$

The root estimate before substituting its selected flatness is $r_*(z) \le C h_{B_R(z_0)} R$ on $\mathcal{Z} \cap B_{8R}(z_0)$. For every branching ball, a center $z \in \mathcal{Z} \cap 4B$ gives $r_{\min} \le C_{\mathrm{nr}}\delta r_B$. Consequently every nonroot active ball, being a child of such a ball, has $r_B \ge r_{\min}/(2C_{\mathrm{nr}}\delta)$. The dyadic generations therefore stop at finite depth, and each generation is finite by the net packing bound. Thus the entire active tree is finite as well.

For completeness, the recentering estimate used for terminal necks is also independent of the selection. Fix an active B, a center $z \in \mathcal{Z} \cap 4B$, put $T = 32r_B$, and choose a minimizing direction e for $E = \mathcal{E}_z(T)$. Then

$$\int_{B_T(z)} |u - V_{z,e}|^2 \le T^7 E^2.$$

If E exceeds a fixed small dimensional constant, the estimate $h_B \le CE$ follows directly from $u(x_B) = 0$, $|\nabla u| \le 1$ and $|V_B(x)| \le |x - x_B|$ on $16B$. In the small-E case, Cauchy–Schwarz gives the exact normalization, with $r' = T/16$,

$$\frac{1}{r'} \fint_{B_T(z)} |u - V_{z,e}| \le \frac{16}{T} \left(\frac{T^7 E^2}{|B_1| T^5} \right)^{1/2} = \frac{16}{|B_1|^{1/2}} E.$$

Thus [6, Corollary 9.2] applies with $\eta = 16|B_1|^{-1/2}E \in (0, 1)$ once the fixed small-E threshold is chosen below $|B_1|^{1/2}/16$; its zero case follows by a positive tolerance. This uses the corollary at radius $T/16$. Since $|x_B - z| \le 4r_B < T/4$, its lower bound at x_B gives $|e \cdot (x_B - z)| \le CET$.

Moreover $16B \subset B_{20r_B}(z) \subset B_T(z)$. The centered vee $\widehat{V}(x) = |e \cdot (x - x_B)|$ is an admissible competitor for h_B, and $|\widehat{V} - V_{z,e}| \le CET$. Therefore

$$\begin{aligned} h_B^2 &\le 2r_B^{-7} \int_{16B} |u - V_{z,e}|^2 + 2r_B^{-7} |16B|\, C^2E^2T^2 \\ &\le CE^2, \end{aligned}$$

where the factor $T/r_B = 32$ is fixed. When $E = 0$, the same conclusion follows either directly from equality with the vee on $B_T(z)$ or by the same estimate with arbitrary positive tolerance. We have established $h_B \le C_{\mathrm{rec}}\mathcal{E}_z(32r_B)$ for every such pair (B, z).

Finally let N be a neck terminal and $z \in \mathcal{Z} \cap 4N$. It satisfies $h_N \ge \delta$, since it meets a neck center and is not branching. By the active-radius and inherited-excess bounds,

$$r_*(z) \le C_{\mathrm{nr}} h_N r_N \le C_{\mathrm{nr}} C_{\mathrm{inh}} \delta r_N.$$

For the reverse comparison use the already established modulus $\mathcal{E}_z(t) \le \omega_*(r_*(z)/t)$ from [6, Lemma 7.8], which concerns every neck center and radius, without a selection exponent. Choose M_δ so large that $C_{\mathrm{rec}}\omega_*(1/(32M_\delta)) < \delta$. If $r_N \ge M_\delta r_*(z)$, recentering would give

$$h_N \le C_{\mathrm{rec}}\mathcal{E}_z(32r_N) \le C_{\mathrm{rec}}\omega_*\left(\frac{1}{32M_\delta}\right) < \delta,$$

contrary to the stopping rule. Hence $r_N < M_\delta r_*(z)$. These two bounds prove $r_N \simeq_\delta r_*(z)$ directly. Since every active center lies in $B_R(z_0)$ and $r_N \le R$, every $z \in \mathcal{Z} \cap 4N$ lies in $B_{5R}(z_0)$. The root radius estimate then gives $r_N \le C_\delta h_{B_R(z_0)} R \le C_{a,\delta}\varepsilon R$. All choices in these local arguments are independent of a except for the already proved constant in $h_{\mathrm{root}} \le C_a\varepsilon$. They therefore remain valid under the later reselection with $a = 5/8$.

For a regular terminal B, $4B \cap \mathcal{Z} = \varnothing$, so $|D^2u| \le C/r_B$ in $(7/2)B \cap \Omega$ by [6, Lemma 5.5]. Rescale by r_B, rotate its minimizing direction to e_5, and use $u \ge V_B - Ch_B r_B$ in $8B$. The local vee-splitting lemma [6, Lemma 9.7] applies with a fixed Hessian bound and height $Ch_B \le CC_{\mathrm{inh}}\delta$. It yields the two regular sheets in $(3/2)B$, with graph Lipschitz norm at most Ch_B and $|\nabla u \mp e_B| \le Ch_B$ on the corresponding sheets. Its order-two conclusion, together with $D_y^2(r_B^{-1}u(x_B + r_B y)) = r_B D_x^2 u$, also gives $|D^2u| \le Ch_B/r_B$ on both sheets. These are precisely the local conclusions needed below; no selection exponent occurs in this application.

Orienting the signed pieces. We use the following precise affine comparison [6, Lemma 9.5]. If L_1, L_2 are real affine functions with $|\nabla L_1| = |\nabla L_2| = 1$, then on every ball $B_s(y)$,

$$\min_{\sigma=\pm1} \left[s|\nabla L_1 - \sigma \nabla L_2| + \|L_1 - \sigma L_2\|_{L^\infty(B_s(y))} \right] \le Cs \left(s^{-7} \int_{B_s(y)} (|L_1| - |L_2|)^2 \right)^{1/2}.$$

This statement concerns arbitrary unit-slope affine functions and has no selected-exponent hypothesis. Every representative ℓ_B has unit gradient by construction. In a parent–child comparison we apply it on $4B$. In the neighboring-ball comparison below we apply it on the radius-$4r_j$ ball in the overlap. The resulting slope and center-value bounds extend to the radius-$8r_j$ ball with the same center by the affine formula, changing only a dimensional constant; that larger ball contains $8D \cap 8E$.

Fix a minimizing vee in each active ball and a representative at the root, and put $e_{\mathrm{root}} = e_{B_R(z_0)}$. If the representative of $A = PB$ has been fixed, compare its vee with that of its child on $4B$. Here $4B \subset 16A$, and the two normalized L^2 errors on $4B$ are bounded by $C(h_B + h_A)$, because $r_A = 2r_B$. The affine comparison [6, Lemma 9.5] therefore gives a sign of the child's representative for which

$$|e_B - e_A| \le C(h_B + h_A), \qquad \|\ell_B - \ell_A\|_{L^\infty(4B)} \le Cr_B(h_B + h_A).$$

By inherited flatness, the direction bound is less than $1/2$ after fixing δ sufficiently small. Consequently $e_B \cdot e_A = 1 - |e_B - e_A|^2/2 > 7/8$, and the opposite representative has negative dot product.

Choosing the positive-dot sign is thus unambiguous, and propagates the root orientation through all chosen predecessors. This argument establishes the representative conclusion of [6, Lemma 9.6] directly from its local affine comparison; no uniqueness of the minimizing vee is asserted.

For a branching ball put $c_* = 1/16$ and define

$$B^{\pm} = B \cap \{\pm \ell_B > c_* r_B\}.$$

For a regular terminal, let $B^{\pm}$ be the intersections with B of the two positive sheets just obtained, labeled by the same orientation.

Same-generation signs. We prove the neighboring-ball comparison with a constant independent of the number of generations. Suppose $D, E \in \mathcal{D}^j$ and $8D \cap 8E \neq \varnothing$. The overlap $16D \cap 16E$ contains a ball of radius $4r_j$ about the midpoint of their centers. Comparing both vees with u on that ball and using the affine comparison gives, for some $\sigma \in \{+1, -1\}$,

$$|e_D - \sigma e_E| + r_j^{-1} \|\ell_D - \sigma \ell_E\|_{L^\infty(8D \cap 8E)} \leq C_2 \delta.$$

The affine bound on the displayed overlap follows from its bound on the midpoint ball and the direction bound, since every point of $8D \cap 8E$ is within Cr_j of that ball. Assume inductively that the oriented conclusion holds at generation $j-1$ with constant $C_0 \geq C_2$. Their parents overlap at eightfold scale, because

$$|x_{PD} - x_{PE}| < \left(\frac{5}{2} + 16 + \frac{5}{2}\right) r_j = 21 r_j < 16 r_{j-1}.$$

The parent conclusion and the two child jumps give the preliminary bound $|e_D - e_E| \leq (C_0 + 2C_1)\delta$. If the preceding affine comparison chose $\sigma = -1$, then

$$2 = |2e_D| \leq |e_D - e_E| + |e_D + e_E| \leq (C_0 + 2C_1 + C_2)\delta < 2,$$

a contradiction once δ is fixed sufficiently small. Thus $\sigma = +1$, and the affine comparison resets the quantitative constant to $C_2 \leq C_0$ at every generation. Starting from the single root proves uniformly in the tree depth

$$|e_D - e_E| + r_j^{-1} \|\ell_D - \ell_E\|_{L^\infty(8D \cap 8E)} \leq C_0 \delta.$$

No holes on a branching ridge. Let B branch, take $y \in B \cap \{\ell_B = 0\} \cap B_{3R/4}(z_0)$, and put $\varrho = 1/64$. If $B_{\varrho r_B}(y)$ meets Γ, a free-boundary point p there lies in the root ball and in $(9/8)B$. The next maximal net has a center $x_{B'}$ with $|p - x_{B'}| < r_B/8$; this is a local child of B, and $|y - x_{B'}| < (\varrho + 1/8) r_B < r_B/4 = r_{B'}/2$. If there were no such child, the ball $B_{\varrho r_B}(y)$ would contain no free boundary. On this connected ball, u is either positive and harmonic, or identically zero, so it is harmonic in either case. Put $s = \varrho r_B/2$. The two averages of u on $B_s(y)$ and $B_{s/2}(y)$ must coincide. The corresponding averages of V_B differ by $5s/32 = 5\varrho r_B/64$. On the other hand, the L^2 vee error on $16B$ bounds each average error by $C\varrho^{-5/2} h_B r_B$. Taking δ sufficiently small contradicts equality. Hence every such ridge point belongs to the half of a local child.

Intersections and signs. Put $\Lambda = 32 = 2/c_*$. If signed pieces D^σ and E^τ intersect, their radii are Λ-comparable. Indeed, if $r_E \leq r_D/\Lambda$ and D branches, then at an intersection point x one has $\sigma \ell_D(x) > c_* r_D$. The free-boundary center x_E lies in $(1 + 1/\Lambda)D$, so $|\ell_D(x_E)| \leq C\delta r_D$, whereas $\sigma \ell_D(x_E) > c_* r_D - r_E \geq c_* r_D/2$, a contradiction. If D is regular, the smaller ball E cannot be the root; hence PE exists and is branching. The intersection and predecessor geometry give

$$4PE \subset B_{r_D + (23/2) r_E}(x_D) \subset 4D,$$

contradicting $4D \cap \mathcal{Z} = \varnothing$. Interchanging D, E proves comparability in both directions.

Their generations therefore differ by at most five. Move the deeper ball to its ancestor in the shallower generation. Its center moves by at most $(5/2)$ times the shallower radius, so it overlaps

the shallower ball at eightfold scale. Combining the preceding neighbor comparison with at most five parent jumps gives

$$|e_D - e_E| \le \eta, \qquad \|\ell_D - \ell_E\|_{L^\infty((3/2)D\cap(3/2)E)} \le \eta \min(r_D, r_E),$$

after choosing a fixed small η and then δ small in terms of it. The number five and Λ are fixed; neither the constant nor this choice depends on the total depth.

If both pieces branch, an intersection of D^+ and E^- would give $\ell_E(x) \ge \ell_D(x) - \eta \min(r_D, r_E) > 0$ for $\eta < c_*$, a contradiction. If D is regular and E branches, then

$$\tau \ell_D(x) > (c_* - \eta) r_E \ge (c_* - \eta) r_D/\Lambda > C h_D r_D.$$

The sheet characterization puts x in the τ sheet of D, so $\sigma = \tau$. If both are regular, their gradient estimates at an intersection point give $\nabla u = \sigma e_D + O(\delta) = \tau e_E + O(\delta)$; $|e_D - e_E| \le \eta$ rules out $\sigma = -\tau$. Thus opposite pieces never intersect.

For completeness, parent–child positive gates have a uniform size. Normalize a parent to $B_2(0)$ and its child to $B_1(\xi)$, with $|\xi| \le 5/2$. At zero error, $\xi_5 = 0$ and both directions are e_5. The point $q_\sigma = (2/3)\xi + \sigma e_5/2$ satisfies

$$|q_\sigma|^2 \le 109/36 < 4, \quad |q_\sigma - \xi|^2 \le 17/18 < 1, \quad \sigma(q_\sigma)_5 = \sigma(q_\sigma - \xi)_5 = 1/2.$$

A sufficiently small fixed ball about it is inside both balls and above height $1/4$ in the corresponding signed coordinates. The actual normal and center errors are $O(\delta)$, so this inclusion persists uniformly after decreasing δ. This ball lies in the branching cores, or in the correctly labeled regular sheet when the child is regular. It supplies a gate of radius $c_g r_B$ in $(PB)^\sigma \cap B^\sigma$.

Set

$$U_\pm = B_{R/2}(z_0) \cap \bigcup_{B\in\mathcal{B}\cup\mathcal{R}} B^\pm, \qquad \mathcal{X}_{\text{neck}} = \overline{B_{R/2}(z_0) \cap \bigcup_{N\in\mathcal{N}} N}. \tag{4.16}$$

The signs label two portions of the same positive phase; they do not denote positive and negative values of u.

The signed cover. Let $x \in \Omega \cap B_{R/2}(z_0)$ lie in a branching ball B but in neither core. Its orthogonal projection $y = x - \ell_B(x) e_B$ remains in B and satisfies $|y - z_0| < R/2 + c_* r_B \le 9R/16 < 3R/4$. The no-hole conclusion gives a local child B' with $|y - x_{B'}| < r_{B'}/2$. Consequently

$$|x - x_{B'}| \le c_* r_B + r_{B'}/2 = 5 r_B/16 < r_{B'}.$$

Starting at the branching root, follow these local children as long as the point is not in a signed core. The finite tree forces a terminal ball. A regular terminal places the point in one of its two regular pieces; a neck terminal places it in $\mathcal{X}_{\text{neck}}$. Thus

$$U_+ \cap U_- = \varnothing, \qquad (\Omega \cap B_{R/2}(z_0)) \setminus \mathcal{X}_{\text{neck}} \subset U_+ \cup U_- \subset \Omega \cap B_{R/2}(z_0). \tag{4.17}$$

This proof follows local children and does not require the independently chosen predecessor to cover the point.

Finite perimeter. Every branching core is the intersection of a ball and a half-space, and hence is a finite-perimeter set. Every regular piece is a ball intersected with a smooth Lipschitz epigraph (or hypograph) supplied by splitting in the larger ball; it also has finite perimeter. Its boundary inside the ball is physical free boundary, while its remaining boundary is spherical. A branching core adds only one flat cutoff face to its spherical boundary. A finite union of finite-perimeter sets has perimeter at most the sum of the perimeters; intersecting with $B_{R/2}(z_0)$ adds only its sphere. The finite tree therefore gives locally finite perimeter for both signed domains.

Put $\mathcal{N}_{\text{in}} = \{N \in \mathcal{N} : 4N \cap B_{R/8}(z_0) \ne \varnothing\}$. The boundary pieces inside the region used below are

$$\Gamma_\pm = \partial^* U_\pm \cap B_{R/8}(z_0),$$
$$\Gamma_\pm^{\text{reg}} = \Gamma_\pm \cap (\Gamma \setminus \mathcal{X}_{\text{neck}}), \qquad \Gamma_\pm^{\text{neck}} = \Gamma_\pm \setminus \Gamma_\pm^{\text{reg}}.$$

Here ∂^* is the reduced boundary. The outward normal is $\nu_\pm$. The regular part is physical free boundary. The neck part contains the remaining boundary, including artificial cuts used to separate the signed pieces. We write $\Gamma^{\mathrm{neck}} = \Gamma^{\mathrm{neck}}_+ \cup \Gamma^{\mathrm{neck}}_-$.

Outside the closed set $\mathcal{X}_{\mathrm{neck}}$, the two disjoint open signed domains cover the positive phase. Hence no point of $\Omega\backslash\mathcal{X}_{\mathrm{neck}}$ can be on their boundary: it lies in one signed domain, whose openness excludes the other. Inside $B_{R/8}(z_0)$ there is no outer cutoff sphere. Thus every nonphysical reduced-boundary point lies in $\mathcal{X}_{\mathrm{neck}}$, and by the definitions the same is true of every point of $\Gamma^{\mathrm{neck}}_\pm$. Finiteness gives the localization recorded in (4.19):

$$\Gamma^{\mathrm{neck}}_\pm \subset \mathcal{X}_{\mathrm{neck}} \cap B_{R/8}(z_0) \subset \bigcup_{N\in\mathcal{N}_{\mathrm{in}}} \overline{N}.$$

At a point of $\Gamma^{\mathrm{reg}}_\sigma$, choose a small one-sided free-boundary chart disjoint from the closed neck region. Its positive side is connected and is partitioned by the two disjoint open signed sets. It must lie wholly in U_σ, because that set approaches the chosen reduced-boundary point. Thus U_σ agrees locally with Ω there, and its outward normal is $\nu_\sigma = -\nabla u$. In particular $\partial_{\nu_\sigma} u = -1$ on the regular physical part.

4.4. **Counting the active balls.** For the inner active family

$$\mathcal{D}^j_{\mathrm{in}} = \{B \in \mathcal{D}^j : 4B \cap B_{R/8}(z_0) \neq \varnothing\},$$

[6, Lemma 10.2] gives

$$\#\mathcal{D}^j_{\mathrm{in}} \le C2^{j(3+2\beta_\circ)}, \qquad \sum_{B\in\mathcal{D}^j_{\mathrm{in}}} h_B^{2/a} \le C_a\varepsilon^{2/a}2^{3j}. \tag{4.18}$$

We derive these bounds from Lemma 4.1 for *all* active balls, keeping the full range of a that will include the later reselection with $a = 5/8$. Choose a fixed constant $K \ge 64$, large enough for the comparisons below and depending only on the already fixed tree parameter δ. For $B \in \mathcal{D}^j_{\mathrm{in}}$, choose a neck center z_B in $4B$ if B is branching or a neck terminal. If B is regular terminal, choose z_B in $4(PB)$, where PB is its branching parent. The active-ball radius estimate, the comparability of terminal neck and threshold radii, and the recentering estimate [6, Lemmas 9.9, 9.14, and 9.15] imply

$$r_*(z_B) \le Kr_j, \qquad |x_B - z_B| \le Kr_j, \qquad h_B \le C_K\mathcal{E}_{z_B}(8Kr_j).$$

For the fine generations, $K2^{-j} < 1/2$, the centers z_B lie in $B_R(z_0)$. Indeed, $4B \cap B_{R/8}(z_0) \neq \varnothing$ gives $|x_B - z_0| \le R/8 + 4r_j$, and hence

$$|z_B - z_0| \le R/8 + (K+4)r_j < R/8 + \frac{K+4}{2K}R < R.$$

The original centers x_B are $r_j/4$-separated. If two balls $B_{Kr_j}(z_B)$ and $B_{Kr_j}(z_{B'})$ intersect, then $|x_B - x_{B'}| \le 4Kr_j$. Packing the mutually disjoint balls $B_{r_j/8}(x_{B'})$ into a ball of radius $(4K+1)r_j$ shows that each ball intersects at most $C(K)$ others, including multiplicities when several B are assigned the same z_B. The intersection graph therefore has a coloring with at most $C(K)+1$ colors, obtained by assigning each vertex a color unused by its already colored neighbors. Each color gives a disjoint family. Thus the number of families depends on K, not on j or k. For one color class $\mathcal{F}$, set $\zeta = K2^{-j}$. Disjointness and (4.8) give

$$\#\mathcal{F}\,|B_1|(\zeta R)^5 \le \left|\bigcup_{z\in\mathcal{Z}_{\zeta R}\cap B_R(z_0)} B_{\zeta R}(z)\right| \le C(\zeta R)^5\zeta^{-3-2\beta_\circ}.$$

Moreover, the recentering bound and (4.9) give

$$\sum_{B\in\mathcal{F}} h_B^{2/a} \le C_{K,a}\sum_{B\in\mathcal{F}} \mathcal{E}_{z_B}(8\zeta R)^{2/a} \le C_{K,a}\zeta^{-3}\varepsilon^{2/a}.$$

Summing over the bounded number of colors and substituting $\zeta = K2^{-j}$ proves both estimates in (4.18) at fine generations. The fixed powers of K are absorbed in the constants.

There are only finitely many coarse generations with $K2^{-j} \geq 1/2$, and each has at most C_K balls. For them, apply the recentering estimate [6, Lemma 9.14] at z_0 with $L = 64K$. Since $r_j \geq R/(2K)$, one has $Lr_j \geq 32R$ and $|x_B - z_0| \leq R/8 + 4r_j \leq Lr_j/4$. Choose in that estimate a direction minimizing $\mathcal{E}_{z_0}(Lr_j)$. It gives $h_B \leq C_K \mathcal{E}_{z_0}(Lr_j)$. The ratio used in fixed-scale propagation is

$$\frac{8R}{Lr_j} \in \left[\frac{1}{8K}, \frac{1}{4}\right].$$

Thus the logarithmic factor in [6, Proposition 7.9] is bounded in terms of K, and (4.7) gives $\mathcal{E}_{z_0}(Lr_j) \leq C_K \mathcal{E}_{z_0}(8R) \leq C_{K,a}\varepsilon$. These generations therefore satisfy the same estimates after an increase of the constants. This proves (4.18) at every generation.

4.5. **The terminal neck region.** Put $\gamma_a = 1 + 2/a$. The localization proved above is

$$\Gamma_{\pm}^{\mathrm{neck}} \subset \bigcup_{N \in \mathcal{N}_{\mathrm{in}}} \overline{N}. \tag{4.19}$$

We now prove the uniform local area bound for its physical and artificial parts, rather than infer it from the stopping rules alone.

Uniform face control at one terminal neck. The reduced boundary of a finite union is, up to a set of $\mathcal{H}^4$ measure zero, contained in the union of the reduced boundaries of its pieces. In the inner ball these consist of physical free boundary and the finitely many flat or spherical artificial faces. Choose $z_N \in \mathcal{Z} \cap 4N$. The already proved terminal comparison gives $r_N \simeq_\delta r_*(z_N)$. If an artificial face of $Q \in \mathcal{B} \cup \mathcal{R}$ meets $\overline{N}$, then $r_Q \simeq_\delta r_N$. For the upper bound, if Q branches and $r_N < r_Q/4$, then $|x_N - x_Q| \leq r_N + r_Q < 5r_Q/4$. Since $x_N \in \Gamma$, the lower-slab estimate for Q bounds $|\ell_Q(x_N)|$. A point y of the artificial face in that intersection therefore satisfies

$$c_* r_Q \leq |\ell_Q(y)| \leq |\ell_Q(x_N)| + r_N \leq C\delta r_Q + r_N,$$

so $r_Q \leq C r_N$; the complementary case is immediate. If Q is regular, $r_N < 3r_Q/5$ would give $|z_N - x_Q| \leq 5r_N + r_Q < 4r_Q$, contrary to its stopping rule. For the reverse inequality suppose $r_Q \leq r_N$ and set $A = Q$ when Q branches and $A = PQ$ otherwise. Pick $z_A \in \mathcal{Z} \cap 4A$. Then

$$r_*(z_A) \leq C\delta r_A \leq 2C\delta r_Q, \qquad |z_A - z_N| \leq C r_N.$$

Since $r_N \simeq_\delta r_*(z_N)$, local neck comparability [6, Lemma 5.7], with a fixed M_δ, gives $r_*(z_A) \geq c_\delta r_N$. Hence $r_Q \geq c_\delta r_N$.

Only $O_\delta(1)$ dyadic generations of Q can occur. At each such generation the separated centers lie in a fixed dilation of N, so elementary packing gives only $O_\delta(1)$ such Q. Each flat or spherical artificial face has area at most $C r_Q^4 \leq C_\delta r_N^4$. The physical part is bounded by $\mathcal{H}^4(\Gamma \cap B_{2r_N}(x_N)) \leq C r_N^4$, using the global upper-perimeter estimate [6, Lemma 3.6]. Consequently

$$\mathcal{H}^4(\Gamma_{\pm}^{\mathrm{neck}} \cap \overline{N}) \leq C_\delta r_N^4, \qquad \mathcal{H}^4(\Gamma_{\pm}^{\mathrm{neck}}) \leq C_\delta \sum_{N \in \mathcal{N}_{\mathrm{in}}} r_N^4.$$

For the radii in this sum, choose $z_N \in \mathcal{Z} \cap 4N$. Since $x_N \in B_R(z_0)$ and $r_N \leq R$, one has $z_N \in B_{5R}(z_0)$. The terminal comparison and root case of (4.15) give

$$r_N \leq C_\delta r_*(z_N) \leq C_a \varepsilon R.$$

The already fixed δ is absorbed into the constants below.

Here is the resulting summation. Since $h_N \geq \delta$ on every neck terminal, (4.18) implies

$$\#(\mathcal{N}_{\mathrm{in}} \cap \mathcal{N}^j) \leq \delta^{-2/a} \sum_{B \in \mathcal{D}_{\mathrm{in}}^j} h_B^{2/a} \leq C_a \varepsilon^{2/a} 2^{3j}.$$

This family is empty unless $2^{-j} \le C_a\varepsilon$. For sufficiently large selected indices, $C_a\varepsilon < 1$. If J is the least integer with $2^{-J} \le C_a\varepsilon$, then

$$\sum_{\{j:2^{-j}\le C_a\varepsilon\}} 2^{-j} = \sum_{j\ge J} 2^{-j} = 2^{1-J} \le 2C_a\varepsilon.$$

Consequently

$$\sum_{N\in\mathcal{N}_{\rm in}} r_N^4 \le C_a R^4 \varepsilon^{2/a} \sum_{\{j:2^{-j}\le C_a\varepsilon\}} 2^{-j} \le C_a R^4 \varepsilon^{1+2/a}.$$

If $\mathcal{N}_{\rm in}$ is empty, both radius sums vanish. Otherwise, the analogous volume sum is bounded by

$$\sum_{N\in\mathcal{N}_{\rm in}} r_N^5 \le \left(\sup_{N\in\mathcal{N}_{\rm in}} r_N\right) \sum_{N\in\mathcal{N}_{\rm in}} r_N^4 \le C_a R^5 \varepsilon^{2+2/a},$$

The finite neck-ball covering and the perimeter estimate now give

$$|\mathcal{X}_{\rm neck} \cap B_{R/8}(z_0)| \le C_a R^5 \varepsilon^{1+\gamma_a}, \qquad \mathcal{H}^4(\Gamma^{\rm neck}) \le C_a R^4 \varepsilon^{\gamma_a}. \tag{4.20}$$

Furthermore, finiteness of the tree implies that each $x \in \mathcal{X}_{\rm neck} \cap B_{R/8}(z_0)$ belongs to some $\overline{N}$ with $N \in \mathcal{N}_{\rm in}$. Since $x_N \in \Gamma$, we have $\operatorname{dist}(x,\Gamma) \le |x - x_N| \le r_N$. The global Lipschitz bound and $u(x_N) = 0$ also give $u(x) \le |x - x_N|$. The terminal radius bound therefore yields

$$\operatorname{dist}(x,\Gamma) \le C_a\varepsilon R, \qquad 0 \le u(x) \le C_a\varepsilon R \quad (x \in \mathcal{X}_{\rm neck} \cap B_{R/8}(z_0)). \tag{4.21}$$

In particular, both bounds hold on $\Gamma^{\rm neck}$ by (4.19).

4.6. **A Hessian estimate on every active covering piece.**

Lemma 4.2 (Hessian control on every covering piece)**.** *The positive phase in $B_{R/8}(z_0)$ is covered, up to a set of five-dimensional measure zero, by pieces associated to inner active balls B, and on each associated piece*

$$w \le C_\delta h_B / r_B. \tag{4.22}$$

Every such piece has volume at most Cr_B^5.

Proof. There are three types of pieces.

For a branching ball B, the signed cores are $B^\pm = B \cap \{\pm\ell_B > r_B/16\}$, where $V_B = |\ell_B|$. The tree parameter δ is fixed sufficiently small. Fix $x \in B^+$ and take $c = 1/128$. Then $B_{cr_B}(x) \subset 2B$ and

$$\ell_B(y) > \frac{r_B}{16} - \frac{r_B}{128} = \frac{7r_B}{128} \qquad (y \in B_{cr_B}(x)).$$

The centered lower-slab estimate, [6, Corollary 9.2], gives $u \ge |\ell_B| - Ch_B r_B$ there. Since $h_B < \delta$, choosing $C\delta < 7/128$ makes the whole ball positive. Hence $u - \ell_B$ is harmonic on that ball. The interior L^2 Hessian estimate and (4.13) give

$$|D^2u(x)| \le Cr_B^{-2-5/2}\|u - \ell_B\|_{L^2(B_{cr_B}(x))} \le Cr_B^{-9/2}(h_B r_B^{7/2}) = Ch_B/r_B.$$

For the second inequality, $V_B = \ell_B$ on this ball, so $\|u - \ell_B\|_{L^2(B_{cr_B}(x))} \le \|u - V_B\|_{L^2(16B)}$. On B^- use the harmonic function $u + \ell_B$ instead.

For a regular terminal, the local splitting argument above already gives $w \le Ch_B/r_B$ on its two positive sheets in B. This is (4.22) for the regular pieces.

For a neck terminal N, choose $z_N \in \mathcal{Z} \cap 4N$. The terminal comparison proved above gives $r_N \simeq_\delta r_*(z_N)$. In particular, $N \subset B_{5r_N}(z_N)$. Since $r_N/r_*(z_N)$ is bounded by a constant depending only on δ, choose a fixed M_δ such that $B_{5r_N}(z_N) \subset B_{M_\delta r_*(z_N)}(z_N)$. Equation (4.3) then gives $w \le C_\delta/r_N$ on $N \cap \Omega$. The stopping rule gives $h_N \ge \delta$: the ball contains a neck and fails to branch precisely because its excess is at least δ. Thus

$$w \le \frac{C_\delta}{r_N} \le \frac{C_\delta}{\delta}\frac{h_N}{r_N},$$

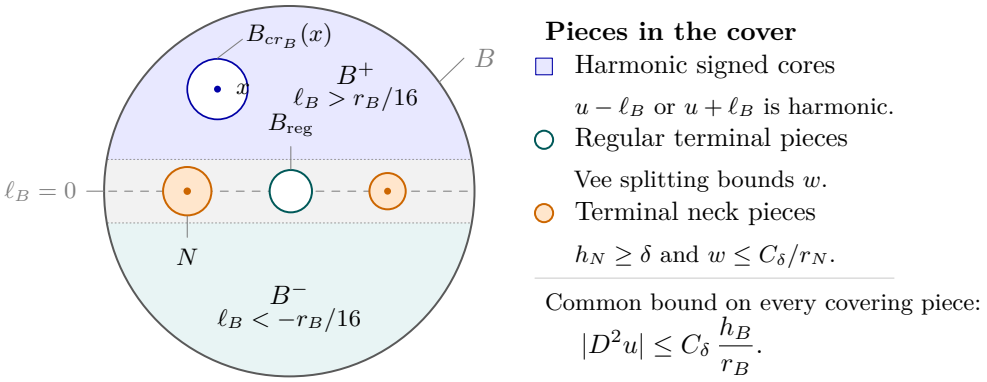


FIGURE 2. A two-dimensional schematic of the hypothetical local cover in the contradiction argument. The dotted lines delimit the signed cores of a branching ball. The central strip may contain free boundary and terminal pieces. Orange discs represent neck neighborhoods; the unfilled disc represents a regular terminal. The relative radii and the strip width are enlarged for clarity. Only the covering geometry is depicted.

which is (4.22) after renaming its fixed constant.

Finally, the proved signed-cover property (4.17) covers the positive phase outside the union of neck terminal balls by branching cores and regular pieces. Adding the neck pieces covers $\Omega \cap B_{R/8}(z_0)$ except possibly for portions of the boundaries of finitely many balls. These have zero five-dimensional measure and do not affect the integrals below. Any piece meeting this region belongs to the indicated inner family. Its volume is at most $|B_1| r_B^5$. Since the integrands are nonnegative, upper bounds obtained by summing the covering pieces do not require them to be disjoint. □

4.7. **Summing over the active generations.**

Lemma 4.3 (Low-power estimate with the full excess amplitude)**.** *For $q = 6/5$ and every fixed $0 < a \leq 101/100$ for which the selected configuration is available,*

$$\int_{\Omega \cap B_{R/8}(z_0)} w^{6/5} \leq C_a \varepsilon^{6/5} R^{19/5}. \tag{4.23}$$

Proof. Put $\theta = qa/2 = 3a/5$. The assumed range of a gives $0 < \theta \leq 303/500 < 1$. For a nonempty generation, apply Hölder's inequality with exponents $1/\theta$ and $1/(1-\theta)$ to $h_B^q = (h_B^{2/a})^\theta \cdot 1$. Equation (4.18) gives

$$\begin{aligned} \sum_{B \in \mathcal{D}_{\mathrm{in}}^j} h_B^q &\leq (\#\mathcal{D}_{\mathrm{in}}^j)^{1-qa/2} \left(\sum_{B \in \mathcal{D}_{\mathrm{in}}^j} h_B^{2/a} \right)^{qa/2} \\ &\leq C_a \varepsilon^q 2^{j(3+2\beta_\circ - \beta_\circ aq)}. \end{aligned}$$

Indeed, the exponent of 2^j is

$$(3 + 2\beta_\circ)(1 - qa/2) + 3qa/2 = 3 + 2\beta_\circ - \beta_\circ aq = \frac{7}{2} - \frac{3a}{10}.$$

An empty generation contributes zero. On each covering piece, Lemma 4.2 bounds the integral by $C_\delta (h_B/r_B)^q r_B^5 = C_\delta h_B^q r_B^{5-q}$. There are at most two signed pieces per ball, which changes only the

fixed constant. Summing the covering integrals consequently yields

$$\int_{\Omega\cap B_{R/8}(z_0)} w^q \le C_\delta \sum_{j\ge 0} r_j^{5-q} \sum_{B\in\mathcal{D}_{\mathrm{in}}^j} h_B^q \le C_a \varepsilon^q R^{5-q} \sum_{j\ge 0} 2^{-j\{2-q-2\beta_\circ+\beta_\circ aq\}}.$$

For $q = 6/5$ and $\beta_\circ = 1/4$, the exponent is

$$2 - \frac{6}{5} - \frac{1}{2} + \frac{3a}{10} = \frac{3}{10}(1+a) > 0.$$

The geometric series can be evaluated independently of the tree depth:

$$\sum_{j\ge 0} 2^{-3(1+a)j/10} = \frac{1}{1-2^{-3(1+a)/10}} \le \frac{1}{1-2^{-3/10}}.$$

Finally $5 - q = 19/5$ and $q = 6/5$, which gives (4.23). □

5. The higher moment and the neck improvement

Lemma 5.1 (Weighted absorption)**.** *Suppose* (4.1) *holds for a power* $P > 2$ *and let* $0 < q < P - 2$. *Then, for every* $x \in \mathbb{R}^5$ *and* $r > 0$,

$$\int_{\Omega\cap B_{r/2}(x)} w^P \le C_{P,q} r^{-(P-q)} \int_{\Omega\cap B_r(x)} w^q. \tag{5.1}$$

Here $C_{P,q}$ *also depends on the fixed constant in* (4.1).

Proof. Choose $\psi \in C_c^\infty(B_r(x))$ with $0 \le \psi \le 1$, $\psi = 1$ on $B_{r/2}(x)$, and $|\nabla\psi| \le C/r$. Set $m = (P-q)/2 > 1$. Then $\eta = \psi^m$ is compactly supported and C^1, with

$$\nabla\eta = m\psi^{m-1}\nabla\psi.$$

The formula also holds where $\psi = 0$, because $m - 1 > 0$. To justify this C^1 test in (4.1), use the smooth approximants

$$\eta_\varepsilon = (\psi+\varepsilon)^m - \varepsilon^m, \qquad \nabla\eta_\varepsilon = m(\psi+\varepsilon)^{m-1}\nabla\psi.$$

They have support contained in $\operatorname{supp}\psi$ and converge with their first derivatives uniformly to ψ^m, because $m > 1$. Since w is locally bounded, both weighted integrals in (4.1) converge by dominated convergence on this common compact support. Define

$$F = \int_\Omega w^P \psi^{P-q}, \qquad L = \int_{\Omega\cap B_r(x)} w^q, \qquad \theta = \frac{P-2-q}{P-q} \in (0,1).$$

Both integrals are finite by local boundedness of w. Substitution in (4.1) gives

$$F \le Cm^2 r^{-2} \int_{\Omega\cap B_r(x)} w^{P-2}\psi^{P-q-2}.$$

The identities

$$P - q - 2 = \theta(P-q), \qquad P - 2 = \theta P + (1-\theta)q$$

give the exact factorization

$$w^{P-2}\psi^{P-q-2} = (w^P\psi^{P-q})^\theta (w^q)^{1-\theta}.$$

Hölder's inequality with conjugate exponents $1/\theta$ and $1/(1-\theta)$, with m^2 absorbed into the constant, therefore yields

$$F \le C_{P,q} r^{-2} F^\theta L^{1-\theta}.$$

If $F = 0$, the inner-ball integral vanishes because $\psi = 1$ there. Otherwise divide by F^θ and raise to the power $1/(1-\theta) = (P-q)/2$. This gives

$$F \le C_{P,q} r^{-2/(1-\theta)} L = C_{P,q} r^{-(P-q)} L.$$

Since $\int_{\Omega \cap B_{r/2}(x)} w^P \le F$, this proves (5.1). □

Proposition 5.2 (Higher moment and neck charge)**.** *For the selected configuration,*

$$\int_{\Omega \cap B_{R/32}(z_0)} w^{39/10} \le C_a \varepsilon^{6/5} R^{11/10}. \tag{5.2}$$

Moreover,

$$\sum_{z_i \in \mathcal{Z}_{R/64} \cap B_{R/64}(z_0)} \left(\frac{r_*(z_i)}{R} \right)^{11/10} \le C_a \varepsilon^{6/5}, \qquad \rho_{z_0}(R/32) \le C_a \varepsilon^{18/11}. \tag{5.3}$$

Every terminal neck meeting $B_{R/256}(z_0)$ satisfies

$$r_N \le C_a \varepsilon^{12/11} R \tag{5.4}$$

for all sufficiently large k. The same estimate holds for the threshold radius of z_0.

Proof. Apply Lemma 5.1 with $P = 39/10$, $q = 6/5$, and $r = R/16$. Its hypothesis holds because $0 < 6/5 < 19/10 = P - 2$, and $P - q = 27/10$. Since $B_{R/16}(z_0) \subset B_{R/8}(z_0)$, Lemma 4.3 gives

$$\begin{aligned} \int_{\Omega \cap B_{R/32}(z_0)} w^{39/10} &\le C (R/16)^{-27/10} \int_{\Omega \cap B_{R/16}(z_0)} w^{6/5} \\ &\le C_a \varepsilon^{6/5} R^{19/5 - 27/10} = C_a \varepsilon^{6/5} R^{11/10}. \end{aligned}$$

The fixed factor $16^{27/10}$ is absorbed into C_a. This proves (5.2).

Write

$$\mathcal{I} = \mathcal{Z}_{R/64} \cap B_{R/64}(z_0), \qquad r_i = r_*(z_i) \quad (z_i \in \mathcal{I}).$$

Equation (4.2) and $5 - P = 11/10 > 0$ imply

$$\eta_0^5 = \int_{B_{r_i}(z_i) \cap \Omega} w^{5-P} w^P \le C^{5-P} r_i^{-(5-P)} \int_{B_{r_i}(z_i) \cap \Omega} w^P.$$

Consequently $\int_{B_{r_i}(z_i) \cap \Omega} w^P \ge c r_i^{11/10}$, where $c = \eta_0^5 C^{-11/10} > 0$ is independent of the center. For $z_i \in \mathcal{I}$, the conditions $|z_i - z_0| < R/64$ and $r_i \le R/64$ imply $B_{r_i}(z_i) \subset B_{R/32}(z_0)$. The disjointness of the threshold balls, established in Section 4, therefore gives

$$c \sum_{z_i \in \mathcal{I}} r_i^{11/10} \le \sum_{z_i \in \mathcal{I}} \int_{B_{r_i}(z_i) \cap \Omega} w^{39/10} \le \int_{B_{R/32}(z_0) \cap \Omega} w^{39/10}.$$

Divide by $R^{11/10}$ and use (5.2) to obtain the first assertion of (5.3).

For the charge estimate put $A_i = (r_i/R)^{11/10}$ and $S = \sum_{z_i \in \mathcal{I}} A_i$. If $S > 0$, the inequality $A_i \le S$ gives

$$\sum_i (r_i/R)^3 = \sum_i A_i^{30/11} \le S^{19/11} \sum_i A_i = S^{30/11} \le C_a \varepsilon^{(6/5)(30/11)} = C_a \varepsilon^{36/11}.$$

When $S = 0$, the sum is zero, so the conclusion is immediate. The index set in (4.4) at $t = R/32$ is exactly $\mathcal{I}$, so

$$\rho_{z_0}(R/32)^2 = 32^3 \sum_{z_i \in \mathcal{I}} (r_i/R)^3 \le C_a \varepsilon^{36/11}.$$

Taking the square root proves the second assertion of (5.3). The first assertion also bounds every individual summand and hence gives

$$\frac{r_i}{R} \le S^{10/11} \le (C_a \varepsilon^{6/5})^{10/11} \le C_a \varepsilon^{12/11} \qquad (z_i \in \mathcal{I}).$$

It remains to place the centers relevant to (5.4) in $\mathcal{I}$. The root estimate (4.15) gives $r_*(z_0) \le C_a\varepsilon R$. For sufficiently large k, this is less than $R/64$, so $z_0 \in \mathcal{I}$ and the preceding single-summand bound proves $r_*(z_0) \le C_a\varepsilon^{12/11}R$.

Now let a terminal neck N meet $B_{R/256}(z_0)$. Then $N \in \mathcal{N}_{\mathrm{in}}$, and the terminal-radius bound from Section 4 gives $r_N \le C_a\varepsilon R$. Choose $z_N \in \mathcal{Z} \cap 4N$, as allowed by the neck stopping rule. Using a point in $N \cap B_{R/256}(z_0)$ gives

$$|z_N - z_0| \le |z_N - x_N| + |x_N - z_0| < 5r_N + R/256 \le 5C_a\varepsilon R + R/256 < R/64$$

once $5C_a\varepsilon < 3/256$. In particular $z_N \in B_{8R}(z_0)$, so (4.15) also gives $r_*(z_N) \le C_a\varepsilon R < R/64$ for large k. Thus $z_N \in \mathcal{I}$, and the same single-summand bound yields $r_*(z_N) \le C_a\varepsilon^{12/11}R$. Finally, the terminal-neck comparison $r_N \simeq_\delta r_*(z_N)$ proved in Section 4 gives (5.4). The parameter δ is fixed throughout, and its dependence is absorbed into C_a. □

Corollary 5.3 (Critical selection exponent)**.** *For the normalized nonflat solution,*

$$\alpha_* \le \frac{11}{18} < \frac{2}{3}. \tag{5.5}$$

Proof. Use the initial selection exponent $a_0 = 101/100$ from Section 4, which is admissible because $\alpha_* \le 1$. Proposition 5.2 therefore applies to this sequence. The lower bound in (4.7) at $t = R/8$ gives

$$\mathcal{E}_{z_0}(R/8) \ge \frac{c_{a_0}}{1+\log 8}\varepsilon.$$

For large k, $z_0 \in \mathcal{Z}_{R/64}$ by the threshold-radius conclusion of Proposition 5.2. Put $t = R/64$, so $8t = R/8$ and $2t = R/32$. The center $z_0 \in \mathcal{Z}_t$ contributes to its own charge, so

$$\rho_{z_0}(2t)^2 \ge \left(\frac{r_*(z_0)}{2t}\right)^3 > 0, \qquad r_*(z_0) \ge r_{\min} > 0.$$

For each fixed $b > 11/18$, the following quotient is therefore well defined. The definition of F_u^b and (5.3) give

$$F_u^b(t) \ge \frac{\mathcal{E}_{z_0}(8t)}{\rho_{z_0}(2t)^b} \ge c_{a_0,b}\varepsilon^{1-18b/11}.$$

Here $c_{a_0,b} > 0$ is independent of the selected index. Since

$$1 - \frac{18b}{11} = -\frac{18}{11}\left(b - \frac{11}{18}\right) < 0,$$

the right side tends to infinity as $\varepsilon \to 0$. Also $t = R/64 \to \infty$. Thus every $b > 11/18$ belongs to the defining divergence set of α_*. Taking the infimum over these b proves (5.5). □

6. Linearization and decay to the amplitude scale

6.1. **Reselection and the numerical reserves.** We now make a new selection. All quantities R, z_0, ε, the tree, and the signed domains in this section refer to that new sequence. Choose

$$\alpha = \frac{5}{8}, \qquad \beta_\circ = \frac{1}{4}, \qquad p_1 = \frac{19}{15}, \qquad p_2 = \frac{41}{20}, \qquad \chi = \frac{1}{1000}, \quad \sigma = \frac{1}{100000}, \quad \tau = \frac{1}{10000000}. \tag{6.1}$$

Corollary 5.3 gives $\alpha - \alpha_* \ge 5/8 - 11/18 = 1/72 > 0$. Lemma 4.1 applies with this value; its proof explains why the original convenient bound $\alpha \ge 3/4$ is unnecessary. Rebuild the tree for the new sequence. The normalized solution u is unchanged, so (4.1) remains available with $P = 39/10$. The estimates in Section 4 and Proposition 5.2 hold with $a = \alpha$.

Following the argument of [6, Proposition 10.4], we prove the higher-power boundary-flux estimate for every p satisfying

$$p > 1, \qquad p\alpha < \frac{4}{3}, \qquad 2p < \gamma_\alpha := 1 + \frac{2}{\alpha}. \tag{6.2}$$

These are the five-dimensional forms of its two strict inequalities. Here

$$\gamma_\alpha = \frac{21}{5}, \qquad p_1\alpha = \frac{19}{24}, \qquad 2p_1 = \frac{38}{15},$$

and

$$p_2\alpha = \frac{41}{32} < \frac{4}{3}, \qquad 2p_2 = \frac{41}{10} < \frac{21}{5}, \qquad p_2 > 2 = \frac{5-1}{2}. \tag{6.3}$$

We also check the summation in the proof of that boundary estimate. Put $d = 3 + 2\beta_\circ = 7/2$ and $s = p\alpha$. At generation j take $t_B = h_B^{2/\alpha}$ for $B \in \mathcal{D}_{\mathrm{in}}^j$; all sums in the following calculation are over this family. Empty generations contribute zero, so assume this family is nonempty. If $0 < s \le 1$, Hölder's inequality and (4.18) give

$$\sum_B h_B^{2p} = \sum_B t_B^s \le (\#\mathcal{D}_{\mathrm{in}}^j)^{1-s} \left(\sum_B t_B\right)^s \le C_{p,\alpha}\varepsilon^{2p} 2^{j\{d-2\beta_\circ s\}}.$$

If $s > 1$, use $\sum_B t_B^s \le (\sum_B t_B)^s$ instead, obtaining $\sum_B h_B^{2p} \le C_{p,\alpha}\varepsilon^{2p}2^{3js}$. Thus the generation exponent is

$$a_{\alpha,p} = \begin{cases} d - 2\beta_\circ p\alpha, & p\alpha \le 1, \\ 3p\alpha, & p\alpha > 1. \end{cases}$$

In the first branch $0 < a_{\alpha,p} < d < 4$; in the second, $0 < a_{\alpha,p} = 3p\alpha < 4$ by (6.2).
The descendant count. Use the finite tree and coherent signed geometry already established, and put $d = 3 + 2\beta_\circ = 7/2$. The family $\mathcal{D}_{\mathrm{in}}^j$ is predecessor-closed. Indeed, if $B \in \mathcal{D}_{\mathrm{in}}^{j+1}$, then

$$\operatorname{dist}(x_{PB}, B_{R/8}(z_0)) < \frac{5}{4}r_{PB} + 4r_B = \frac{13}{4}r_{PB} < 4r_{PB}.$$

Thus every ancestor used below remains in its corresponding inner family, where both proved generation counts apply.

Fix $A \in \mathcal{D}_{\mathrm{in}}^j$ and $l \ge j$, and let $\mathcal{F}_{A,l}$ be the *inner* descendants

$$\mathcal{F}_{A,l} = \{B \in \mathcal{D}_{\mathrm{in}}^l : P^{l-j}B = A\}.$$

The restriction to $\mathcal{D}_{\mathrm{in}}^l$ is needed for the total count bound below. The predecessor geometry gives

$$|x_B - x_A| \le \frac{5}{4}\sum_{m=j}^{l-1} r_m < \frac{5}{2}r_j \qquad (B \in \mathcal{F}_{A,l}).$$

Their balls $B_{r_l/16}(x_B)$ are pairwise disjoint and lie in $B_{3r_j}(x_A)$. We use the precise nondegeneracy input [6, Lemma 4.3, equation (4.1)]: for every global classical stable Bernoulli solution in dimension $n \ge 2$, every $y \in \Gamma$, and every $s > 0$,

$$\mathcal{H}^{n-1}(\Gamma \cap B_s(y)) \ge c(n)s^{n-1}.$$

Its hypotheses hold for our normalized global classical stable solution, independently of the selection exponent. Apply it in these disjoint balls with $n = 5$, and use the upper perimeter bound at $x_A \in \Gamma$. This gives

$$\#\mathcal{F}_{A,l}\, cr_l^4 \le \mathcal{H}^4(\Gamma \cap B_{3r_j}(x_A)) \le Cr_j^4.$$

Together with the already proved total count $\#\mathcal{D}_{\mathrm{in}}^l \le C2^{ld}$, it follows that

$$\#\mathcal{F}_{A,l} \le C\min\{2^{4(l-j)}, 2^{ld}\}. \tag{6.4}$$

The fourth power comes from lower and upper free-boundary area; five-dimensional volume packing alone would not give this bound.

Regular-boundary reduction. Every regular part of $\Gamma_\pm$ is covered, up to $\mathcal{H}^4$-null sets, by the correctly labeled sheets of regular terminal balls. To see this directly, the finite signed-piece union has its boundary contained in the union of the local piece boundaries. The centered L^2-to-L^1 conversion from Section 4 and its lower-slab estimate give $u \geq V_B - Ch_B r_B$ in $12B$. Fix the tree parameter so that $C\delta < c_*/2$. On the closure of a branching core, $V_B \geq c_* r_B$ and $h_B < \delta$, hence $u \geq (c_* - C\delta) r_B \geq c_* r_B/2 > 0$. Its closure therefore lies in Ω and contributes no physical boundary point. Outside the closed neck region, the signed cover makes $U_\pm$ agree locally with the appropriate positive component. Consequently a physical boundary point in $\Gamma_\pm^{\mathrm{reg}}$ has the corresponding regular-sheet trace from at least one regular terminal ball. Assign it to the first such ball in a fixed enumeration; this yields a measurable covering with no multiplicity issue. Each assigned part has area at most Cr_B^4, by the perimeter bound on a fixed enlargement of B.

Let $B \in \mathcal{R}^l \cap \mathcal{D}^l_{\mathrm{in}}$. On its correctly labeled physical sheet, the regular splitting estimate gives $|\pm \nabla u - e_B| \leq Ch_B$, while $|\nabla u| = |e_B| = |e_{\mathrm{root}}| = 1$. For unit vectors g, e, $1 - e \cdot g = |g - e|^2/2$. Hence

$$|1 \mp e_{\mathrm{root}} \cdot \nabla u| \leq C(h_B^2 + |e_B - e_{\mathrm{root}}|^2).$$

If $A_j = P^{l-j}B$ is its generation-j ancestor, parent jumps and the exact root orientation imply

$$h_B + |e_B - e_{\mathrm{root}}| \leq C \sum_{j=0}^{l} h_{A_j}.$$

For $p > 1$, the finite-sum Hölder inequality therefore gives

$$(h_B^2 + |e_B - e_{\mathrm{root}}|^2)^p \leq C_p (l+1)^{2p-1} \sum_{j=0}^{l} h_{A_j}^{2p}.$$

On $\Gamma_\pm^{\mathrm{reg}}$ the outward normal is $\nu_\pm = -\nabla u$, so the left side to be estimated is exactly $|\partial_{\nu_\pm}(u \mp e_{\mathrm{root}} \cdot (x - z_0))|^p = |1 \mp e_{\mathrm{root}} \cdot \nabla u|^p$. Integrating the sheetwise estimates, summing both signs, and grouping by ancestors yields the following sum. Every assigned terminal ball belongs to $\mathcal{D}^l_{\mathrm{in}}$, so predecessor closure puts its ancestor in $\mathcal{D}^j_{\mathrm{in}}$; the multiplicity is exactly bounded by the restricted family $\mathcal{F}_{A,l}$ just defined.

$$\begin{aligned} I_{\mathrm{reg}} &:= \sum_{\pm} \int_{\Gamma_\pm^{\mathrm{reg}}} |\partial_{\nu_\pm}(u \mp e_{\mathrm{root}} \cdot (x - z_0))|^p \, d\mathcal{H}^4 \\ &\leq C_p \sum_{l \geq 0} r_l^4 (l+1)^{2p-1} \sum_{j=0}^{l} \sum_{A \in \mathcal{D}^j_{\mathrm{in}}} \#\mathcal{F}_{A,l} \, h_A^{2p}. \end{aligned}$$

The tree is finite; extending the nonnegative sums to infinity is only an upper bound.

The preceding generation interpolation gives

$$\sum_{A \in \mathcal{D}^j_{\mathrm{in}}} h_A^{2p} \leq C_{p,\alpha} \varepsilon^{2p} 2^{ja}, \qquad a = a_{\alpha,p}.$$

For every flux-admissible p, $0 < a < 4$ by the explicit inequalities already checked at $\alpha = 5/8$. Substituting (6.4), $r_l = 2^{-l}R$, and then $s = l - j$ gives

$$I_{\mathrm{reg}} \leq C_{p,\alpha} R^4 \varepsilon^{2p} \sum_{l \geq 0} (l+1)^{2p-1} 2^{-4l} \sum_{s=0}^{l} \min\{2^{4s}, 2^{ld}\} \, 2^{(l-s)a}.$$

To bound the inner sum, put $m = \lfloor ld/4 \rfloor$ and factor out 2^{la}. For $s \leq m$, sum the increasing geometric series $2^{(4-a)s}$; for $s > m$, sum the decreasing series $2^{ld - as}$. Since $0 < a < 4$, these give

respectively

$$\sum_{s=0}^{m} 2^{(4-a)s} \le C_a 2^{(4-a)ld/4}, \qquad \sum_{s=m+1}^{l} 2^{ld-as} \le C_a 2^{ld-ald/4}.$$

Thus the inner sum is at most $C_a 2^{l\{d+(1-d/4)a\}}$, and

$$I_{\mathrm{reg}} \le C_{p,\alpha} R^4 \varepsilon^{2p} \sum_{l\ge 0} (l+1)^{2p-1} 2^{l(a-4)/8} \le C_{p,\alpha} R^4 \varepsilon^{2p}.$$

This proves the regular-boundary bound for the rebuilt tree.
The neck contribution. The globally one-Lipschitz solution and unit root direction give $|\partial_{\nu_\pm}(u \mp e_{\mathrm{root}} \cdot (x - z_0))| \le 2$ on the artificial boundary, using its positive-side normal traces. The proved neck-boundary area estimate and $\gamma_\alpha = 1 + 2/\alpha > 2p$ imply

$$I_{\mathrm{neck}} \le C_p \mathcal{H}^4(\Gamma^{\mathrm{neck}}) \le C_{p,\alpha} R^4 \varepsilon^{\gamma_\alpha} \le C_{p,\alpha} R^4 \varepsilon^{2p}.$$

Combining the regular and neck terms proves exactly the higher-power signed boundary estimate needed by the local flux and energy arguments. The proof depends on α only through the established generation moment and the explicit strict inequalities $a < 4$ and $2p < \gamma_\alpha$, not through the source's initial choice $\alpha \ge 3/4$.

For p_1 and p_2, respectively, the exact values are

$$a_{\alpha,p_1} = \frac{149}{48}, \qquad d - 4 + (1 - d/4)a_{\alpha,p_1} = -\frac{43}{384},$$
$$a_{\alpha,p_2} = \frac{123}{32}, \qquad d - 4 + (1 - d/4)a_{\alpha,p_2} = -\frac{5}{256}.$$

For the neck term, the strict margins are

$$\gamma_\alpha - 2p_1 = 5/3, \qquad \gamma_\alpha - 2p_2 = 1/10.$$

Set

$$R_{\mathrm{f}} = \varepsilon^{\chi} R, \qquad A_{\#} = \varepsilon^{1+\sigma} R_{\mathrm{f}}, \qquad \rho = \varepsilon^{\tau} R_{\mathrm{f}}.$$

The first affine iteration uses $\gamma = 2/5$. The two error exponents needed there are strictly positive:

$$1 - \chi(9/5 + 4/p_1) = \frac{94529}{95000} > 0, \qquad 2p_1 - 1 - \chi(5 + 4/5) = \frac{22913}{15000} > 0.$$

Increase the selected index so that $8R_{\mathrm{f}} \le R/256$, which is possible because $R_{\mathrm{f}}/R = \varepsilon^{\chi} \to 0$. The individual-radius estimate in Proposition 5.2 gives $r_*(z_0) \le C\varepsilon^{12/11} R$ and, for every terminal neck meeting $B_{8R_{\mathrm{f}}}(z_0)$,

$$r_N + r_*(z_0) \le C\varepsilon^{12/11} R =: r_{\mathrm{neck}}, \qquad \frac{r_{\mathrm{neck}}}{A_{\#}} \le C\varepsilon^{1/11-\chi-\sigma} \longrightarrow 0. \tag{6.5}$$

Indeed, $1/11 - \chi - \sigma = 98889/1100000 > 0$. Since $r_*(z_0) \ge r_{\min} > 0$, the same estimate also gives

$$A_{\#} = \varepsilon^{1+\sigma+\chi} R \ge c r_{\min} \varepsilon^{-(1/11-\chi-\sigma)} \longrightarrow \infty.$$

Thus the physical radii $A_{\#}, \rho, R_{\mathrm{f}}, R$ all diverge, even though their successive ratios tend to zero:

$$\frac{A_{\#}}{\rho} = \varepsilon^{1+\sigma-\tau}, \qquad \frac{\rho}{R_{\mathrm{f}}} = \varepsilon^{\tau}, \qquad \frac{R_{\mathrm{f}}}{R} = \varepsilon^{\chi}.$$

The constants are fixed before the index of the selected sequence is increased. In particular, all terminal necks meeting the balls used below are much smaller than those balls, even when their radii are below εR.

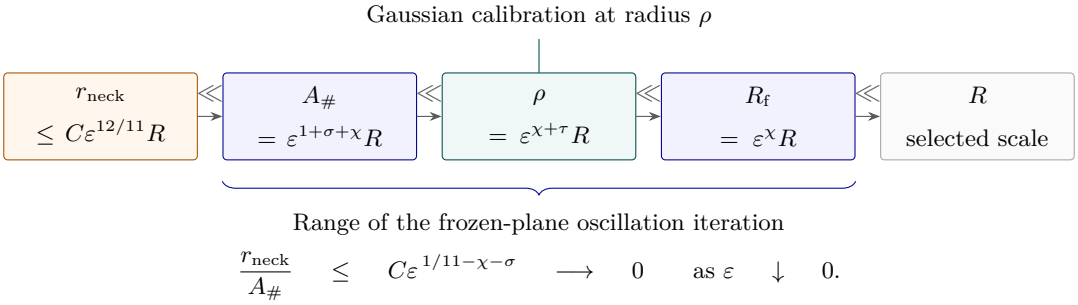


FIGURE 3. The scale hierarchy in the five-dimensional closing argument. The diagram is not to scale. Every $\ll$ denotes an asymptotic ratio tending to zero for the fixed parameter choices in the proof. In particular, $\chi+\sigma<1/11$ places all relevant necks below the amplitude scale $A_\#$. The iteration runs between $A_\#$ and R_{f}; its estimates are calibrated at the intermediate radius ρ.

6.2. **Signed slabs above the neck scale.** We prove the geometric inclusions used below, following [6, Lemma 10.1], with the cone threshold supplied by Lemma 2.1. For $z\in\mathcal{Z}$ and $r>0$ such that

$$B_r(z)\subset B_{R/2}(z_0), \qquad r\ge Cr_*(z),$$

every direction $e\in S^4$ can be oriented so that

$$\begin{aligned} B_r(z)\cap\{\pm e\cdot(x-z)>C\mathcal{E}_z(8r;e)r\}\subset U_\pm\cap B_r(z)\\ \subset\{\pm e\cdot(x-z)>-C\mathcal{E}_z(8r;e)r\}. \end{aligned} \tag{6.6}$$

At the root scale the orientation is e_{root}, and

$$\begin{aligned} B_{R/8}(z_0)\cap\{\pm e_{\mathrm{root}}\cdot(x-z_0)>C\varepsilon R\}\subset U_\pm\cap B_{R/8}(z_0)\\ \subset\{\pm e_{\mathrm{root}}\cdot(x-z_0)>-C\varepsilon R\}. \end{aligned} \tag{6.7}$$

We use the local estimates and the previously constructed signed tree. Write R_0 for the radius of the selected root ball and z_0 for its center, to distinguish that radius from the local radius r below. The constants may depend on the already fixed tree parameter δ. Let $z\in\mathcal{Z}$,

$$B_r(z)\subset B_{R_0/2}(z_0), \qquad r\ge C_{\mathrm{sep}}r_*(z), \qquad E=\mathcal{E}_z(8r;e).$$

The constant C_{sep} will be fixed below. Replacing e by $-e$ does not change E, since both its ridge plane and its vee are unchanged. Also $E>0$: if $E=0$, its nonnegative square-integral term and continuity would give $u(x)=|e\cdot(x-z)|$ throughout $B_{8r}(z)$, contrary to the locally one-sided classical free boundary at z.

Fix a small positive number τ_{flat}. When $E\ge\tau_{\mathrm{flat}}$, the desired inclusions are vacuous for an eventual slab constant greater than $\tau_{\mathrm{flat}}^{-1}$. Hence suppose $0<E<\tau_{\mathrm{flat}}$. Fixed-direction propagation [6, Proposition 7.9], with outer radius $16r$ and ratio $1/2$, gives $\mathcal{E}_z(16r;e)\le CE$. The normalized L^1 error on $B_{16r}(z)$ is therefore at most

$$r^{-1}\fint_{B_{16r}(z)}|u-V_{z,e}|\le 16|B_1|^{-1/2}\mathcal{E}_z(16r;e)\le C_0E.$$

Choose τ_{flat} so that $C_0\tau_{\mathrm{flat}}<1$. Corollary 9.2 of [6] then yields, with a fixed constant C_{phys},

$$\begin{aligned} u\ge V_{z,e}-C_{\mathrm{phys}}Er \quad \text{in } B_{12r}(z),\\ \Gamma\cap B_{12r}(z)\subset\{|e\cdot(x-z)|\le C_{\mathrm{phys}}Er\}. \end{aligned}$$

In particular, the parts of $B_r(z)$ whose signed heights exceed $C_{\mathrm{phys}}Er$ lie in the positive phase.

We next bound the radii of the deleted terminal necks; the physical slab estimate alone would not imply this step. Let N be a terminal neck with $N \cap B_r(z) \neq \varnothing$, and let C_δ be the already proved constant such that

$$r_N \le C_\delta r_*(q) \qquad (q \in \mathcal{Z} \cap 4N).$$

If $r_N \ge r/3$, intersection with $B_r(z)$ gives $|z - x_N| < r + r_N \le 4r_N$, so $z \in \mathcal{Z} \cap 4N$. Consequently $r/3 \le r_N \le C_\delta r_*(z)$, contradicting $r \ge C_{\mathrm{sep}} r_*(z)$ once $C_{\mathrm{sep}} > 3C_\delta$. Thus $r_N < r/3$. Choose $z_N \in \mathcal{Z} \cap 4N$. Then

$$|z_N - z| < r + 5r_N < 3r.$$

We use the exact upper-nearby-radius statement [6, Lemma 5.8]: there is a universal $M^\circ > 0$ such that, for any $z \in \mathcal{Z}$ and $T \ge M^\circ r_*(z)$,

$$r_*(q) \le T/8 \qquad (q \in \mathcal{Z} \cap B_{3T/4}(z)).$$

Its hypotheses concern the normalized solution and its threshold centers, and contain no selected exponent. If also $C_{\mathrm{sep}} \ge M^\circ/4$, that lemma applies at center z with radius $4r$, and gives

$$s_N := r_*(z_N) \le r/2, \qquad B_{4s_N}(z_N) \subset B_{5r}(z).$$

The physical trapping at z_N implies $|e \cdot (z_N - z)| \le C_{\mathrm{phys}} E r$. Therefore on this last threshold ball the same lower bound, recentered at z_N, becomes

$$u \ge |e \cdot (x - z_N)| - 2C_{\mathrm{phys}} E r.$$

The already proved one-sided trapping consequence of the local splitting estimate (the argument of [6, Lemma 9.8]) now gives $s_N \le CEr$. The terminal comparison then proves

$$r_N \le C_{\mathrm{neck}} E r \qquad (N \cap B_r(z) \neq \varnothing).$$

All constants have been fixed independently of the selected index and of the selected exponent.

For such an N, its free-boundary center satisfies $|x_N - z| < r + r_N < 4r/3$. Hence the physical trapping also gives $|e \cdot (x_N - z)| \le C_{\mathrm{phys}} E r$. For every $x \in \overline{N} \cap B_r(z)$ it follows that

$$|e \cdot (x - z)| \le C_{\mathrm{phys}} E r + r_N \le (C_{\mathrm{phys}} + C_{\mathrm{neck}}) E r.$$

The finite tree makes the closure in the definition of $\mathcal{X}_{\mathrm{neck}}$ a closure of a finite union. A closed terminal ball meeting the open ball $B_r(z)$ has its open ball meeting it as well. We have therefore proved

$$\mathcal{X}_{\mathrm{neck}} \cap B_r(z) \subset \{|e \cdot (x - z)| \le C_X E r\}, \qquad C_X = C_{\mathrm{phys}} + C_{\mathrm{neck}}.$$

It remains to identify the signs of the two caps; physical positivity does not identify these labels by itself. Decrease τ_{flat} so that $C_{\mathrm{neck}} \tau_{\mathrm{flat}} < 1/16$. Starting with the branching root, iterate the local-child free-boundary cover through z. The first child and each subsequent chosen child contain z in their quarter ball. Such a ball cannot be a regular terminal, because its fourfold dilation contains $z \in \mathcal{Z}$. The finite chain therefore ends at a neck terminal containing z. By the local terminal-radius bound, its radius is less than $r/16$. Since $r \le R_0/2$, the dyadic chain consequently contains a branching ball B such that

$$z \in \tfrac{1}{4}B, \qquad r/8 < r_B \le r/4.$$

In particular $B \subset B_r(z)$, while $16B \subset B_{8r}(z)$, since $16r_B + |x_B - z| \le (65/4) r_B \le 65r/16 < 8r$.

The physical trapping gives $|e \cdot (x_B - z)| \le C_{\mathrm{phys}} E r$. Use the centered vee $\widehat{V}(x) = |e \cdot (x - x_B)|$ as a competitor in the definition of h_B. Since $r/r_B < 8$,

$$\begin{aligned} h_B^2 &\le 2r_B^{-7} \int_{16B} |u - V_{z,e}|^2 + 2r_B^{-7} |16B|\, C_{\mathrm{phys}}^2 E^2 r^2 \\ &\le CE^2. \end{aligned}$$

The triangle inequality thus bounds the normalized L^2 difference of $V_B = |\ell_B|$ and $\widehat{V}$ on B by CE. The elementary vee comparison [6, Lemma 9.5] permits a choice of the sign of e for which $|e - e_B| \le CE$. After decreasing τ_{flat} further, $e \cdot e_B \ge 3/4$.

Fix $K > \max\{C_{\mathrm{phys}}, C_X\}$ and define the connected caps

$$H_\pm = B_r(z) \cap \{\pm e \cdot (x - z) > KEr\}.$$

They lie in $\Omega \setminus \mathcal{X}_{\mathrm{neck}}$ by the two trapping estimates. The explicit points $x_\pm = x_B \pm (r_B/2)e_B$ belong to the corresponding branching cores $B^\pm$ and to $B_r(z)$, and satisfy

$$\pm e \cdot (x_\pm - z) \ge -C_{\mathrm{phys}}Er + 3r_B/8 > -C_{\mathrm{phys}}Er + 3r/64.$$

Choose τ_{flat} sufficiently small that this is greater than KEr. Then $x_\pm \in H_\pm \cap B^\pm \subset H_\pm \cap U_\pm$. The already proved signed covering and disjointness say that $H_\pm$ is covered by the two disjoint relatively open sets $H_\pm \cap U_+$ and $H_\pm \cap U_-$. Connectedness and the displayed witnesses force $H_\pm \subset U_\pm$. Disjointness also implies that a point of $U_\pm \cap B_r(z)$ cannot have opposite signed height greater than KEr. Thus its corresponding signed height is at least $-KEr$. Enlarging the common slab constant to $2K$ gives the strict outer inequality and only shrinks the inner caps. Enlarging that constant once more to exceed C_{sep} and $\tau_{\mathrm{flat}}^{-1}$ proves the claimed mesoscopic inclusions in all cases.

The root estimate can be obtained directly, without identifying a directional combined excess from $h_{B_{R_0}(z_0)}$ alone. The proved root bound $h_{B_{R_0}(z_0)} \le C_a\varepsilon$ and Corollary 9.2 of [6] give

$$u \ge V_{z_0,e_{\mathrm{root}}} - C_a\varepsilon R_0, \qquad \Gamma \cap B_{12R_0}(z_0) \subset \{|e_{\mathrm{root}} \cdot (x - z_0)| \le C_a\varepsilon R_0\}.$$

Every terminal neck has radius at most $C_{a,\delta}\varepsilon R_0$: its associated center lies in $B_{5R_0}(z_0)$, and this is the root active-radius bound followed by the terminal comparison. Its free-boundary center, and hence its entire portion meeting $B_{R_0/8}(z_0)$, lie in a slab of width $C_a\varepsilon R_0$ in the direction e_{root}. The two caps of $B_{R_0/8}(z_0)$ outside a sufficiently large such slab consequently lie in $\Omega \setminus \mathcal{X}_{\mathrm{neck}}$. The points

$$z_0 \pm \frac{3R_0}{32} e_{\mathrm{root}}$$

lie in those caps for large selected indices, and also in the corresponding root cores, because $1/16 < 3/32 < 1/8$. Connectedness, the signed covering and disjointness prove the two root inclusions with their prescribed orientation, exactly as above.

Localized John domains. We record the connectivity consequence needed for the energy estimate. Write R_0 for the root radius. There is a fixed $c_J > 0$ such that, if $z \in \mathcal{Z} \cap B_{R_0/8}(z_0)$, $r \le R_0/16$, and every neck terminal meeting $B_{3r/4}(z)$ has radius at most $c_J r$, then for each sign there is a c_J-John domain $O_\pm$ satisfying

$$U_\pm \cap B_{r/2}(z) \subset O_\pm \subset U_\pm \cap B_{3r/4}(z), \qquad |O_\pm| \ge c_J r^5.$$

Here a c-John domain with center q means that every point can be joined to q by an arclength-parametrized curve γ for which $\operatorname{dist}(\gamma(t), \partial O) \ge ct$. We prove the assertion for the positive sign.

The local-child free-boundary cover can be followed through z. Each chosen child contains z in its quarter ball. The chain cannot stop at a regular terminal, because that ball's fourfold dilation contains $z \in \mathcal{Z}$. It therefore ends at a neck terminal N containing z, and the terminal comparison gives

$$r_*(z) \le C_\delta r_N \le C_\delta c_J r.$$

The neck modulus (4.5), equivalently [6, Lemma 7.8], gives

$$\mathcal{E}_z(6r) \le \omega_*\left(\frac{r_*(z)}{6r}\right) \le \omega_*\left(\frac{C_\delta c_J}{6}\right).$$

Choose a minimizing direction for this excess. Once $C_\delta c_J \le 3/(4C_{\mathrm{sep}})$, the generic signed-slab estimate is admissible at radius $3r/4$; its error is bounded by $C(3r/4)\omega_*(C_\delta c_J/6)$. This can be

made as small a multiple of r as required, since $\omega_*(s) \to 0$. The final choice of c_J is made below after the geometric tolerance is fixed.

For each raw signed piece D^+, put $p_D = x_D + (r_D/2)e_D$. There is a fixed $c > 0$ such that

$$B_{ctr_D}((1-t)x + tp_D) \subset D^+ \qquad (x \in D^+,\ 0 < t \le 1).$$

For a branching core this follows by interpolating balls inside the convex intersection of a ball and a half-space: p_D has fixed relative room from both boundary pieces. For a regular piece, use coordinates $x_D = 0$, $e_D = e_5$, and write its sheet as $\{x_5 > g(x')\} \cap B_{r_D}$. The splitting estimate gives $|g| \le Lr_D$ and $\operatorname{Lip} g \le L$, with $L = C\delta$ as small as required. For $x_t = (1-t)x + t(0, r_D/2)$,

$$(x_t)_5 - g(x_t') \ge (1-t)(x_5 - g(x')) + t(1/2 - 2L)r_D \ge tr_D/4.$$

Also $r_D - |x_t| \ge tr_D/2$. Decreasing the fixed c in terms of these two margins proves the displayed inclusion. The lower sheet has the identical property with the signs reversed.

For each nonroot piece D^+, the already constructed gate is a ball $G_D = B_{c_g r_D}(g_D) \subset D^+ \cap (PD)^+$. The ball inclusion connects a point of D^+ to p_D and then to g_D. The latter connection has a tube of width cr_D: apply that formula to every point of a smaller concentric gate ball, and interpolate the resulting balls. More generally, the incoming and outgoing gate balls in a branching piece of radius s can be joined through p_D by curves of length at most $C_1 s$ and balls of radius at least $c_1 s$ along the connector. Indeed the incoming gate has radius $c_g s/2$, the outgoing one has radius $c_g s$, and interpolation with the fixed ball at p_D keeps a positive fixed fraction of the smaller radius. These constants do not depend on the generation.

Fix $\lambda > 0$ so small that all connectors up to a top dyadic radius λr have total length and spatial displacement less than $r/16$. This is possible because their bounds are a fixed constant times the sum of their radii. Choose m so that

$$\lambda r/2 < r_m \le \lambda r, \qquad \lambda r/4 < r_{m+1} \le \lambda r/2.$$

Take $x \in U_+ \cap B_{r/2}(z)$ and a raw piece $D^+ \ni x$, with D at generation l. If $l \ge m+1$, follow the predecessor gates from D up to the gate of its generation-$(m+1)$ ancestor. At a connector of radius s, the length previously accumulated is bounded by $C\sum_{i\ge 0} 2^{-i}s \le 2Cs$, while the connector balls have radius at least cs. The initial segment supplied by the ball inclusion has radius proportional to its travelled length. These facts give a uniform John chain to the top gate. Its total displacement, including the radii of its balls, is at most $C\lambda r < r/8$ after the fixed choice of λ. If $l \le m$, stop this construction when $t = r_{m+1}/r_D$. This gives a chain of length at most Cr_{m+1} ending in a ball of radius at least cr_{m+1}, with the same displacement bound. Consequently in either case there is an open ball-chain domain O_x containing x such that

$$O_x \subset U_+ \cap B_{5r/8}(z), \qquad B_{\kappa r}(q_x) \subset O_x,$$

where $\kappa > 0$ is fixed, for example a sufficiently small fixed multiple of $\min(c, c_g)\lambda/4$. The chain from x has length at most $C\lambda r$. Every point of a chain ball joins its center by a segment, whose distance from the ball boundary grows at least as fast as the segment length. It then follows the remaining ball chain. The starting ball's radius is comparable to, or greater than, the length already accumulated at that point of the chain. This proves the same John bound for every point of O_x, with center q_x. An arbitrarily small initial ball about x is included if needed. This ball-chain description also proves the claim when x is close to the boundary of its raw piece.

Only now fix a flatness tolerance $0 < \eta < \kappa/8$. Decrease c_J so that the signed slabs at radius $3r/4$ give an orientation e with

$$H_+ := B_{3r/4}(z) \cap \{e \cdot (y-z) > 2\eta r\} \subset U_+ \cap B_{3r/4}(z) \subset \{e \cdot (y-z) > -\eta r\}.$$

We may also require $\eta < 1/32$. Since the full endpoint ball $B_{\kappa r}(q_x)$ lies in the outer slab, $e\cdot(q_x - z) \ge (\kappa - \eta)r$. Its concentric ball of radius $\kappa r/4$ therefore lies in H_+. This quantitative shrinking is necessary; the outer slab alone does not put the entire original endpoint ball in the inner cap.

The convex set H_+ contains a fixed ball $B_{c_H r}(q_H)$ with $q_H = z + (r/2)e$ and a fixed $c_H > 0$. Interpolation of balls in this convex set joins each shrunken endpoint ball to that common ball. The radius at parameter t is at least $(1-t)\kappa r/4 + t c_H r$, and the newly added length is at most Crt. The preceding length is at most $C\lambda r \le C_\kappa \kappa r$. Thus these appended connections preserve a uniform John constant. The union of H_+ and all the domains O_x is an open connected John domain with center q_H, contains $U_+ \cap B_{r/2}(z)$, and lies inside $U_+ \cap B_{3r/4}(z)$. Its volume is at least $|H_+| \ge cr^5$. Decreasing one final constant c_J makes all the stated conclusions hold with the same symbol. The negative sign is treated in the same way. In particular, for $r/2 \le s \le r$ the signed slabs also imply $|U_\pm \cap B_s(z)| \simeq r^5$ in this regime.

6.3. **Local flux and signed energy estimates.** We state the analytic estimates with their normalizations. The radius condition in the following lemma is needed when the iteration passes below εR.

Lemma 6.1 (Flux and local energy)**.** *Fix $C_0 \ge 1$ and let p satisfy (6.2). On U_+ let $L(x) = a \cdot (x - z_0) + b$ satisfy*

$$|a| = 1, \qquad |a - e_{\mathrm{root}}| \le C_0 \varepsilon, \qquad |b| \le C_0 \varepsilon R,$$

and put $v = u - L$. For $0 < r \le R/16$, define

$$\Phi_r(v) = r^{-3} \varepsilon R \int_{\Gamma_+ \cap B_r(z_0)} |\partial_{\nu_+} v| \, d\mathcal{H}^4. \tag{6.8}$$

Then

$$\int_{\Gamma_+ \cap B_r(z_0)} |\partial_{\nu_+} v| \le C r^{4-4/p} R^{4/p} \varepsilon^2 + C R^4 \varepsilon^{2p}. \tag{6.9}$$

Suppose also that every terminal neck meeting $B_{3r/4}(z_0)$ has radius at most $c_J r$, for a fixed sufficiently small geometric constant $c_J > 0$. For $2 < q \le 10/3$, one then has

$$r^2 \fint_{U_+ \cap B_{3r/4}(z_0)} |\nabla v|^2 + \left(\fint_{U_+ \cap B_{r/2}(z_0)} |v|^q \right)^{2/q} \le C_q \fint_{U_+ \cap B_r(z_0)} |v|^2 + C \Phi_r(v). \tag{6.10}$$

The same statements hold on U_- when a is close to $-e_{\mathrm{root}}$. The functional Φ_r is invariant under a constant shift of v. The intercept condition must still hold when the energy estimate is applied after such a shift. The constants are independent of r and the selected index; C_q is independent of C_0, whereas C may depend on C_0, p, q and the fixed parameters.

Proof. We give the flux and energy arguments directly at the parameters above, following [6, Lemma 10.5 and Proposition 10.6]. This avoids using their standing selection range. Write $\Omega_s = U_+ \cap B_s(z_0)$ and $\Gamma_s = \Gamma_+ \cap B_s(z_0)$.

On the regular part of Γ_r put $g = \nabla u$, so $|g| = 1$ and $\nu_+ = -g$. Hence

$$|\partial_{\nu_+} v| = 1 - a \cdot g, \qquad 1 - a \cdot g = \tfrac{1}{2}|a - g|^2 \le |a - e_{\mathrm{root}}|^2 + 2(1 - e_{\mathrm{root}} \cdot g).$$

The higher-power regular-boundary estimate proved above therefore gives

$$\int_{\Gamma_r \cap \Gamma_+^{\mathrm{reg}}} |\partial_{\nu_+} v|^p \le C R^4 \varepsilon^{2p}.$$

Indeed, the term $|a - e_{\mathrm{root}}|^{2p}$ contributes at most $C(C_0\varepsilon)^{2p} \mathcal{H}^4(\Gamma \cap B_r(z_0))$, and the perimeter bound is $\mathcal{H}^4(\Gamma \cap B_r(z_0)) \le C r^4 \le C R^4$. Hölder's inequality on this regular part yields

$$\int_{\Gamma_r \cap \Gamma_+^{\mathrm{reg}}} |\partial_{\nu_+} v| \le (C r^4)^{1-1/p} (C R^4 \varepsilon^{2p})^{1/p} \le C r^{4-4/p} R^{4/p} \varepsilon^2.$$

On the neck part, $|\nabla u| \le 1$ and $|a| = 1$ give $|\partial_{\nu_+} v| \le 2$. Its area is at most $C R^4 \varepsilon^{\gamma_\alpha}$. Since $\gamma_\alpha > 2p$ and $0 < \varepsilon < 1$, its contribution is bounded by $C R^4 \varepsilon^{2p}$. Adding the two parts proves (6.9). This

derivation uses only the higher-power estimate and the already proved boundary decomposition; it has no further restriction on the selection exponent.

We next prove the energy estimate. The localized John construction above applies under the stated terminal-radius hypothesis and gives

$$\Omega_{r/2} \subset \mathcal{O} \subset \Omega_{3r/4}, \qquad |\mathcal{O}| \geq cr^5.$$

Its constants are geometric and its conclusion does not require $r \geq \varepsilon R$. For clarity, volume comparability also follows from these same local hypotheses. The local-child chain through z_0 ends in a neck terminal N with $z_0 \in \frac{1}{4}N$. Thus the terminal comparison gives $r_*(z_0) \leq C_\delta r_N \leq C_\delta c_J r$. Choose c_J small enough. The neck modulus and the signed-cap estimate at radius $r/2$ then put a cap of fixed relative volume in $\Omega_{r/2}$. Consequently

$$cr^5 \leq |\Omega_{r/2}| \leq |\Omega_{3r/4}| \leq |\Omega_r| \leq Cr^5.$$

In particular all averages below have positive denominators.

The boundary values of v are uniformly small at the root amplitude. On $\Gamma_r \cap \Gamma_+^{\mathrm{reg}}$ one has $u = 0$ and $|e_{\mathrm{root}} \cdot (x - z_0)| \leq C\varepsilon R$ by the root slab. On the neck part the proved neck-height and deleted-neck slab bounds give

$$u(x) + |e_{\mathrm{root}} \cdot (x - z_0)| \leq C\varepsilon R.$$

Since $|x - z_0| \leq r \leq R/16$, $|a - e_{\mathrm{root}}| \leq C_0\varepsilon$ and $|b| \leq C_0\varepsilon R$ imply

$$|v| = |u - a \cdot (x - z_0) - b| \leq C\varepsilon R \quad \text{for } \mathcal{H}^4\text{-almost every } x \in \Gamma_r.$$

Only the constant in this boundary estimate depends on C_0.

Choose $\eta \in C_c^\infty(B_r(z_0))$, equal to one on $B_{3r/4}(z_0)$, with $0 \leq \eta \leq 1$ and $|\nabla\eta| \leq C/r$. The finite-perimeter construction and the positive-side smoothness identify the normal trace with $\partial_{\nu_+} v$ on the physical and artificial faces. The flux bound makes the boundary terms integrable. Green's formula for the harmonic function v and the compactly supported test $v\eta^2$ gives

$$\int_{U_+} \eta^2 |\nabla v|^2 = -2 \int_{U_+} v\eta \nabla v \cdot \nabla\eta + \int_{\Gamma_r} v\eta^2 \partial_{\nu_+} v.$$

There is no spherical boundary term. Using $2|v\eta\nabla v \cdot \nabla\eta| \leq \frac{1}{2}\eta^2|\nabla v|^2 + 2v^2|\nabla\eta|^2$ and the boundary bound, and absorbing the half of the gradient term, we obtain

$$\int_{\Omega_{3r/4}} |\nabla v|^2 \leq Cr^{-2} \int_{\Omega_r} |v|^2 + C\varepsilon R \int_{\Gamma_r} |\partial_{\nu_+} v|.$$

Volume comparability and the definition of Φ_r yield

$$r^2 \fint_{\Omega_{3r/4}} |\nabla v|^2 \leq C \fint_{\Omega_r} |v|^2 + C\Phi_r(v).$$

The coefficient of the first term is geometric and independent of C_0.

The Sobolev–Poincaré inequality on the uniformly John domain $\mathcal{O}$, whose diameter is at most $3r/2$, gives, for $2 < q \leq 10/3$,

$$\left(\fint_{\Omega_{r/2}} |v|^q \right)^{2/q} \leq C_q \left[\fint_{\mathcal{O}} |v|^2 + r^2 \fint_{\mathcal{O}} |\nabla v|^2 \right].$$

Here we used $|\Omega_{r/2}| \simeq |\mathcal{O}| \simeq r^5$ to pass from the domain average to the smaller set. The inclusions $\mathcal{O} \subset \Omega_{3r/4} \subset \Omega_r$, followed by the preceding gradient estimate, bound the right side by $C_q \fint_{\Omega_r} |v|^2 + C\Phi_r(v)$. Adding the gradient bound proves (6.10). The Sobolev and volume constants do not involve C_0, as claimed.

For U_- reverse the root orientation throughout. Finally, $\partial_{\nu_+}(v-c)=\partial_{\nu_+}v$, whereas subtracting c changes the affine intercept to $b+c$. Thus Φ_r is invariant under constant shifts, with the stated intercept condition for the energy estimate. □

Multiplying (6.9) by $r^{-3}\varepsilon R$ gives the form used in both iterations:

$$\Phi_r(v)\le CR^2\left[\varepsilon^3(r/R)^{1-4/p}+\varepsilon^{2p+1}(R/r)^3\right]. \tag{6.11}$$

The scaling of the weak residual will be used twice. Let $\widetilde{v}(y)=v(z_0+ry)/H$ and $\widetilde{U}=r^{-1}(U_+-z_0)$, where $H>0$. The signed-cover construction gives $U_+\subset\Omega$ and locally finite perimeter, with boundary assembled from the physical boundary and the artificial faces of the finite tree. Thus $v=u-L$ is harmonic in U_+. On a compact test support, the positive-side smoothness of u permits a smooth extension across the physical boundary; at an artificial face inside Ω, u is already smooth. Consequently the Gauss–Green normal trace is $\nabla v\cdot\nu_+=\partial_{\nu_+}v$ almost everywhere on the reduced boundary. These traces are integrable on the test support by Lemma 6.1. For $0<tr\le S\le R/16$ and $\varphi\in C_c^\infty(B_t)$, the change of variables and Green's formula give

$$\left|\int_{\widetilde{U}}\nabla\widetilde{v}\cdot\nabla\varphi\right|\le\left(\frac{S}{r}\right)^3\frac{\Phi_S(v)}{H\varepsilon R}\|\varphi\|_{L^\infty}. \tag{6.12}$$

To check the scaling explicitly, set $\psi_r(x)=\varphi((x-z_0)/r)$. It is supported in $B_{tr}(z_0)\subset B_S(z_0)$. The Jacobian and the two gradient factors give

$$\int_{\widetilde{U}}\nabla\widetilde{v}\cdot\nabla\varphi=\frac{r^{-3}}{H}\int_{U_+}\nabla v\cdot\nabla\psi_r=\frac{r^{-3}}{H}\int_{\Gamma_+}\psi_r\partial_{\nu_+}v.$$

There is no spherical boundary term because ψ_r is compactly supported. Taking absolute values and substituting $\int_{\Gamma_+\cap B_S}|\partial_{\nu_+}v|=S^3\Phi_S(v)/(\varepsilon R)$ proves (6.12).

6.4. **The Neumann compactness step.** We record the dimension-five version of [6, Lemma 11.1]. We use degree one for the first affine iteration and degree zero for the subsequent constant-shift iteration. The compactness tolerance δ_{nc} below is distinct from the tree parameter δ fixed in Section 4.

Lemma 6.2 (Neumann compactness)**.** *Fix $q>2$, $d\in\{0,1\}$, and $\theta\in(0,1)$. For every $\eta>0$ there are $\delta_{\mathrm{nc}}\in(0,1/4)$ and $T=\delta_{\mathrm{nc}}^{-1}$ with the following property. Let $D\subset\mathbb{R}^5$ be open, and let $v\in H^1(D\cap B_T)\cap L^q(D\cap B_T)$ be harmonic there. Suppose that some $a\in S^4$ satisfies, for $1\le t\le T$,*

$$B_t\cap\{a\cdot y>\delta_{\mathrm{nc}}t\}\subset D\cap B_t\subset B_t\cap\{a\cdot y>-\delta_{\mathrm{nc}}t\},$$

$$\left(\fint_{D\cap B_t}|v|^2\right)^{1/2}+t\left(\fint_{D\cap B_t}|\nabla v|^2\right)^{1/2}+\left(\fint_{D\cap B_t}|v|^q\right)^{1/q}\le t^{d+\theta},$$

and, for $\varphi\in C_c^\infty(B_t)$,

$$\left|\int_D\nabla v\cdot\nabla\varphi\right|\le\delta_{\mathrm{nc}}t^{3+d+\theta}\|\varphi\|_{L^\infty}.$$

Then a harmonic polynomial P of degree at most d, even across $a^\perp$, satisfies

$$\left(\fint_{D\cap B_{1/2}}|v-P|^2\right)^{1/2}\le\eta.$$

For $d=0$ this polynomial is constant. For $d=1$ it has the form $P(y)=q_0\cdot y+d_0$ with $q_0\cdot a=0$.

Proof. Apply [6, Lemma 11.1] with its parameters $(m, q, N, \vartheta, \varepsilon_{\mathrm{cmp}}) = (5, q, d, \theta, \eta)$. Let δ_{cmp} be its tolerance and set

$$\delta_{\mathrm{nc}} = \delta_{\mathrm{cmp}}/2, \qquad T = 2\delta_{\mathrm{cmp}}^{-1} = \delta_{\mathrm{nc}}^{-1}.$$

For $1 \le t \le \delta_{\mathrm{cmp}}^{-1}$, the source's closed inner cap $\{a \cdot y \ge \delta_{\mathrm{cmp}} t\}$ is contained in the strict inner cap $\{a \cdot y > \delta_{\mathrm{nc}} t\}$ used here. Our outer cap is contained in $\{a \cdot y \ge -\delta_{\mathrm{cmp}} t\}$, and our weak residual bound is stronger because $\delta_{\mathrm{nc}} \le \delta_{\mathrm{cmp}}$. The required function spaces and growth bounds hold on the larger ball B_T, so all hypotheses of the cited lemma hold on $B_{\delta_{\mathrm{cmp}}^{-1}}$. Its conclusion gives the asserted approximation without changing its normalization.

For $d = 0$ the polynomial is constant. For $d = 1$, reflection of $P(y) = q_0 \cdot y + d_0$ across $a^\perp$ changes its value by $-2(q_0 \cdot a)(a \cdot y)$, so evenness is equivalent to $q_0 \cdot a = 0$. □

6.5. **Affine approximation at the flat scale.** The next statement is the dimension-five form of [6, Proposition 11.2]. We include the iteration because its normalization differs from the later degree-zero iteration.

Proposition 6.3. *There are unit vectors $a_\pm^{\mathrm{f}}$ and affine maps $L_\pm^{\mathrm{f}}(x) = a_\pm^{\mathrm{f}} \cdot (x - z_0) + b_\pm^{\mathrm{f}}$ such that*

$$|a_\pm^{\mathrm{f}} \mp e_{\mathrm{root}}| \le C\varepsilon, \qquad |b_\pm^{\mathrm{f}}| \le C\varepsilon R,$$

and

$$\sum_{\pm} \left[\fint_{U_\pm \cap B_{8R_{\mathrm{f}}}(z_0)} |u - L_\pm^{\mathrm{f}}|^2 + R_{\mathrm{f}}^2 \fint_{U_\pm \cap B_{8R_{\mathrm{f}}}(z_0)} |\nabla u - a_\pm^{\mathrm{f}}|^2 \right] \le C\varepsilon^{2+2\chi/3} R_{\mathrm{f}}^2. \tag{6.13}$$

Proof. We work on U_+; the other sign is identical. Put

$$\gamma = \frac{2}{5}, \qquad q_{\mathrm{nc}} = \frac{5}{2}, \qquad h(r) = \varepsilon (r/R)^\gamma r, \qquad r_i = 2^{-i} R, \quad h_i = h(r_i).$$

The exponent q_{nc} is below $10/3$. Because $\varepsilon R / R_{\mathrm{f}} = \varepsilon^{1-\chi} \to 0$, the root slabs give $|U_+ \cap B_r(z_0)| \simeq r^5$ for $R_{\mathrm{f}} \le r \le R/16$. Every terminal neck meeting $B_{3r/4}(z_0)$ on this range belongs to $\mathcal{N}_{\mathrm{in}}$. Section 4 gives $r_N \le C\varepsilon R$, and hence

$$\frac{r_N}{r} \le C\varepsilon^{1-\chi} \longrightarrow 0 \qquad (R_{\mathrm{f}} \le r \le R/16).$$

Thus the terminal-radius condition in Lemma 6.1 holds uniformly on this range.

Inductive estimates. We construct unit-slope affine maps $L_i(x) = a_i \cdot (x - z_0) + b_i$, beginning with $L_4 = e_{\mathrm{root}} \cdot (x - z_0)$, for which

$$\left(\fint_{U_+ \cap B_{r_i}(z_0)} |u - L_i|^2 \right)^{1/2} \le C_{\mathrm{ind}} h_i. \tag{6.14}$$

Comparing two successive affine maps on a fixed-size cap in the smaller ball gives

$$r_i |a_{i+1} - a_i| + |b_{i+1} - b_i| \le C C_{\mathrm{ind}} h_i.$$

Here the root slabs place a ball of radius cr_i in $U_+ \cap B_{r_{i+1}}(z_0)$. Restrict both error estimates to that ball and use the triangle inequality. The required affine coefficient estimate follows, after translation to the ball center, from

$$\fint_{B_s} |A \cdot x + B|^2 = \frac{s^2}{7} |A|^2 + B^2.$$

Translating the intercept back to z_0 costs only a fixed multiple of $r_i|A|$, since the cap center is at distance $O(r_i)$ from z_0. Summing the geometric series gives, for $4 \le i \le j$,

$$\begin{aligned} r_i|a_j - a_i| + |b_j - b_i| &\le CC_{\mathrm{ind}} h_i, \\ |a_j - e_{\mathrm{root}}| + R^{-1}|b_j| &\le CC_{\mathrm{ind}}\varepsilon. \end{aligned} \tag{6.15}$$

Apply (6.11) with $p = p_1$. If $t = r/R$, then $h(r)^2 = \varepsilon^2 R^2 t^{2+2\gamma}$, so

$$\frac{\Phi_r(u - L_j)}{h(r)^2} \le C(C_{\mathrm{ind}}) \left[\varepsilon t^{-1-4/p_1-2\gamma} + \varepsilon^{2p_1-1} t^{-5-2\gamma}\right] \le C(C_{\mathrm{ind}})\zeta_k^2,$$

where

$$\zeta_k^2 = \varepsilon^{1-\chi(9/5+4/p_1)} + \varepsilon^{2p_1-1-\chi(5+4/5)} \longrightarrow 0.$$

The last limit follows from the positive exponents checked in Section 6.1.

We describe precisely the compactness window. For the constants T and j_0 to be chosen, assume $2^{j_0} \ge 64T$ and $j \ge j_0$. For $1 \le t \le T$, choose $i \ge 4$ with $2tr_j \le r_i < 4tr_j$. This is possible because $4Tr_j \le R/16$. Define

$$v_j(y) = \frac{u(z_0 + r_j y) - L_j(z_0 + r_j y)}{h_j}, \qquad D_j = r_j^{-1}(U_+ - z_0).$$

Use (6.14) at r_i, (6.15), and the local energy estimate at r_i. In detail, the affine jumps first give

$$\left(\fint_{U_+ \cap B_{r_i}(z_0)} |u - L_j|^2 \right)^{1/2} \le CC_{\mathrm{ind}} h_i.$$

The energy estimate then controls the gradient on $B_{3r_i/4}$ and the $L^{5/2}$ norm on $B_{r_i/2}$, both of which contain B_{tr_j}. The relevant volume ratios are bounded because $2tr_j \le r_i < 4tr_j$. Since $h_i/h_j \le (4t)^{1+\gamma}$, they give

$$\begin{aligned} &\left(\fint_{D_j \cap B_t} |v_j|^2 \right)^{1/2} + t \left(\fint_{D_j \cap B_t} |\nabla v_j|^2 \right)^{1/2} + \left(\fint_{D_j \cap B_t} |v_j|^{5/2} \right)^{2/5} \\ &\qquad \le (KC_{\mathrm{ind}} + C(C_{\mathrm{ind}})\zeta_k) t^{1+\gamma}. \end{aligned} \tag{6.16}$$

The constant K is independent of C_{ind} and T. Equation (6.12), with $S = r_i$ and $H = h_j$, gives, for $\varphi \in C_c^\infty(B_t)$,

$$\begin{aligned} \left| \int_{D_j} \nabla v_j \cdot \nabla \varphi \right| &\le \left(\frac{r_i}{r_j} \right)^3 \frac{\Phi_{r_i}(u - L_j)}{h_j \varepsilon R} \|\varphi\|_\infty \\ &\le C(C_{\mathrm{ind}})\zeta_k^2 t^3 \left(\frac{h_i}{h_j} \right)^2 \frac{h_j}{\varepsilon R} \|\varphi\|_\infty \\ &\le C(C_{\mathrm{ind}})\zeta_k^2 t^{5+2\gamma} \|\varphi\|_\infty. \end{aligned}$$

We used $h_j/(\varepsilon R) \le 1$ and $h_i/h_j \le (4t)^{1+\gamma}$. Finally, replacing the root normal by a_j in its slabs costs $|a_j - e_{\mathrm{root}}| tr_j$. Their relative width is therefore at most

$$C\frac{\varepsilon R}{tr_j} + |a_j - e_{\mathrm{root}}| \le CC_{\mathrm{ind}} \varepsilon^{1-\chi},$$

uniformly for $1 \le t \le T$ and $r_j \ge R_{\mathrm{f}}$.

Order of choices and improvement. Choose $\eta > 0$ so that $(K+1)\eta \le 2^{-3-\gamma}$. Obtain δ_{nc}, T from Lemma 6.2 with $(q, d, \theta) = (5/2, 1, \gamma)$. Fix $j_0 \ge 4$ with $2^{j_0} \ge 64T$. For $4 \le i \le j_0$, take $L_i = L_4$. The root vee estimate and (6.7) imply

$$\int_{U_+ \cap B_{R/8}(z_0)} |u - L_4|^2 \le C\varepsilon^2 R^7.$$

Indeed, the root slabs imply $(L_4)_- \le C\varepsilon R$ on U_+, and

$$|u - L_4| \le |u - |L_4|| + 2(L_4)_-.$$

For the first term, $V_{B_R(z_0)} = |L_4|$ and the root bound from Section 4 give

$$\int_{B_{R/8}(z_0)} |u - |L_4||^2 \le \int_{B_{16R}(z_0)} |u - V_{B_R(z_0)}|^2 = h_{B_R(z_0)}^2 R^7 \le C\varepsilon^2 R^7.$$

The slab height controls the second term, giving the displayed integral bound for $u - L_4$ on U_+. Thus a sufficiently large fixed $C_{\mathrm{ind}} = C(j_0)$ initializes (6.14). Fix C_0 in Lemma 6.1 larger than the coefficient bound in (6.15); it depends only on these already fixed constants.

Only now increase the selected index. Require $r_{j_0} \ge 32R_{\mathrm{f}}$, all geometric smallness conditions, and

$$C(C_{\mathrm{ind}})\zeta_k \le C_{\mathrm{ind}}, \qquad CC_{\mathrm{ind}}\varepsilon^{1-\chi} \le \delta_{\mathrm{nc}}, \qquad \frac{C(C_{\mathrm{ind}})\zeta_k^2 T^{1+\gamma}}{(K+1)C_{\mathrm{ind}}} \le \delta_{\mathrm{nc}}.$$

The last inequality compares the residual exponent $5 + 2\gamma$ with the required exponent $4 + \gamma$ on the fixed window. Consequently $v_j/((K+1)C_{\mathrm{ind}})$ satisfies every hypothesis of Lemma 6.2. It is approximated on $B_{1/2}$ by $P_j(y) = q_j \cdot y + d_j$, where $q_j \cdot a_j = 0$. Comparison on a fixed cap gives $|q_j| + |d_j| \le C$.

Put $A_j = (K+1)C_{\mathrm{ind}}h_j$ and first set

$$\overline{L}_{j+1}(x) = L_j(x) + A_j P_j((x - z_0)/r_j).$$

Normalize its slope to length one while preserving its value at z_0. Write $\lambda_j = A_j/r_j$ and $\bar{a}_{j+1} = a_j + \lambda_j q_j$. Orthogonality gives

$$|\bar{a}_{j+1}|^2 = 1 + \lambda_j^2 |q_j|^2 \ge 1, \qquad \left| \frac{\bar{a}_{j+1}}{|\bar{a}_{j+1}|} - \bar{a}_{j+1} \right| = |\bar{a}_{j+1}| - 1 \le \frac{1}{2}\lambda_j^2 |q_j|^2.$$

Thus the normalized slope is well defined, and its change on $B_{r_j/2}(z_0)$ is at most $A_j^2|q_j|^2/(4r_j)$. Relative to the inductive amplitude, this is bounded by

$$\frac{A_j^2|q_j|^2}{4r_j C_{\mathrm{ind}} h_j} \le C(K+1)^2 C_{\mathrm{ind}} \frac{h_j}{r_j} \le C(K+1)^2 C_{\mathrm{ind}}\varepsilon \longrightarrow 0.$$

All constants are fixed before increasing the selected index, so this limit is uniform in j. The compactness error and this quadratic error are therefore at most $2^{-2-\gamma}C_{\mathrm{ind}}h_j$ after increasing the selected index. This is smaller than $C_{\mathrm{ind}}h_{j+1}$ and proves the next step. The affine comparison again gives (6.15), so the induction closes as long as $r_{j+1} \ge R_{\mathrm{f}}$.

Extraction of the affine map. Choose an index i with $16R_{\mathrm{f}} \le r_i < 32R_{\mathrm{f}}$ and take $L_+^{\mathrm{f}} = L_i$. The local energy estimate at r_i, restricted to $B_{8R_{\mathrm{f}}}(z_0)$, gives the left-hand side of (6.13) on U_+ bounded by

$$C\left(C_{\mathrm{ind}}^2 h_i^2 + \Phi_{r_i}(u - L_i)\right) \le Ch_i^2.$$

Indeed, $r_i/2 \ge 8R_{\mathrm{f}}$ and all relevant volume ratios are bounded. Since

$$h_i^2 \le C\varepsilon^{2+2\gamma\chi} R_{\mathrm{f}}^2 \le C\varepsilon^{2+2\chi/3} R_{\mathrm{f}}^2,$$

the required estimate follows, since $\gamma = 2/5 > 1/3$ and $0 < \varepsilon < 1$. The normalization follows from (6.15). Repeat on U_-. □

We now compare the two affine maps, following [6, Lemma 11.3]. We give the argument at our parameters; the last step uses the mean-value property of harmonic functions.

Lemma 6.4 (Comparison of the two affine maps)**.** *The maps in Proposition 6.3 satisfy*

$$u \geq \max\{L_+^{\mathrm{f}}, L_-^{\mathrm{f}}\} - C\varepsilon^{1+\chi/6} R_{\mathrm{f}} \quad \text{in } B_{5R_{\mathrm{f}}}(z_0). \tag{6.17}$$

Moreover, either $|a_+^{\mathrm{f}} + a_-^{\mathrm{f}}| \leq C\varepsilon^{1+\chi/6}$*, or*

$$\{L_+^{\mathrm{f}} > 0\} \cap \{L_-^{\mathrm{f}} > 0\} \cap B_{4R_{\mathrm{f}}}(z_0) = \varnothing.$$

Proof. Write $\ell = R_{\mathrm{f}}$. Lemma 2.1 gives $5 < n_{\mathrm{cone}}$, so the dimension-general excess and one-sided estimates of [6] apply to the normalized solution. Choose e_0 minimizing $\mathcal{E}_{z_0}(8\ell;\cdot)$. The neck modulus and (6.5), followed by fixed-direction propagation [6, Proposition 7.9] with outer radius 64ℓ and ratio $1/8$, give

$$\delta_k := \mathcal{E}_{z_0}(64\ell; e_0) \leq C\mathcal{E}_{z_0}(8\ell; e_0) \leq C\omega_*\Big(C\varepsilon^{12/11-\chi}\Big) \longrightarrow 0.$$

The signed slabs (6.6) at radius 8ℓ are admissible: $B_{8\ell}(z_0) \subset B_{R/2}(z_0)$ and $8\ell/r_*(z_0) \to \infty$. Orient e_0 toward U_+.

We first obtain rough alignment on explicit balls. Put

$$Q_\pm = B_{\ell/4}(z_0 \pm 2\ell e_0).$$

These balls lie in $B_{8\ell}(z_0)$, and their signed heights are at least $7\ell/4$. The slab threshold is $8C\delta_k\ell = o(\ell)$, so $Q_\pm \subset U_\pm$ for large k. The vee equals $\pm e_0 \cdot (x - z_0)$ there. The nonnegativity of the Weiss deficit, proved in Section 4, implies $M^{e_0}_{64\ell, z_0} \leq \delta_k^2$. Thus

$$\int_{B_{64\ell}(z_0)} |u - V_{z_0,e_0}|^2 \leq (64\ell)^7 \delta_k^2.$$

Also, (6.13) and $|U_\pm \cap B_{8\ell}(z_0)| \leq C\ell^5$ give

$$\int_{U_\pm \cap B_{8\ell}(z_0)} |u - L_\pm^{\mathrm{f}}|^2 \leq C\varepsilon^{2+2\chi/3}\ell^7.$$

Since $|Q_\pm| = |B_1^5|(\ell/4)^5$, restriction and the triangle inequality yield

$$\left(\fint_{Q_\pm} |L_\pm^{\mathrm{f}} \mp e_0 \cdot (x - z_0)|^2 \right)^{1/2} \leq C(\varepsilon^{1+\chi/3} + \delta_k)\ell = o(\ell).$$

For an affine function $A \cdot x + B$, the identity

$$\fint_{B_s} |A \cdot x + B|^2 = \frac{s^2}{7}|A|^2 + B^2$$

controls its slope and its value at the ball center. Apply it with $s = \ell/4$ and then translate from $z_0 \pm 2\ell e_0$ back to z_0. We obtain

$$|a_+^{\mathrm{f}} - e_0| + |a_-^{\mathrm{f}} + e_0| + \ell^{-1}(|b_+^{\mathrm{f}}| + |b_-^{\mathrm{f}}|) = o(1). \tag{6.18}$$

For each sign set

$$y_\pm = z_0 - b_\pm^{\mathrm{f}} a_\pm^{\mathrm{f}}, \qquad D_\pm = B_{7\ell}(y_\pm) \cap \{L_\pm^{\mathrm{f}} > 7\ell/8\}.$$

The unit-slope condition gives $L_\pm^{\mathrm{f}}(x) = a_\pm^{\mathrm{f}} \cdot (x - y_\pm)$. Consequently

$$|D_\pm| = c_{\mathrm{cap}}(7\ell)^5, \qquad c_{\mathrm{cap}} = |B_1^5 \cap \{x_5 > 1/8\}| > 0,$$

independently of the sign and direction. If $x \in D_\pm$, (6.18) gives

$$|x - z_0| < 7\ell + |b_\pm^{\mathrm{f}}| < 8\ell,$$

$$(\pm e_0)\cdot(x-z_0) > \frac{7\ell}{8} - |b_\pm^{\mathrm{f}}| - 8\ell|a_\pm^{\mathrm{f}} \mp e_0| \geq \frac{3\ell}{4}$$

for large k. This height exceeds the $o(\ell)$ signed-slab threshold, so $D_\pm \subset U_\pm \cap B_{8\ell}(z_0)$. Cauchy–Schwarz and the preceding affine-error integral therefore give

$$\int_{D_\pm} (u - L_\pm^{\mathrm{f}})_- \leq |D_\pm|^{1/2} \left(\int_{D_\pm} |u - L_\pm^{\mathrm{f}}|^2 \right)^{1/2} \leq C\varepsilon^{1+\chi/3}\ell^6.$$

In particular,

$$\frac{1}{7\ell} \fint_{D_\pm} (u - L_\pm^{\mathrm{f}})_- \leq C\varepsilon^{1+\chi/3} =: \eta_k < 1$$

for large k. This is exactly the normalized cap hypothesis in [6, Lemma 5.9], with center $y_\pm$, unit direction $a_\pm^{\mathrm{f}}$, radius 7ℓ, and tolerance $\eta_k > 0$. We choose its displayed upper bound as the tolerance, even when the negative part is identically zero. The conclusion is

$$u \geq L_\pm^{\mathrm{f}} - C\varepsilon^{1+\chi/3}\ell \quad \text{on } B_{21\ell/4}(y_\pm).$$

Since $|y_\pm - z_0| = |b_\pm^{\mathrm{f}}| < \ell/4$ eventually, this ball contains $B_{5\ell}(z_0)$. Since $0 < \varepsilon < 1$ and $\chi > 0$, one has $\varepsilon^{1+\chi/3} \leq \varepsilon^{1+\chi/6}$. Taking the maximum of the two lower bounds proves (6.17).

It remains to prove the alternative. Translate z_0 to zero and rescale ℓ to one, writing

$$\widetilde{u}(y) = \ell^{-1}u(z_0 + \ell y), \qquad \widetilde{L}_\pm(y) = a_\pm^{\mathrm{f}}\cdot y + b_\pm^{\mathrm{f}}/\ell, \qquad g = a_+^{\mathrm{f}} + a_-^{\mathrm{f}}, \qquad \theta = \varepsilon^{1+\chi/6}.$$

If $g = 0$, the first alternative holds. Otherwise suppose that the two positive half-spaces meet at $y' \in B_4$, and put $v = g/|g|$. Their strict positivity and (6.18) imply $|e_0 \cdot y'| = o(1)$ and $|g| = o(1)$. The unit-slope identity $a_+^{\mathrm{f}} \cdot g = |g|^2/2$ gives

$$|e_0 \cdot v| \leq |e_0 - a_+^{\mathrm{f}}| + |g|/2 = o(1).$$

This estimate is uniform even when $|g|$ is small. Set $y_1 = y' + v/2$. For $y \in B_{1/4}(y_1)$, the sum of the two affine maps is strictly larger than $|g|/4$, so

$$\max\{\widetilde{L}_+(y), \widetilde{L}_-(y)\} > |g|/8.$$

This ball lies in B_5, and $e_0 \cdot y_1 = o(1)$. Its projection y_0 onto $e_0^\perp$ therefore satisfies $|y_0 - y_1| = o(1)$, so

$$B_{1/8}(y_0) \subset B_{1/4}(y_1) \subset B_5$$

for large k. If $|g| \geq M\theta$, where M is a fixed constant larger than $16(C_{\mathrm{lower}} + 1)$ and C_{lower} is the constant in (6.17), that lower bound makes this whole ball positive. Thus $\widetilde{u}$ is harmonic on $B_{1/8}(y_0)$.

Put $r_0 = 1/8$. Harmonicity would give equal averages of $\widetilde{u}$ over $B_{r_0}(y_0)$ and $B_{r_0/2}(y_0)$. On the other hand, the vee estimate gives

$$\int_{B_8} |\widetilde{u} - |e_0 \cdot y||^2 \leq 64^7 \delta_k^2 = o(1).$$

Both balls lie in B_8, and their fixed positive volumes make the errors in their averages $o(1)$ by Cauchy–Schwarz, uniformly in y_0 and e_0. Since $e_0 \cdot y_0 = 0$, translation, rotation, and a one-variable integral give, for $s > 0$,

$$\fint_{B_s(y_0)} |e_0 \cdot y|\, dy = \frac{2|B_1^4|}{|B_1^5|s^5} \int_0^s t(s^2 - t^2)^2\, dt = \frac{5}{16}s.$$

The difference of these two vee averages is therefore $5r_0/32 = 5/256 > 0$, contradicting equality of the harmonic averages. Hence $|g| < M\theta$ whenever the half-spaces meet, which proves the stated alternative. □

6.6. **Oscillation decay with a fixed affine slope.** We now keep each affine slope fixed and change only the additive constant. The improved neck estimate (6.5) is the reason that this iteration can reach the amplitude scale.

Proposition 6.5. *There is $e_k \in S^4$, with $|e_k - e_{\mathrm{root}}| \le C\varepsilon$, such that*

$$\fint_{U_\pm \cap B_r(z_0)} |u \mp e_k \cdot (x - z_0)|^2 + r^2 \fint_{U_\pm \cap B_{3r/4}(z_0)} |\nabla(u \mp e_k \cdot (x - z_0))|^2 \le C A_\#^2 \tag{6.19}$$

for every $A_\# \le r \le R_{\mathrm{f}}$. In addition,

$$u \ge |e_k \cdot (x - z_0)| - C A_\# \quad \text{in } B_{5R_{\mathrm{f}}}(z_0). \tag{6.20}$$

Proof. Set $\beta = \beta_{\mathrm{osc}} = 1/1000$ and

$$H(r) = A_\#(r/R_{\mathrm{f}})^\beta, \qquad r_j = 2^{-j} R_{\mathrm{f}}, \qquad H_j = H(r_j).$$

We first work on U_+ with the fixed map $L = L_+^{\mathrm{f}}$ and slope $a = a_+^{\mathrm{f}}$. The error in (6.13) is $o(A_\#)$ because $\chi/3 - \sigma = 97/300000 > 0$.

Step 1: Flux on the entire interval. Use (6.11) with $p = p_2$. For $r = \varepsilon^s R$, the interval $A_\# \le r \le R_{\mathrm{f}}$ is $\chi \le s \le 1 + \chi + \sigma$. Dividing by $H(r)^2$ gives the two powers of ε

$$A(s) = 1 + s(1 - 4/p_2) - 2\chi - 2\sigma - 2\beta(s - \chi), \tag{6.21}$$

$$B(s) = 2p_2 - 1 - 3s - 2\chi - 2\sigma - 2\beta(s - \chi). \tag{6.22}$$

Their derivatives are

$$A'(s) = 1 - \frac{4}{p_2} - 2\beta = -\frac{19541}{20500} < 0, \qquad B'(s) = -3 - 2\beta = -\frac{1501}{500} < 0.$$

Thus both attain their minimum at $s = 1 + \chi + \sigma$, the lower-radius endpoint. The values there are

$$\min A = \frac{89789459}{2050000000} > \chi, \qquad \min B = \frac{4647499}{50000000} > \chi.$$

Thus, for every constant c,

$$\frac{\Phi_r(u - L - c)}{H(r)^2} \le C\varepsilon^\chi \qquad (A_\# \le r \le R_{\mathrm{f}}). \tag{6.23}$$

The invariance under c concerns the flux only; its use in the energy estimate will be justified below.

Step 2: Consequences of the induction hypothesis. We construct constants c_j for which

$$\left(\fint_{U_+ \cap B_{r_i}(z_0)} |u - L - c_i|^2 \right)^{1/2} \le C_{\mathrm{ind}} H_i \qquad (0 \le i \le j). \tag{6.24}$$

The initial block has $c_i = 0$; its length and C_{ind} will be fixed before increasing the selected index.

For $A_\# \le s \le R_{\mathrm{f}}$, choose a direction ν_s minimizing $\mathcal{E}_{z_0}(8s; \cdot)$ and orient it toward U_+. Equations (4.5), (6.5), and (6.6) give signed slabs with relative widths at most

$$E_s = C\mathcal{E}_{z_0}(8s; \nu_s) \le C\omega_*(Cr_{\mathrm{neck}}/s) \le C\omega_*(Cr_{\mathrm{neck}}/A_\#) = o(1).$$

Consequently $|U_\pm \cap B_s(z_0)| \simeq s^5$ throughout the interval. Restricting two consecutive estimates in (6.24) to their common smaller ball gives

$$|c_i - c_{i-1}| \le CC_{\mathrm{ind}} H_{i-1}, \qquad |c_j - c_i| \le C_\beta C_{\mathrm{ind}} H_i, \qquad |c_j| \le C_\beta C_{\mathrm{ind}} A_\#. \tag{6.25}$$

The second and third bounds follow by summing a geometric series. For example,

$$\sum_{l=i}^{j-1} H_l \le \frac{H_i}{1 - 2^{-\beta}},$$

so the constants in these two bounds may contain $(1-2^{-\beta})^{-1}$. This is fixed because $\beta = 1/1000$. In particular, $|c_j|/(\varepsilon R) \le CC_{\mathrm{ind}}\varepsilon^{\chi+\sigma} \to 0$. Fix the intercept constant C_0 at least one larger than the flat-map coefficient bound. Once the selected index is large enough that $|c_j| \le \varepsilon R$ for all inductive indices, the intercept hypothesis of Lemma 6.1 is preserved after subtracting c_j.

We need the normal direction of the slabs to agree with the fixed slope a. For $r_j \le s \le R_{\mathrm{f}}$, choose $i \le j$ with $s \le r_i < 2s$. The ball

$$Q_s = B_{s/8}(z_0 + s\nu_s/4)$$

lies in $U_+ \cap B_s(z_0)$ for large selected index. On it the positive branch of the optimal centered vee is $\nu_s \cdot (x - z_0)$. The vee error and (6.24) give

$$\left(\fint_{Q_s} |L + c_i - \nu_s \cdot (x - z_0)|^2 \right)^{1/2} \le CC_{\mathrm{ind}} H_i + CE_s s.$$

Here the vee contribution is bounded using

$$\int_{B_{8s}(z_0)} |u - V_{z_0,\nu_s}|^2 \le (8s)^7 \mathcal{E}_{z_0}(8s;\nu_s)^2$$

and $|Q_s| \simeq s^5$. The inductive error is restricted from B_{r_i}, with $s \le r_i < 2s$. Affine coefficient comparison on Q_s, followed by (6.25), yields

$$|a - \nu_s| + \frac{|L(z_0) + c_j|}{s} \le CC_{\mathrm{ind}} \frac{H(s)}{s} + CE_s \le \vartheta_k \longrightarrow 0, \tag{6.26}$$

where one may take

$$\vartheta_k = CC_{\mathrm{ind}}\varepsilon^{\beta(1+\sigma)} + C\omega_*\Big(C\varepsilon^{1/11-\chi-\sigma}\Big).$$

Here $\sup_{[A_\#,R_{\mathrm{f}}]} H(s)/s = \varepsilon^{\beta(1+\sigma)}$. Indeed, $H(s)/s = (A_\#/R_{\mathrm{f}})^\beta (s/A_\#)^{\beta-1}$ is largest at $s = A_\#$, and $A_\#/R_{\mathrm{f}} = \varepsilon^{1+\sigma}$. Thus all the domains in the compactness step are trapped between half-spaces normal to the same vector a, with relative error ϑ_k: changing ν_s to a changes the defining linear function by at most $|a - \nu_s|s$ on $B_s(z_0)$.

Step 3: The fixed compactness window. Assume $2^{j_0} \ge 32T$ and $j \ge j_0$, where T will be fixed below. For $1 \le t \le T$, choose $i \le j$ with

$$2tr_j \le r_i < 4tr_j.$$

Then $i \ge 0$ and all these radii are at most R_{f}. By (6.24), (6.25), and volume comparability,

$$\left(\fint_{U_+ \cap B_{2tr_j}(z_0)} |u - L - c_j|^2 \right)^{1/2} \le CC_{\mathrm{ind}} H_i \le KC_{\mathrm{ind}} t^\beta H_j.$$

Apply Lemma 6.1 at $S = 2tr_j$. Its terminal-radius hypothesis follows from (6.5); the intercept condition was just checked. With

$$v_j(y) = \frac{u(z_0 + r_j y) - L(z_0 + r_j y) - c_j}{H_j}, \qquad D_j = r_j^{-1}(U_+ - z_0),$$

we obtain

$$\begin{aligned} \left(\fint_{D_j \cap B_t} |v_j|^2 \right)^{1/2} + t \left(\fint_{D_j \cap B_t} |\nabla v_j|^2 \right)^{1/2} + \left(\fint_{D_j \cap B_t} |v_j|^3 \right)^{1/3} \\ \le (KC_{\mathrm{ind}} + C(C_{\mathrm{ind}})\varepsilon^{\chi/2}) t^\beta. \end{aligned} \tag{6.27}$$

The constant K is independent of C_{ind} and T. The exponent 3 is allowed because $3 < 10/3$. For the weak residual, use (6.12) at the same $S = 2tr_j$. Since $H(S)/H_j = (2t)^\beta$ and $H_j/(\varepsilon R) \le \varepsilon^{\chi+\sigma} \le 1$, it gives

$$\left| \int_{D_j} \nabla v_j \cdot \nabla \varphi \right| \le C(C_{\mathrm{ind}}) \varepsilon^\chi t^{3+2\beta} \|\varphi\|_{L^\infty}, \qquad \varphi \in C_c^\infty(B_t). \tag{6.28}$$

Explicitly, the bound before discarding the last factor is

$$\left(\frac{S}{r_j}\right)^3 \frac{\Phi_S(u - L - c_j)}{H_j \varepsilon R} \le C\varepsilon^\chi (2t)^3 \left(\frac{H(S)}{H_j}\right)^2 \frac{H_j}{\varepsilon R} \le C\varepsilon^\chi t^{3+2\beta}.$$

Equation (6.26), applied at $s = tr_j$, gives the required slabs on this whole window.

Step 4: Closing the induction. Choose $\eta > 0$ so that $(K+1)\eta \le 2^{-2-\beta}$. Let δ_{nc}, T be given by Lemma 6.2 with $(q, d, \theta) = (3, 0, \beta)$. Fix j_0 with $2^{j_0} \ge 32T$ and then fix $C_{\mathrm{ind}} \ge 1$. For the finite initial block, (6.13) gives

$$\max_{0 \le i \le j_0} \frac{1}{H_i} \left(\fint_{U_+ \cap B_{r_i}(z_0)} |u - L|^2 \right)^{1/2} \le C 2^{(5/2+\beta) j_0} \varepsilon^{\chi/3 - \sigma} \longrightarrow 0.$$

Thus $c_i = 0$ initializes (6.24) after increasing the selected index. At the same time require $r_{j_0+1} \ge A_\#$, all earlier geometric conditions, and

$$\vartheta_k \le \delta_{\mathrm{nc}}, \qquad C(C_{\mathrm{ind}}) \varepsilon^{\chi/2} \le C_{\mathrm{ind}}, \qquad \frac{C(C_{\mathrm{ind}}) \varepsilon^\chi T^\beta}{(K+1) C_{\mathrm{ind}}} \le \delta_{\mathrm{nc}}.$$

These requirements are compatible because all their constants have already been fixed. The factor T^β is necessary: it converts the exponent $3 + 2\beta$ in (6.28) into the exponent $3 + \beta$ required by compactness.

Suppose the induction has reached $j \ge j_0$ and $r_{j+1} \ge A_\#$. Equations (6.27) and (6.28), and the slab bound, show that $v_j/((K+1)C_{\mathrm{ind}})$ satisfies Lemma 6.2. Its approximating polynomial is a constant d_j. The normalized L^2 bound on $B_{1/2}$ gives $|d_j| \le C$. Set

$$c_{j+1} = c_j + (K+1) C_{\mathrm{ind}} H_j d_j.$$

Then

$$|c_{j+1} - c_j| \le C C_{\mathrm{ind}} H_j, \qquad \left(\fint_{U_+ \cap B_{r_{j+1}}(z_0)} |u - L - c_{j+1}|^2 \right)^{1/2} \le \tfrac{1}{2} C_{\mathrm{ind}} H_{j+1}.$$

This closes the induction.

Step 5: Removing shifts and matching the two sides. The geometric sum in (6.25) gives $|c_j| \le C A_\#$. Together with (6.24), this proves

$$\left(\fint_{U_+ \cap B_r(z_0)} |u - L|^2 \right)^{1/2} \le C A_\# \qquad (A_\# \le r \le R_{\mathrm{f}}).$$

To include all radii and the lower endpoint, let $J = \max\{j : r_j \ge A_\#\}$, so $r_J \in [A_\#, 2A_\#)$. For every $r \in [A_\#, R_{\mathrm{f}}]$, choose $i \le J$ with $r \le r_i < 2r$. Restricting the error bound from B_{r_i} and using volume comparability proves the displayed estimate at r. At the last dyadic radius r_J, the facts

$u(z_0)=0$, $|\nabla u|\le 1$, and $|a|=1$ give

$$|L(z_0)+c_J| \le \left(\fint_{U_+\cap B_{r_J}} |u-L-c_J|^2 \right)^{1/2} + 2r_J \le CA_{\#}.$$

In fact $|u(x)-a\cdot(x-z_0)|\le 2r_J$ on that ball, which explains the term $2r_J$. Since $|c_J|\le CA_{\#}$, we obtain $|L(z_0)|\le CA_{\#}$. Apply the local energy inequality to $u-L$ at each radius r. Its intercept condition holds, and all relevant terminal radii are $o(A_{\#})$. Together with (6.23) and $H(r)\le A_{\#}$, it gives

$$r^2 \fint_{U_+\cap B_{3r/4}(z_0)} |\nabla u - a|^2 \le CA_{\#}^2 + C\Phi_r(u-L) \le CA_{\#}^2.$$

Repeat the construction on U_-.

It remains to replace the two maps by an exactly centered opposite pair. Put $g=a_+^{\mathrm{f}}+a_-^{\mathrm{f}}$. Their values at z_0 have just been bounded by $CA_{\#}$. If the first alternative of Lemma 6.4 holds, then

$$|g| \le C\varepsilon^{1+\chi/6} \le CA_{\#}/R_{\mathrm{f}},$$

because $\chi/6-\sigma = 47/300000 > 0$. In the other alternative, if $g\neq 0$, evaluate both affine maps at $z_0+3R_{\mathrm{f}}g/|g|$. Each value is $\frac{3}{2}R_{\mathrm{f}}|g|$ plus its value at z_0. They cannot both be positive, so again $|g|\le CA_{\#}/R_{\mathrm{f}}$. Choose $e_k=a_+^{\mathrm{f}}$. On $B_{5R_{\mathrm{f}}}(z_0)$ the change from $L_\pm^{\mathrm{f}}$ to $\pm e_k\cdot(x-z_0)$ is $O(A_{\#})$, and the change in slope is $O(A_{\#}/R_{\mathrm{f}})$. For example, on the minus side the correction is $b_-^{\mathrm{f}}+g\cdot(x-z_0)$, whose squared value and scaled squared gradient on B_r are bounded by

$$C(|b_-^{\mathrm{f}}|^2 + r^2|g|^2) \le CA_{\#}^2 \qquad (r\le R_{\mathrm{f}}).$$

This proves (6.19). Finally, (6.17) and $\varepsilon^{1+\chi/6}R_{\mathrm{f}}\le A_{\#}$ prove (6.20). □

7. Gaussian calibration and rigidity

We now estimate both terms in the combined excess. The estimates in the preceding section are centered at z_0. The Weiss part of the excess also involves nearby centers on the ridge. We therefore keep all constants uniform in that center.

Proposition 7.1 (Improvement of the combined excess)**.** *Let the parameters be as in* (6.1)*, and set* $\rho=\varepsilon^{\tau}R_{\mathrm{f}}$*. For all sufficiently large selected indices,*

$$\mathcal{E}_{z_0}(\rho;e_k)^2 \le C\varepsilon^{2+\sigma}. \tag{7.1}$$

Proof. Step 1: Geometry and endpoint estimates. Put $T=R_{\mathrm{f}}/4$, and fix

$$s\in (z_0+e_k^{\perp})\cap B_\rho(z_0).$$

Since $\rho/R_{\mathrm{f}}=\varepsilon^{\tau}\to 0$, we may assume $\rho\le R_{\mathrm{f}}/4$. The triangle inequality then gives, uniformly in s,

$$B_T(s)\subset B_{R_{\mathrm{f}}/2}(z_0)\subset B_{R/8}(z_0).$$

For $*\in\{+,-\}$ write

$$e_+=e_k,\quad e_-=-e_k,\qquad P_*(x)=e_*\cdot(x-s),\quad w_*=u-P_*,\quad H_*=\{P_*>0\}.$$

The orthogonality of $s-z_0$ and e_k gives $P_*(x)=e_*\cdot(x-z_0)$. Thus these affine maps are independent of the choice of s. Proposition 6.5, used at $r=R_{\mathrm{f}}$, gives

$$\sum_{*=\pm}\int_{U_*\cap B_{R_{\mathrm{f}}/2}(z_0)} (w_*^2+R_{\mathrm{f}}^2|\nabla w_*|^2) \le CA_{\#}^2R_{\mathrm{f}}^5. \tag{7.2}$$

Indeed, the value estimate there is on $U_* \cap B_{R_{\mathrm{f}}}(z_0)$ and the gradient estimate is on $U_* \cap B_{3R_{\mathrm{f}}/4}(z_0)$. Both contain the integration region above; converting their averages to integrals costs at most CR_{f}^5. The centered lower bound also gives

$$|P_*| \le u + CA_{\#} \qquad \text{in } B_{5R_{\mathrm{f}}}(z_0). \tag{7.3}$$

Step 2: The fixed-radius identity. For almost every $r \in (0,T)$ define

$$\mathcal{W}_{*,s}(r) = r^{-5}\int_{U_*\cap B_r(s)} (|\nabla u|^2 + 1) - r^{-6}\int_{U_*\cap \partial B_r(s)} u^2 \, d\mathcal{H}^4.$$

All boundary integrals in this proof are with respect to $\mathcal{H}^4$. The normal ν_* is outward from U_*. On its regular physical boundary, $u = 0$ and $\partial_{\nu_*} u = -1$. The signed domains have locally finite perimeter, with boundary assembled from smooth physical pieces and the artificial faces of the finite tree. The three vector fields used below are

$$u\nabla u, \qquad P_* e_*, \qquad P_*\nabla u - u e_*.$$

Their divergences in the positive phase are respectively $|\nabla u|^2$, 1, and 0, because $\Delta u = \Delta P_* = 0$ and $\nabla P_* = e_*$. They have the positive-side boundary traces supplied by classical regularity. Gauss–Green on the finite-perimeter signed domains, followed by coarea for their intersections with $B_r(s)$, gives the following identities for almost every r.

Green's formula for u and the identity $\operatorname{div}(P_* e_*) = 1$ give

$$\begin{aligned}\int_{U_*\cap B_r(s)} |\nabla u|^2 &= \int_{U_*\cap \partial B_r(s)} u\,\partial_r u + \int_{\Gamma_*^{\mathrm{neck}}\cap B_r(s)} u\,\partial_{\nu_*} u,\\ |U_* \cap B_r(s)| &= r^{-1}\int_{U_*\cap\partial B_r(s)} P_*^2 + \int_{\Gamma_*\cap B_r(s)} P_* e_*\cdot\nu_*.\end{aligned}$$

Here $\partial_r u = r^{-1}(x-s)\cdot\nabla u$ on the sphere. Since both u and P_* are harmonic, a second Green identity is

$$\begin{aligned}0 = r^{-1}&\int_{U_*\cap\partial B_r(s)} P_*((x-s)\cdot\nabla u - u) - \int_{\Gamma_*^{\mathrm{reg}}\cap B_r(s)} P_*\\ &+ \int_{\Gamma_*^{\mathrm{neck}}\cap B_r(s)} (P_*\partial_{\nu_*} u - u e_*\cdot\nu_*).\end{aligned}$$

Subtract this identity, multiplied by r^{-5}, from the previous expression for $\mathcal{W}_{*,s}(r)$. For the algebra, abbreviate $D = (x-s)\cdot\nabla u$ and $P = P_*$, so $w_* = u - P$. The spherical integrand becomes

$$uD - u^2 + P^2 - P(D-u) = (D-u)w_* + P^2.$$

The regular boundary term becomes $P(1 + e_*\cdot\nu_*)$, and the neck term becomes

$$\begin{aligned}&u\partial_{\nu_*}u + Pe_*\cdot\nu_* - (P\partial_{\nu_*}u - ue_*\cdot\nu_*)\\ &\quad = w_*\partial_{\nu_*}u + (u+P)e_*\cdot\nu_*.\end{aligned}$$

Finally, symmetry of the sphere gives

$$\int_{H_*\cap\partial B_r(s)} P_*^2 = \frac{1}{2}\frac{r^2}{5}\mathcal{H}^4(\partial B_r) = \frac{|B_1^5|}{2}r^6.$$

This proves

$$\begin{aligned}\mathcal{W}_{*,s}(r) - \frac{|B_1^5|}{2} &= r^{-6}\int_{U_*\cap\partial B_r(s)} ((x-s)\cdot\nabla u - u)w_*\\ &\quad + r^{-6}\int_{\partial B_r(s)} P_*^2(\mathbf{1}_{U_*} - \mathbf{1}_{H_*})\\ &\quad + r^{-5}\int_{\Gamma_*^{\mathrm{reg}}\cap B_r(s)} P_*(1 + e_*\cdot\nu_*)\end{aligned}$$

$$+ r^{-5} \int_{\Gamma_*^{\mathrm{neck}} \cap B_r(s)} \big[w_* \partial_{\nu_*} u + (u + P_*) e_* \cdot \nu_* \big]. \tag{7.4}$$

In particular, the artificial-boundary terms have not been discarded.

Step 3: Passage to the Gaussian functional. Define the full classical Weiss functional by

$$\mathbf{W}_s(u, r) = r^{-5} \int_{B_r(s)} (|\nabla u|^2 + \mathbf{1}_\Omega) - r^{-6} \int_{\partial B_r(s)} u^2 \, d\mathcal{H}^4.$$

Fubini's theorem and the coarea formula imply

$$W_{\rho,s} = \rho^{-7} \int_0^\infty r^6 e^{-r^2/\rho^2} \mathbf{W}_s(u, r) \, dr. \tag{7.5}$$

Indeed, Tonelli's theorem gives, for every nonnegative locally integrable density f,

$$\rho^{-7} \int_0^\infty r e^{-r^2/\rho^2} \int_{B_r(s)} f(x) \, dx \, dr = \frac{1}{2\rho^5} \int_{\mathbb{R}^5} f(x) e^{-|x-s|^2/\rho^2} \, dx.$$

The boundary term in (7.5) becomes $-\rho^{-7} \int u^2 e^{-|x-s|^2/\rho^2}$. These are exactly the two terms in the definition of $W_{\rho,s}$. Both Gaussian integrals are finite, uniformly in the allowed s. The energy density is bounded by 2. For the other term, the global Lipschitz bound and $u(z_0) = 0$ give $u(s + \rho y) \le \rho(1 + |y|)$ and hence

$$\rho^{-7} \int_{\mathbb{R}^5} u(x)^2 e^{-|x-s|^2/\rho^2} \, dx \le \int_{\mathbb{R}^5} (1 + |y|)^2 e^{-|y|^2} \, dy < \infty.$$

Thus the two nonnegative terms can be averaged separately.

The weight in (7.5) has total mass

$$\rho^{-7} \int_0^\infty r^6 e^{-r^2/\rho^2} \, dr = \frac{\Gamma(7/2)}{2} = \frac{15\sqrt{\pi}}{16}.$$

Since $|B_1^5| = 8\pi^2/15$, the classical vee value $|B_1^5|$ therefore becomes

$$W_\infty = \frac{15\sqrt{\pi}}{16} |B_1^5| = \frac{\pi^{5/2}}{2}.$$

For terms integrated on $B_r(s)$, the truncated averaging kernel is

$$\begin{aligned} K_{\rho,T}(x - s) &= \rho^{-7} \mathbf{1}_{\{|x-s|<T\}} \int_{|x-s|}^T r e^{-r^2/\rho^2} \, dr \\ &= \frac{1}{2\rho^5} (e^{-|x-s|^2/\rho^2} - e^{-T^2/\rho^2}) \mathbf{1}_{\{|x-s|<T\}} \le C\rho^{-5} e^{-|x-s|^2/\rho^2}. \end{aligned} \tag{7.6}$$

We also justify the truncated averages of the signed phase terms. The center s need not lie on Γ, but $u(x) \le |x - s| + \rho$ implies

$$|\mathcal{W}_{*,s}(r)| \le C(1 + \rho^2/r^2).$$

Multiplication by $\rho^{-7} r^6 e^{-r^2/\rho^2}$ leaves an integrable bound $C\rho^{-7}(r^6 + \rho^2 r^4)$ near zero. The spherical terms in (7.4) are absolutely integrable by coarea, since their factors are bounded on $B_T(s)$. For a bounded boundary factor b_∂ in that identity, Tonelli applied to its absolute value gives

$$\begin{aligned} &\rho^{-7} \int_0^T r e^{-r^2/\rho^2} \int_{\Gamma_* \cap B_r(s)} |b_\partial| \, d\mathcal{H}^4 \, dr \\ &\quad = \int_{\Gamma_*} K_{\rho,T}(x - s) |b_\partial| \, d\mathcal{H}^4 \\ &\quad \le \frac{1}{2\rho^5} \int_{\Gamma_* \cap B_T(s)} |b_\partial| \, d\mathcal{H}^4 < \infty. \end{aligned}$$

The last inequality uses finite local perimeter. These bounds are uniform in s, since all the integration regions lie in the same fixed ball $B_{R_{\mathrm{f}}/2}(z_0)$.

Step 4: The four averaged phase terms. Multiply (7.4) by $\rho^{-7}r^6e^{-r^2/\rho^2}$, integrate from 0 to T, and sum over the signs. Denote the four resulting terms by I, II, III, IV, in their displayed order. By coarea, the first term is

$$I = \rho^{-7}\sum_{*}\int_{U_*\cap B_T(s)}((x-s)\cdot\nabla u - u)w_* e^{-|x-s|^2/\rho^2}\,dx.$$

Here $(x-s)\cdot\nabla P_* = P_*$, and hence

$$(x-s)\cdot\nabla u - u = (x-s)\cdot\nabla w_* - w_*.$$

Young's inequality, $|x-s|\le R_{\mathrm{f}}$, and (7.2) give

$$|I| \le C\rho^{-7}\sum_{*}\int_{U_*\cap B_{R_{\mathrm{f}}/2}(z_0)}(w_*^2 + R_{\mathrm{f}}^2|\nabla w_*|^2) \le CA_{\#}^2R_{\mathrm{f}}^5\rho^{-7}. \tag{7.7}$$

For II, the identities $P_-^2 = P_+^2$ and $\mathbf{1}_{H_+} + \mathbf{1}_{H_-} = 1$ almost everywhere yield

$$II = -\rho^{-7}\int_{B_T(s)\setminus(U_+\cup U_-)} P_+^2 e^{-|x-s|^2/\rho^2}\,dx \le 0.$$

Let $Y = \Omega\setminus(U_+\cup U_-)$. The signed covering property gives $Y\cap B_{R_{\mathrm{f}}/2}(z_0)\subset\mathcal{X}_{\mathrm{neck}}$. On $B_{R_{\mathrm{f}}/2}(z_0)\setminus\Omega$, (7.3) gives $|P_+|\le CA_{\#}$. On $(Y\cup\Gamma^{\mathrm{neck}})\cap B_{R_{\mathrm{f}}/2}(z_0)$, terminal-neck localization and (4.21) give $u\le C\varepsilon R$. Thus (7.3) and $A_{\#}\le\varepsilon R$ give the local bound

$$u(x) + |P_*(x)| \le C\varepsilon R \quad (x\in(Y\cup\Gamma^{\mathrm{neck}})\cap B_{R_{\mathrm{f}}/2}(z_0)). \tag{7.8}$$

More explicitly, the zero-phase contribution to $-II$ is bounded by $CA_{\#}^2\rho^{-7}\int_{\mathbb{R}^5}e^{-|x-s|^2/\rho^2}\,dx = C(A_{\#}/\rho)^2$. The contribution of Y is bounded by $C(\varepsilon R/\rho)^2\rho^{-5}|\mathcal{X}_{\mathrm{neck}}\cap B_{R_{\mathrm{f}}/2}(z_0)|$. Since $\varepsilon R/\rho = \varepsilon^{1-\chi-\tau}\to 0$, this proves

$$0 \le -II \le C(A_{\#}/\rho)^2 + C\rho^{-5}|\mathcal{X}_{\mathrm{neck}}\cap B_{R_{\mathrm{f}}/2}(z_0)|. \tag{7.9}$$

On $\Gamma_*^{\mathrm{reg}}\cap B_T(s)$ one has $u=0$, so $|P_*|\le CA_{\#}$ by (7.3). We use the flux part of Lemma 6.1 with $p=p_2$, local radius $R/16$, and $v=w_*$. Its slope condition holds because $|e_k - e_{\mathrm{root}}|\le C\varepsilon$, and its centered intercept is zero. Since $B_T(s)\subset B_{R_{\mathrm{f}}/2}(z_0)\subset B_{R/16}(z_0)$ for large k, that lemma gives

$$\begin{aligned}\sum_{*}\int_{\Gamma_*^{\mathrm{reg}}\cap B_T(s)}|1+e_*\cdot\nu_*| &= \sum_{*}\int_{\Gamma_*^{\mathrm{reg}}\cap B_T(s)}|\partial_{\nu_*}w_*| \\ &\le CR^4(\varepsilon^2+\varepsilon^{2p_2}) \le CR^4\varepsilon^2.\end{aligned}$$

Here $\partial_{\nu_*}u = -1$ on the regular part, and $p_2>1$. This uses the quadratic direction estimate already incorporated in the preceding flux lemma. Thus (7.6) gives

$$|III| \le CA_{\#}\rho^{-5}R^4\varepsilon^2. \tag{7.10}$$

On the artificial boundary, $|\partial_{\nu_*}u|\le 1$ and $|w_*|\le u + |P_*|$. Hence (7.8) gives

$$|IV| \le C\varepsilon R\rho^{-5}\mathcal{H}^4(\Gamma^{\mathrm{neck}}\cap B_{R_{\mathrm{f}}/2}(z_0)), \tag{7.11}$$

where $\Gamma^{\mathrm{neck}} = \Gamma_+^{\mathrm{neck}}\cup\Gamma_-^{\mathrm{neck}}$.

Step 5: The omitted positive phase and the Gaussian tail. Since u is Lipschitz and vanishes on the zero phase, its weak gradient is zero almost everywhere there. Thus, using the disjointness of U_+, U_- and $Y = \Omega\setminus(U_+\cup U_-)$, the energy partition is

$$\mathbf{W}_s(u,r) = \mathcal{W}_{+,s}(r) + \mathcal{W}_{-,s}(r) + r^{-5}\int_{Y\cap B_r(s)}(|\nabla u|^2+1)$$

$$- r^{-6} \int_{Y \cap \partial B_r(s)} u^2.$$

Average the last two nonnegative integrals over $0 < r < T$ with weight $\rho^{-7} r^6 e^{-r^2/\rho^2}$, and call the results J_{bulk} and J_{bdry}, respectively. Also put

$$J_{\mathrm{tail}} = \rho^{-7} \int_T^\infty r^6 e^{-r^2/\rho^2} \left(|B_1^5| - \mathbf{W}_s(u, r) \right) dr.$$

The normalization in Step 3 gives the exact deficit formula

$$W_\infty - W_{\rho,s} = -I - II - III - IV - J_{\mathrm{bulk}} + J_{\mathrm{bdry}} + J_{\mathrm{tail}}.$$

We may omit $-J_{\mathrm{bulk}}$ in an upper bound because $J_{\mathrm{bulk}} \geq 0$. This sign is useful: the energy of the positive phase omitted by the signed domains cannot increase the deficit. Its spherical u^2 term has the opposite sign and must still be estimated. By coarea and (7.8),

$$\begin{aligned} J_{\mathrm{bdry}} &= \rho^{-7} \int_{Y \cap B_T(s)} u^2 e^{-|x-s|^2/\rho^2} \, dx \\ &\leq C \left(\frac{\varepsilon R}{\rho} \right)^2 \rho^{-5} |\mathcal{X}_{\mathrm{neck}} \cap B_{R_{\mathrm{f}}/2}(z_0)| \\ &\leq C \rho^{-5} |\mathcal{X}_{\mathrm{neck}} \cap B_{R_{\mathrm{f}}/2}(z_0)|. \end{aligned}$$

To estimate the tail, use $u(z_0) = 0$ and the global Lipschitz bound. For $r \geq T$ and $x \in \partial B_r(s)$, $u(x) \leq r + \rho$. As $\rho/T \to 0$, this gives

$$|B_1^5| - \mathbf{W}_s(u, r) \leq |B_1^5| + r^{-6} \int_{\partial B_r(s)} u^2 \leq C.$$

Consequently

$$J_{\mathrm{tail}} \leq C \int_{T/\rho}^\infty t^6 e^{-t^2} \, dt \leq C e^{-c(R_{\mathrm{f}}/\rho)^2}.$$

We used $t = r/\rho$ and absorbed the polynomial factor into a Gaussian with a smaller positive exponent. All constants are independent of s. Combining these estimates with Step 4, and absorbing $(A_\#/\rho)^2$ into $A_\#^2 R_{\mathrm{f}}^5 \rho^{-7}$ because $\rho \leq R_{\mathrm{f}}$, we obtain

$$\begin{aligned} W_\infty - W_{\rho,s} \leq C \big[& A_\#^2 R_{\mathrm{f}}^5 \rho^{-7} + A_\# \rho^{-5} R^4 \varepsilon^2 + \rho^{-5} |\mathcal{X}_{\mathrm{neck}} \cap B_{R_{\mathrm{f}}/2}(z_0)| \\ & + \varepsilon R \rho^{-5} \mathcal{H}^4(\Gamma^{\mathrm{neck}} \cap B_{R_{\mathrm{f}}/2}(z_0)) + e^{-c(R_{\mathrm{f}}/\rho)^2} \big]. \end{aligned} \tag{7.12}$$

Step 6: Every error has a strict power gain. Use (4.20) with $\gamma_\alpha = 1 + 2/\alpha = 21/5$. The first two terms in (7.12) have the exact forms

$$A_\#^2 R_{\mathrm{f}}^5 \rho^{-7} = \left(\frac{A_\#}{R_{\mathrm{f}}} \right)^2 \left(\frac{R_{\mathrm{f}}}{\rho} \right)^7 = \varepsilon^{2+2\sigma-7\tau} =: \varepsilon^{\lambda_1},$$

$$A_\# \rho^{-5} R^4 \varepsilon^2 = \frac{A_\#}{R_{\mathrm{f}}} \varepsilon^2 \left(\frac{R}{R_{\mathrm{f}}} \right)^4 \left(\frac{R_{\mathrm{f}}}{\rho} \right)^5 = \varepsilon^{3+\sigma-4\chi-5\tau} =: \varepsilon^{\lambda_2}.$$

The volume and surface neck terms are both bounded by

$$C R^5 \rho^{-5} \varepsilon^{1+\gamma_\alpha} = C \varepsilon^{1+\gamma_\alpha-5\chi-5\tau} =: C \varepsilon^{\lambda_3}.$$

The exact reserves above the desired exponent are

$$\begin{aligned} \lambda_1 - (2+\sigma) &= \sigma - 7\tau = \frac{93}{10000000} > 0, \\ \lambda_2 - (2+\sigma) &= 1 - 4\chi - 5\tau = \frac{1991999}{2000000} > 0, \\ \lambda_3 - (2+\sigma) &= \gamma_\alpha - 1 - \sigma - 5\chi - 5\tau = \frac{6389979}{2000000} > 0. \end{aligned}$$

The Gaussian tail has the same required bound. Indeed, putting $y = \varepsilon^{-2\tau} \to \infty$, for every fixed $N > 0$ one has

$$\varepsilon^{-N} e^{-c\varepsilon^{-2\tau}} = y^{N/(2\tau)} e^{-cy} \longrightarrow 0.$$

Thus

$$W_\infty - W_{\rho,s} \le C\varepsilon^{2+\sigma}$$

uniformly over the ridge centers s under consideration.

Step 7: The Monneau term. Unlike the Weiss term, this is an ordinary integral over $B_\rho(z_0)$. On U_*, nonnegativity of u implies

$$|u - |P_*|| \le |u - P_*| = |w_*|.$$

On the zero phase use (7.3); on $Y \cap B_\rho(z_0)$ use (7.8). Splitting the ball into these four portions gives

$$\begin{aligned} M^{e_k}_{\rho,z_0} &\le C\rho^{-7}\left[A_\#^2 R_{\mathrm{f}}^5 + A_\#^2\rho^5 + (\varepsilon R)^2 |\mathcal{X}_{\mathrm{neck}} \cap B_{R_{\mathrm{f}}/2}(z_0)|\right] \\ &\le C A_\#^2 R_{\mathrm{f}}^5 \rho^{-7} + C(A_\#/\rho)^2 + C\left(\frac{\varepsilon R}{\rho}\right)^2 \rho^{-5} |\mathcal{X}_{\mathrm{neck}} \cap B_{R_{\mathrm{f}}/2}(z_0)| \\ &\le C\varepsilon^{2+\sigma}. \end{aligned}$$

For the last inequality, $\rho \le R_{\mathrm{f}}$ absorbs the second term into the first, $\varepsilon R/\rho \le 1$, and Step 6 bounds the two remaining errors. Taking the supremum over s in the Weiss estimate and adding this bound proves (7.1). □

7.1. **Proof of the rigidity theorem.**

Proof of Theorem 1.1. Assume first that a solution has $D^2u \not\equiv 0$ in its positive phase. The normalization lemma in Section 4 reduces this to a normalized nonflat global classical stable solution. The first selected sequence and Lemma 4.3, Proposition 5.2, and Corollary 5.3 give $\alpha_* \le 11/18$. We may therefore select a new sequence with $\alpha = 5/8$ and the parameters (6.1). The same moment argument gives the individual-neck bound (6.5) for this sequence.

Propositions 6.5 and 7.1 now yield

$$\mathcal{E}_{z_0}(\rho) \le C\varepsilon^{1+\sigma/2}, \qquad \rho = \varepsilon^{\chi+\tau} R.$$

This radius lies in $(0, R]$. The selected logarithmic lower bound (4.7) consequently gives

$$\frac{c\varepsilon}{1 + (\chi+\tau)|\log\varepsilon|} \le \mathcal{E}_{z_0}(\rho) \le C\varepsilon^{1+\sigma/2}.$$

Dividing by $\varepsilon > 0$ and multiplying by the positive logarithmic denominator gives

$$0 < c \le C\varepsilon^{\sigma/2}\left[1 + (\chi+\tau)|\log\varepsilon|\right] \longrightarrow 0.$$

Here c, C are independent of the selected index, and a positive power dominates the logarithm as $\varepsilon \downarrow 0$. This is impossible. Thus every solution in the theorem satisfies $D^2u = 0$ in its positive phase.

It remains to identify the resulting functions. If $\Gamma = \varnothing$, the open set Ω is also closed, so connectedness of $\mathbb{R}^5$ gives $\Omega = \varnothing$ or $\Omega = \mathbb{R}^5$. In the first case $u = 0$. In the second, the already proved vanishing Hessian makes u affine on $\mathbb{R}^5$, and nonnegativity on every line forces its slope to be zero. Thus u is constant in either case. Assume from now on that $\Gamma \neq \varnothing$. Every positive component is then a proper subset of $\mathbb{R}^5$ and has nonempty boundary. On each connected component Ω_i of Ω, the function is affine: $u(x) = a_i \cdot x + b_i$. Indeed, the vanishing Hessian makes each first derivative constant on the connected open set Ω_i. We have $\partial\Omega_i \subset \Gamma$: if a point of $\partial\Omega_i$ had $u > 0$, a small connected ball in Ω around it would meet Ω_i and hence lie in that same component, contrary to its being a boundary point. Choose a point of $\partial\Omega_i$. Continuity gives $u = 0$ there, and smoothness from the positive side and the Bernoulli condition give $|a_i| = 1$. Positivity gives $\Omega_i \subset H_i := \{a_i \cdot x + b_i > 0\}$. The set Ω_i is relatively closed in H_i. Indeed, if $x_j \in \Omega_i$ tends to $x \in H_i$, continuity gives $u(x) = a_i \cdot x + b_i > 0$. A small ball around x lies in Ω and contains x_j for

large j. Its connectedness puts the whole ball, including x, in Ω_i. It is also relatively open. Since H_i is connected, $\Omega_i = H_i$.

Two disjoint open half-spaces have opposite normal directions. Indeed, if $a_j \neq -a_i$, set $v = a_i + a_j$. Then

$$a_i \cdot v = a_j \cdot v = 1 + a_i \cdot a_j > 0,$$

so tv belongs to both half-spaces for sufficiently large t, regardless of their intercepts. Thus, once two components have been chosen, their normals are opposite; the normal of a third component would have to be opposite to both, which is impossible. There are therefore at most two positive components. If there is one, we obtain $u = (e \cdot x - a)_+$. If there are two, they have the form $\{e \cdot x > a\}$ and $\{e \cdot x < b\}$, with $b \leq a$. Equality would make the positive phase locally two-sided along the common hyperplane, contrary to the classical hypothesis. Hence $b < a$, and $u = (e \cdot x - a)_+ + (b - e \cdot x)_+$. These formulas hold everywhere because $u = 0$ outside its positive components. This completes the classification. □

Appendix A. Analytic estimates and the interior certificate

Appendix B defines the multiplier by ordinary polynomial and tensor formulas. The scalar weight and the boundary inequality admit the elementary proofs given here. The interior coercivity inequality follows from the exact factorization supplied electronically and described below.

A.1. **An elementary lower bound for the scalar weight.**

Lemma A.1. *On the normalized domain $A \in \mathrm{STF}^2(\mathbb{R}^5)$, $|A| = 1$, and $|p| \leq 1$, the scalar field in Appendix B satisfies*

$$\frac{1}{6} \leq f(A, p) \leq 5.$$

Proof. Put $s = |p|^2$, $t = p \cdot Ap$, and $r = p \cdot A^2 p = |Ap|^2$. Thus $0 \leq s \leq 1$, $|t| \leq s$, $0 \leq r \leq s$, and $t^2 \leq sr$, by Cauchy–Schwarz and $\|A\|_{\mathrm{op}} \leq |A| = 1$. Write the scalar polynomial of Appendix B as

$$f = F_0(s) + tF_1(s) + rF_2(s) + t^2 F_3(s) - c\,tr, \qquad c = \frac{1294671023}{10^{10}}.$$

Here $F_0, \ldots, F_3$ are the four coefficient polynomials obtained from the displayed formula for F there after division by 10^{10}. All estimates below are comparisons of the displayed integers.

First, F_1 and F_3 are nonnegative on $[0, 1]$, and

$$F_2(s) - cs = \frac{482004749 + 3555523883s - 357668087s^2}{10^{10}} \geq \frac{482004749 + 3197855796s}{10^{10}} > 0.$$

We shall use the following three bounds:

$$F_0(s) \geq \frac{5}{6}, \qquad 0 \leq F_1(s) \leq \frac{61}{40}, \qquad F_3(s) + \frac{F_2(s)}{s} \geq \frac{7}{8} \quad (0 < s \leq 1). \tag{A.1}$$

To verify the first, differentiate twice and subtract the indicated quadratic:

$$F_0''(s) - \left(\frac{37}{100} - \frac{7}{10}s + \frac{23}{10}s^2\right) = \frac{46029088 + 275018074s + 964420312s^2}{10^{10}} \geq 0.$$

Moreover,

$$\frac{37}{100} - \frac{7}{10}s + \frac{23}{10}s^2 = \frac{23}{10}\left(s - \frac{7}{46}\right)^2 + \frac{1457}{4600} > \frac{3}{10}.$$

Moreover,

$$F_0(7/10) = \frac{20959290075749}{25000000000000} \geq \frac{167}{200}, \qquad F_0'(7/10) = -\frac{20665795799}{5000000000000}, \quad |F_0'(7/10)| \leq \frac{1}{100}.$$

The lower bound for the second derivative implies

$$F_0(s) \geq F_0(7/10) + F_0'(7/10)(s-7/10) + \frac{3}{20}(s-7/10)^2.$$

For real x, m, completing the square gives $3x^2/20 + mx \geq -5m^2/3$. Apply this with $x = s - 7/10$ and $m = F_0'(7/10)$ to obtain

$$F_0(s) \geq \frac{167}{200} - \frac{5}{3}\left(\frac{1}{100}\right)^2 = \frac{5009}{6000} > \frac{5}{6}.$$

The nonnegative coefficients of F_1 give

$$F_1(s) \leq F_1(1) = \frac{1887717781}{1250000000} < \frac{61}{40}.$$

Finally,

$$F_3(s) + \frac{F_2(s)}{s} = a_0 + b_0 s + \frac{c_0}{s},$$

where

$$a_0 = \frac{7757774591}{10^{10}}, \qquad b_0 = \frac{130013659}{2000000000}, \qquad c_0 = \frac{482004749}{10^{10}}.$$

These positive fractions satisfy

$$a_0 - \frac{31}{40} = \frac{7774591}{10^{10}} > 0, \qquad 4b_0c_0 - \frac{1}{100} = \frac{12667201072866591}{5000000000000000000} > 0.$$

The arithmetic–geometric mean inequality therefore gives $a_0 + b_0 s + c_0/s \geq 31/40 + 1/10 = 7/8$. This proves (A.1).

If $s = 0$, then $p = 0$ and $f = 1$. Suppose now that $s > 0$. If $t \geq 0$, then $t \leq s$ and $r(F_2 - ct) \geq 0$. The other terms except F_0 are nonnegative, so $f \geq F_0 \geq 5/6$. If $t < 0$, then $-ctr \geq 0$ and $F_2 > 0$. Using $r \geq t^2/s$, we obtain

$$\begin{aligned} f &\geq F_0 + tF_1 + t^2\left(F_3 + \frac{F_2}{s}\right) \\ &\geq \frac{5}{6} - \frac{61}{40}|t| + \frac{7}{8}|t|^2 \\ &= \frac{7}{8}\left(|t| - \frac{61}{70}\right)^2 + \frac{2837}{16800} > \frac{1}{6}. \end{aligned}$$

For the upper bound, every monomial in s, t, r has absolute value at most one. The sum of the absolute values of the coefficients of f is $5593542551/1250000000 < 5$. Thus $f \leq 5$. □

A.2. **Elementary tensor bounds.**

Lemma A.2. *Let $A \in \mathrm{STF}^2(\mathbb{R}^5)$, $|A| = 1$, and let $X \in \mathrm{STF}^3(\mathbb{R}^5)$. For $q_l = A_{ij}X_{lij}$, one has*

$$\|A\|_{\mathrm{op}}^2 \leq \frac{4}{5}, \qquad |\operatorname{tr}(A^3)|^2 \leq \frac{9}{20}, \qquad |q|^2 \leq \frac{5}{7}|X|^2. \tag{A.2}$$

Proof. Let $\lambda_1, \ldots, \lambda_5$ be the eigenvalues of A. Since their sum is zero and the sum of their squares is one,

$$\lambda_i^2 = \left(\sum_{j\neq i}\lambda_j\right)^2 \leq 4\sum_{j\neq i}\lambda_j^2 = 4(1-\lambda_i^2).$$

This proves the operator-norm estimate.

We record the cubic inequality in a form also used below. If $y_1 + \cdots + y_m = 0$, where $m \geq 2$, then

$$\left|\sum_{i=1}^m y_i^3\right| \leq \frac{m-2}{\sqrt{m(m-1)}}\left(\sum_{i=1}^m y_i^2\right)^{3/2}. \tag{A.3}$$

For a proof, the zero vector is immediate, and homogeneity reduces the question to $\sum y_i^2 = 1$. The maximum exists on a compact set. The gradients of the two constraints are independent because y is nonzero and orthogonal to $(1, \dots, 1)$. At an extremum, Lagrange multipliers give $3y_i^2 = \alpha + 2\beta y_i$, so the entries take at most two values. They cannot all be equal, since their sum is zero and their squared norm is one. If one value has multiplicity k, where $1 \le k \le m-1$, and the other has multiplicity $m-k$, the constraints give the two values

$$a = \pm\sqrt{\frac{m-k}{mk}}, \qquad b = \mp\sqrt{\frac{k}{m(m-k)}}.$$

Indeed, $ka + (m-k)b = 0$ and $ka^2 + (m-k)b^2 = 1$ determine these values. Substitution yields

$$\sum y_i^3 = \pm\frac{m-2k}{\sqrt{mk(m-k)}}.$$

The square of this expression is $m/[k(m-k)] - 4/m$. Since $k(m-k) \ge m-1$, its largest value is $(m-2)^2/[m(m-1)]$. Replacing y by $-y$ handles the minimum. This proves (A.3); taking $m = 5$ proves the second estimate in (A.2).

To prove the last estimate, fix a unit vector ξ and put

$$W_{ijk} = \frac{1}{3}(A_{ij}\xi_k + A_{ik}\xi_j + A_{jk}\xi_i), \qquad v = A\xi.$$

For any vector v, define the trace-insertion tensor

$$J(v)_{ijk} = \delta_{ij}v_k + \delta_{ik}v_j + \delta_{jk}v_i.$$

Its trace is $7v$. Its three diagonal norm terms sum to $15|v|^2$, and its six cross terms sum to $6|v|^2$, so $|J(v)|^2 = 21|v|^2$. The orthogonal projection of W onto symmetric trace-free three-tensors is therefore

$$\widehat{W} = W - \frac{2}{21}J(v).$$

Indeed, the trace of W is $2v/3$, and a trace-insertion tensor is orthogonal to every trace-free tensor. In the nine terms of $|W|^2$, the three diagonal terms give $3|A|^2|\xi|^2$ and the six cross terms give $6|A\xi|^2$. Consequently

$$|W|^2 = \frac{3|A|^2|\xi|^2 + 6|A\xi|^2}{9} = \frac{1}{3} + \frac{2}{3}|v|^2,$$
$$|W - \widehat{W}|^2 = \frac{4}{21^2}|J(v)|^2 = \frac{4}{21}|v|^2.$$

The two projected parts are orthogonal, so

$$|\widehat{W}|^2 = \frac{1}{3} + \frac{10}{21}|A\xi|^2 \le \frac{1}{3} + \frac{10}{21}\frac{4}{5} = \frac{5}{7}.$$

Symmetry and the vanishing traces of X give $\xi \cdot q = \langle X, W\rangle = \langle X, \widehat{W}\rangle$. Cauchy–Schwarz and then the supremum over unit ξ prove the result. □

For completeness, the same operator-norm estimate implies

$$|\operatorname{tr}(A^k)| \le (4/5)^{(k-2)/2} \quad (k \ge 4), \qquad |X : A^k| \le (4/5)^{(k-1)/2}|X| \quad (k \ge 2),$$

where $(X : A^k)_l = X_{lij}(A^k)_{ij}$. The first bound follows by summing $|\lambda_i|^k$; the second follows from $|A^k|^2 \le \|A\|_{\mathrm{op}}^{2k-2}|A|^2$ and Cauchy–Schwarz. The three dimension-five bounds are also stated in [9, Lemma 6.3]; its Frobenius bound for A^k, combined with Cauchy–Schwarz, likewise gives the power estimates above.

A.3. **The exact interior certificate.** We prove Proposition 3.2, including the endpoints $p = 0$ and $|p| = 1$. Fix an arbitrary tuple

$$A \in \mathrm{STF}^2(\mathbb{R}^5), \quad |A| = 1, \quad p \in \mathbb{R}^5, \quad |p| \le 1, \quad w \in \mathbb{R}, \quad X \in \mathrm{STF}^3(\mathbb{R}^5), \quad \ell \in \mathbb{R}^5,$$

and set $b = 19/20$ and $Y = (|w|, |\ell|, |X|)$. The purpose of the Python calculation is to establish an exact polynomial decomposition of I_b into nonnegative squared norms, a positive reserve $10^{-6}|Y|^2$, and a remainder bounded in absolute value by $16 \cdot 10^{-8}|Y|^2$. The estimates below apply to this arbitrary tuple, which is why the calculation proves the inequality on the whole unit ball.

The single self-contained supplement `verify_interior_certificate.py` contains all rational coefficients, tensor keys, factor matrices, and reconstruction code. It requires Python 3.8 or later and uses only the standard library. From the directory containing the file, run:

```
python3 -I -S verify_interior_certificate.py
```

From another directory, replace the filename by its path. All coefficients are stored as ratios of integers and all calculations use exact integer or rational arithmetic; there is no floating-point roundoff. The program rejects Python's `-O` and `-OO` modes, which would disable its assertions. We explain the mathematical meaning of the checks in their order of use.

Reconstructing the polynomial to be estimated. The program first expands the 99 encoded coefficient rows into the 13 ordinary scalar polynomials defining f, U, K, M in Appendix B. The function `coefficient_match` compares every rational monomial coefficient with the corresponding Appendix B expression, also embedded in the file. Thus the reconstruction uses exactly the multiplier printed in this paper.

The left side of the certificate is then built from (3.3)–(3.4). In this step D_l differentiates only the coefficient fields, which are polynomials in (A, p); the quantities w, X, ℓ are independent parameters. To see explicitly what differentiation means, write $X_l = (X_{lij})_{ij}$ and recall

$$q_l = A_{ij}X_{lij}, \qquad D_lA = X_l - q_lA, \qquad D_lp = wAe_l.$$

Here e_l is the lth coordinate vector. The product rule gives, for $k \ge 1$,

$$D_l(A^k) = \sum_{a=0}^{k-1} A^a X_l A^{k-1-a} - kq_lA^k, \qquad D_l \operatorname{tr}(A^k) = kX_l : A^{k-1} - kq_l \operatorname{tr}(A^k).$$

For instance, if $s = |p|^2$ and $t = p \cdot Ap$, then

$$D_ls = 2w(Ap)_l, \qquad D_lt = X_{lij}p_ip_j - q_lt + 2w(A^2p)_l.$$

These derivative rules also respect the normalizations of A: $D_l \operatorname{tr} A = 0$ and $D_l \operatorname{tr}(A^2) = 2A : X_l - 2q_l \operatorname{tr}(A^2) = 0$. Thus simplifying by the trace identities is compatible with differentiation. The function `gradient` implements the product rule; its Euclidean terms, retained by `flat` as explained below, are precisely the terms just displayed. The function `row_interior` forms $G_{lijk} = \delta_{li}U_{jk} + K_{lijk} - K_{iljk}$ and inserts the remaining terms of (3.3)–(3.4), with $b = 19/20$. For example, one scalar-field term contributes precisely $-(D_lf)\ell_l - 2bfq_l\ell_l + f|\ell|^2$. The function `flat` sets the two auxiliary spherical bookkeeping variables to zero. This is polynomial evaluation at zero: every Euclidean term is retained. These auxiliary variables are distinct from the scalar invariant $z = p \cdot A^3p$ in Appendix B, which is retained. Consequently the constructed left side is exactly the Euclidean polynomial I_b of (3.4).

Throughout this calculation a tensor polynomial is represented by a finite dictionary: a key describes an index contraction and its value is that contraction's rational coefficient. The simplifications in `canonical` are tensor identities. They concatenate matrix paths using $A^aA^c = A^{a+c}$, reverse paths using the symmetry of A^k, rename summation indices and copies of the symmetric tensor X, and use

$$\operatorname{tr} I = 5, \qquad \operatorname{tr} A = 0, \qquad \operatorname{tr}(A^2) = 1, \qquad X_{iik} = 0.$$

In particular, a closed identity path contributes the factor 5 to the rational coefficient. Each rewriting preserves the value at every admissible tuple. Therefore any identity obtained by these finite coefficient operations remains an identity when the keys are evaluated at our arbitrary tuple. The keys need not be linearly independent: equality of their coefficient dictionaries is sufficient for equality of the evaluated expressions. No values of A or p are sampled in this argument.

The squared tensor norms. We next specify the projections and their normalizations, since these explain the signs of all the Gram terms. For a rank-two tensor T, set

$$P_2T = \frac{T+T^{\mathsf{T}}}{2} - \frac{\operatorname{tr} T}{5} I, \qquad A_2T = \frac{T-T^{\mathsf{T}}}{2}.$$

For a rank-three tensor T, let $\operatorname{Sym} T$ denote the average over all six permutations of its slots, let $\tau = \operatorname{tr}_{12}(\operatorname{Sym} T)$, and put

$$(P_3T)_{ijk} = (\operatorname{Sym} T)_{ijk} - \frac{1}{7}(\delta_{ij}\tau_k + \delta_{ik}\tau_j + \delta_{jk}\tau_i).$$

For the hook component, define

$$T^0 = \operatorname{Sym}_{12} T - \operatorname{Sym} T, \qquad a = 2\operatorname{tr}_{12} T - \operatorname{tr}_{13} T - \operatorname{tr}_{23} T,$$
$$(H_3T)_{ijk} = T^0_{ijk} - \frac{1}{24}(2\delta_{ij}a_k - \delta_{ik}a_j - \delta_{jk}a_i).$$

Here $\operatorname{Sym}_{12} T = (T_{ijk} + T_{jik})/2$, and a trace contracts the named slots, leaving a vector in $\mathbb{R}^5$. The tensor H_3T is symmetric in its first two slots, has zero full symmetrization, and has zero trace in every pair. Indeed, writing $t_{ij} = \operatorname{tr}_{ij} T$, one has

$$\operatorname{tr}_{12} T^0 = t_{12} - \frac{t_{12}+t_{13}+t_{23}}{3} = \frac{a}{3},$$
$$\operatorname{tr}_{13} T^0 = \frac{t_{13}+t_{23}}{2} - \frac{t_{12}+t_{13}+t_{23}}{3} = -\frac{a}{6}.$$

The third trace equals the second by symmetry in the first two slots. The displayed correction has exactly these traces. It is orthogonal to every trace-free hook tensor. Its squared norm is $|a|^2/24$, since the squared norm of its numerator is

$$(4\cdot 5 + 5 + 5 - 4 - 4 + 2)|a|^2 = 24|a|^2.$$

The bilinear contractions used by the certificate are

$$\langle T, S\rangle, \quad 10\langle P_2T, P_2S\rangle, \quad 2\langle A_2T, A_2S\rangle, \quad 42\langle P_3T, P_3S\rangle, \quad 48\langle H_3T, H_3S\rangle.$$

For example, the two rank-two expressions expand as

$$10\langle P_2T, P_2S\rangle = 5(T:S + T:S^{\mathsf{T}}) - 2\operatorname{tr} T \operatorname{tr} S, \qquad 2\langle A_2T, A_2S\rangle = T:S - T:S^{\mathsf{T}}.$$

For rank three, if $a_{\mathrm{sym}}(T)$ is the sum of the three pairwise traces of T, then

$$42\langle P_3T, P_3S\rangle = 7\sum_{\pi\in S_3}\langle T, S^\pi\rangle - 2a_{\mathrm{sym}}(T)\cdot a_{\mathrm{sym}}(S).$$

Indeed, $\operatorname{tr}_{12}(\operatorname{Sym} T) = a_{\mathrm{sym}}(T)/3$. The trace-insertion calculation in the proof of Lemma A.2 shows that the squared norm removed by P_3 is $3|\tau|^2/7$. Thus its coefficient after multiplication by 42 is $42(3/7)(1/9) = 2$, as in the displayed trace correction. For the hook expression, both symmetrizations are orthogonal projections, and the fully symmetric subspace is contained in the subspace symmetric in the first two slots. Their difference $R = \operatorname{Sym}_{12} - \operatorname{Sym}$ is therefore an orthogonal projection and satisfies

$$48\langle RT, RS\rangle = 24(\langle T, S\rangle + \langle T, S^{(12)}\rangle) - 8\sum_{\pi\in S_3}\langle T, S^\pi\rangle.$$

Removing its trace part subtracts $2a(T)\cdot a(S)$. Thus the hook formula has coefficient 16 for the identity permutation and the interchange of the first two slots, coefficient -8 for each of

the other four permutations, and trace correction $-2a(T)\cdot a(S)$. These are exactly the finite contraction formulas implemented by `projection_matchings` and `projected_inner`. In particular, the bilinear forms are positive constants times Euclidean squared norms when $S=T$, even for raw tensors without any imposed symmetry.

The rational factors and their exact expansion. Each factor record specifies a list of raw polynomial tensors $T_{j,1},\dots,T_{j,n_j}$ and a rational matrix L_j. A retained tensor key has the form $(0,e,0,\mathbf{t},\mathbf{E})$. It supplies a factor w^e and $\prod_{k\in\mathbf{t}}\operatorname{tr}(A^k)$. Each edge (u,v,k) in $\mathbf{E}$ represents an A^k path joining its two endpoint indices. The labels $0,\dots,\nu-1$ are the free slots for a rank-ν tensor; labels 5 and 6 supply p and ℓ, and each three-legged vertex 7 or 8 supplies a copy of X. Each incidence of a vector label has its own summation index; the three incidences of an X vertex are its three tensor slots. For example, the rank-one key

$$(0,1,0,(),((0,5,2),))$$

represents the vector wA^2p. This specifies the raw tensors by ordinary index contraction, with $A^0=I$. The retained keys contain no spherical variables and are linear in (w,X,ℓ). Before using each key, the checker verifies its rank, the number of free slots, its linearity, and its canonical contraction form. No spanning or linear independence property is needed: the argument uses these explicit tensors only to form and expand squares.

Let Π_j be the appropriate projection from the table below, with the identity projection for ranks zero and one, and put $B_{j,i}=\Pi_jT_{j,i}$. The factor matrices all have denominator 10^{10}; their integer numerators are embedded in the file.

Write $s=|p|^2$. The eleven blocks have the following parameters:

j	1	2	3	4	5	6	7	8	9	10	11
n_j	1	12	3	43	17	33	10	54	52	21	24
Π_j	Id	Id	Id	P_2	A_2	P_2	A_2	P_3	H_3	P_3	H_3
κ_j	1	1	1	10	2	10	2	42	48	42	48
e_j	0	0	1	0	0	1	1	0	0	1	1

For each block define the tensor polynomial

$$Q_j=\kappa_j(1-s)^{e_j}\sum_{k=1}^{n_j}\left|\sum_{i=1}^{n_j}(L_j)_{ik}B_{j,i}\right|^2.$$

Its exact expansion is

$$Q_j=\kappa_j(1-s)^{e_j}\sum_{i,i'=1}^{n_j}\left(\sum_{k=1}^{n_j}(L_j)_{ik}(L_j)_{i'k}\right)\langle B_{j,i},B_{j,i'}\rangle.$$

This explains both the matrix convention $L_jL_j^{\mathsf T}$ and the factor 2 for off-diagonal pairs in `reconstruct`. The function `projected_inner` already includes the factor κ_j from the projection formulas above; it is used once. Blocks with $e_j=1$ are multiplied by $1-s$ by subtracting their s-multiple. Thus every term of the squared norms, including every mixed term, is reconstructed by rational coefficient operations. Their nonnegativity follows directly from their displayed form; no numerical test of eigenvalues is needed.

The program subtracts these expanded polynomials and the reserve $\delta|Y|^2$, where $\delta=10^{-6}$, from the independently constructed I_b. It collects the resulting remainder $\mathcal{R}$ into 731 nonzero contraction keys. In particular, the following is an exact identity:

$$I_b=\sum_{j=1}^{11}Q_j+\frac{1}{10^6}(w^2+|\ell|^2+|X|^2)+\mathcal{R}. \tag{A.4}$$

Every coefficient of $\mathcal{R}$ is computed by this subtraction. All identities used in collecting it hold at our arbitrary admissible tuple, so (A.4) holds at that tuple as well. It remains to bound the computed remainder uniformly.

A uniform bound for each residual contraction. Write

$$\mathcal{R} = \sum_{\alpha} c_\alpha m_\alpha, \qquad c_\alpha \in \mathbb{Q},$$

where m_α is the contraction described by one of the 731 keys. Each coefficient field depends only on (A, p), and its derivative is linear in (w, X). Hence I_b is homogeneous of degree two in (w, X, ℓ). Each raw tensor $T_{j,i}$ is linear in these variables, so every Q_j, and hence $\mathcal{R}$, is also homogeneous of degree two. The checker verifies term by term that

$$d_w + d_\ell + d_X = 2,$$

where d_w is the power of w, d_ℓ counts occurrences of ℓ, and d_X counts copies of the three-tensor X. Consequently the magnitude of each monomial is bounded by a constant times one of $Y_i Y_{i'}$, $1 \le i, i' \le 3$.

We give the constants explicitly. Put $\theta = 4/5$. Closed matrix paths are trace factors. By Lemma A.2 and its consequences,

$$|\operatorname{tr}(A^3)|^2 \le \frac{9}{20}, \qquad |\operatorname{tr}(A^k)|^2 \le \theta^{k-2} \quad (k \ge 4), \qquad \|A^k\|_{\mathrm{op}}^2 \le \theta^k \quad (k \ge 0).$$

If an edge joins two slots of the same X, contract those slots first. An identity edge gives a zero trace of X; for the other powers one has

$$|X : A|^2 \le \frac{5}{7}|X|^2, \qquad |X : A^k|^2 \le \theta^{k-1}|X|^2 \quad (k \ge 2).$$

This contraction leaves a vector. There is at most one such edge at each three-slot tensor vertex.

We give the remaining graph argument explicitly, following [9, proof of Lemma 6.4]. After the closed matrix paths have been separated as trace factors, each occurrence of p or ℓ is a separate one-valent leaf, and each X vertex has three incidences. The checker verifies these incidence conditions, the absence of free indices, and the nonnegative integer path powers for every residual key. Its degree-two check gives at most two copies of X. First reduce every self-edge at X by the bounds just proved; the corresponding cubic vertex becomes an ordinary vector leaf. The norm operations used for all remaining edges are

$$|T \lrcorner v| \le |T|\,|v|, \qquad |(B \otimes I \otimes \cdots \otimes I)T| \le \|B\|_{\mathrm{op}}|T|.$$

Here $T \lrcorner v$ means contraction in one slot. Now consider a connected component of the reduced graph. If it has no cubic vertex, it is one matrix edge between two vector leaves, so the operator-norm bound applies. If it has one cubic vertex, all three paths end at vector leaves. Move each matrix onto its leaf and apply Cauchy–Schwarz to the cubic tensor and the tensor product of the three resulting vectors. If it has two cubic vertices, they are joined by $r = 1, 2$, or 3 edges, and every other slot ends at a vector leaf. Contract those leaves first, obtaining rank-r tensors V, W. The remaining contraction, after a possible permutation of slots, is

$$\langle V, (A^{k_1} \otimes \cdots \otimes A^{k_r})W\rangle, \qquad \left|\langle V, (A^{k_1} \otimes \cdots \otimes A^{k_r})W\rangle\right| \le |V|\,|W| \prod_{a=1}^{r} \|A^{k_a}\|_{\mathrm{op}}.$$

This includes all three parallel edges when $r = 3$ and introduces no dimension factor. These cases are exhaustive, since all other vertices are leaves. If the two cubic vertices are disconnected, apply the one-vertex estimate to their separate components. Multiplying the component bounds counts each $|\ell|$ and $|X|$ once per occurrence. All p factors have norm at most one; include also the scalar power of $|w|$ and the separated trace bounds. The total degree two therefore gives exactly one of the products $Y_i Y_{i'}$.

In precise terms, let $\mathbf{t}_\alpha$ be the list of trace powers in a residual key, let $\mathbf{E}_{\mathrm{self}}$ be its edges joining two slots of one X, and let $\mathbf{E}_{\mathrm{other}}$ be its other edges. Define the positive rational number

$$L_\alpha^2 = \prod_{k\in\mathbf{t}_\alpha} \gamma_k \prod_{(u,v,k)\in\mathbf{E}_{\mathrm{self}}} \beta_k \prod_{(u,v,k)\in\mathbf{E}_{\mathrm{other}}} \theta^k,$$

where

$$\gamma_k = \begin{cases} 9/20, & k=3,\\ \theta^{k-2}, & k\ge 4, \end{cases} \qquad \beta_k = \begin{cases} 5/7, & k=1,\\ \theta^{k-1}, & k\ge 2. \end{cases}$$

The trace powers $0,1,2$ and the zero self-edges have already been removed by the exact identities above. The preceding estimates prove

$$|m_\alpha| \le L_\alpha |w|^{d_w} |\ell|^{d_\ell} |X|^{d_X} = L_\alpha Y_{i_\alpha} Y_{i'_\alpha} \tag{A.5}$$

for suitable $i_\alpha, i'_\alpha \in \{1,2,3\}$, uniformly on $|p|\le 1$. This is the number L_α^2 computed by `key_factors`. For example,

$$|w\ \mathrm{tr}(A^3)\, p\cdot A^2\ell| \le \sqrt{\frac{9}{20}\theta^2}\,|w|\,|\ell|, \qquad |\ell_l A_{ij} X_{lij}| \le \sqrt{\frac{5}{7}}\,|\ell|\,|X|.$$

Thus the constants used by the program come from inequalities valid for arbitrary tensors on the stated domain.

Exact rounding and the residual matrix. To keep every subsequent comparison rational, set

$$\widehat{L}_\alpha = 10^{-10}\left\lceil 10^{10}\sqrt{L_\alpha^2}\right\rceil, \qquad a_\alpha = |c_\alpha|\widehat{L}_\alpha.$$

Then $\widehat{L}_\alpha \ge L_\alpha$. The ceiling is computed without approximating a square root: if $L_\alpha^2 = a/d$ with positive integers a,d, the program finds the least integer n such that $n^2 d \ge 10^{20} a$. Explicitly, it first sets

$$n_0 = \mathrm{isqrt}\left(\left\lfloor \frac{10^{20}a}{d}\right\rfloor\right),$$

where $\mathrm{isqrt}(N)$ is the greatest integer whose square is at most N. If $n_0^2 d = 10^{20}a$, it takes $n=n_0$; otherwise it takes $n=n_0+1$. Indeed, $n_0^2 \le 10^{20}a/d < (n_0+1)^2$, so these are exactly the two possibilities for the ceiling. It returns $\widehat{L}_\alpha = n/10^{10}$. Every rounding therefore enlarges the bound in (A.5).

Construct a symmetric 3×3 matrix D by

$$D_{ii} = \sum_{i_\alpha = i'_\alpha = i} a_\alpha, \qquad D_{ii'} = \frac{1}{2}\sum_{\{i_\alpha, i'_\alpha\} = \{i,i'\}} a_\alpha \quad (i\ne i').$$

The triangle inequality and (A.5) now give

$$|\mathcal{R}| \le \sum_\alpha a_\alpha Y_{i_\alpha} Y_{i'_\alpha} = Y^{\mathsf{T}} D Y.$$

The factor $1/2$ in the off-diagonal entries accounts for the two occurrences of $D_{ii'}Y_iY_{i'}$ in this quadratic form. The function `reconstruct` computes this matrix from the residual just obtained and verifies, by rational comparisons, that

$$0 \le D_{ii'} \le H_{ii'}, \qquad H = \frac{1}{10^8}\begin{pmatrix} 2 & 2 & 3\\ 2 & 5 & 5\\ 3 & 5 & 8\end{pmatrix}. \tag{A.6}$$

Thus D is recomputed from the 731 contractions; the stored expected entries of D are only an additional consistency check. All entries of Y are nonnegative, so the entrywise comparison (A.6)

implies $Y^{\mathsf{T}}DY \le Y^{\mathsf{T}}HY$. Since H is symmetric with nonnegative entries, $2Y_iY_{i'} \le Y_i^2 + Y_{i'}^2$ yields

$$\begin{aligned} Y^{\mathsf{T}}HY &= \sum_i H_{ii}Y_i^2 + 2\sum_{i<i'} H_{ii'}Y_iY_{i'} \\ &\le \sum_i \left(\sum_{i'} H_{ii'}\right) Y_i^2 \le \frac{16}{10^8}|Y|^2. \end{aligned}$$

The last constant is the largest row sum of H, whose row sums are $7/10^8$, $12/10^8$, and $16/10^8$. We have therefore proved the uniform estimate $|\mathcal{R}| \le 16 \cdot 10^{-8}|Y|^2$.

Deduction on the whole unit ball. For our arbitrary tuple, $s = |p|^2 \in [0,1]$ and every κ_j is positive. If $e_j = 0$, its polynomial factor is identically one; if $e_j = 1$, its factor $1-s$ is nonnegative. Consequently every Q_j is nonnegative, and (A.4) gives

$$\begin{aligned} I_b &\ge \frac{1}{10^6}|Y|^2 - |\mathcal{R}| \\ &\ge \left(\frac{1}{10^6} - \frac{16}{10^8}\right)|Y|^2 = \frac{21}{25000000}|Y|^2 \ge \frac{1}{1250000}(w^2 + |\ell|^2 + |X|^2). \end{aligned}$$

At $|p| = 1$ the blocks with $e_j = 1$ simply vanish, while the reserve and its residual estimate remain valid. At $p = 0$ the same identities and bounds apply directly. There is no division by $|p|$, $1-|p|^2$, or $|Y|$, so neither endpoint nor the case $Y = 0$ requires a limiting argument. No boundary eigenvector condition on (A, p) has been used.

The tuple was arbitrary, and every coefficient identity and magnitude bound above holds throughout its stated algebraic domain. Hence the exact finite Python checks prove the full interior inequality for every $|p| \le 1$. The computer-assisted work is the rational expansion and comparison; the reason those checks imply uniform coercivity is the tensor identity and the estimates just proved.

A.4. **An elementary proof of the boundary inequality.** We prove (3.6) directly from the ordinary tensor formulas in Appendix B:

$$S(A,p) \ge \frac{1}{10^6}. \tag{A.7}$$

The hypotheses are $A \in \mathrm{STF}^2(\mathbb{R}^5)$, $|A| = 1$, $|p| = 1$, and $Ap = \lambda p$ with $\lambda \le 0$. The quantity $\theta = -\lambda$ below is a dimensionless eigenvalue of A; it is not the physical mean curvature H.

Step 1: reduction to two scalar invariants. Since A is symmetric and p is an eigenvector, its orthogonal complement is invariant under A. Rotate first to put $p = e_5$ and then diagonalize the restriction of A to $p^{\perp}$. This gives

$$p = e_5, \qquad A = \mathrm{diag}(x_1, x_2, x_3, x_4, -\theta), \qquad \theta \ge 0.$$

The trace and norm constraints imply

$$\sum_{i=1}^4 x_i = \theta, \qquad \sum_{i=1}^4 x_i^2 = 1 - \theta^2, \qquad 0 \le \theta \le \frac{2}{\sqrt{5}} < \frac{9}{10}. \tag{A.8}$$

The last inequality follows by applying Cauchy–Schwarz to the first sum. Set $T = \mathrm{tr}(A^3)$. Substitution into the multiplier gives

$$10^{10}S(A,p) = P(\theta) - C(\theta)T, \tag{A.9}$$

where

$$\begin{aligned} P(\theta) &= 12946710023\theta^4 - 61288054730\theta^3 + 121332736720\theta^2 - 129060173270\theta + 4935627631, \\ C(\theta) &= 1913784625\theta^2 - 53213313900\theta + 3927806547. \end{aligned}$$

For clarity, this reduction can be checked without expanding any rank-four tensor. Evaluate the scalar polynomials of Appendix B at $(s,t,r,z)=(1,\lambda,\lambda^2,\lambda^3)$ and leave T unchanged. Then

$$\begin{aligned}10^{10}S = &-\lambda F + U_A(2\lambda - T) + U_0 + \lambda U_1 + K_2(\lambda - T)\\ &+ M_0 + \lambda M_1 + \lambda^2 M_2 + \lambda^3 M_3.\end{aligned}$$

Here is the contraction calculation, with $d = 10^{10}$. The eigenvector condition gives $S_0 p = 0$, and the definition of V therefore gives

$$p_i V_{ijk} = (S_0)_{jk} + p_j p_k, \qquad V(p,p,p) = 1.$$

Since $A : S_0 = \lambda \operatorname{tr}(A^2) - \operatorname{tr}(A^3) = \lambda - T$, the U contribution is

$$\begin{aligned}d\, U_{jk} p_i V_{ijk} &= \big(U_A A + U_0 p \otimes p + U_1 p \otimes Ap\big) : (S_0 + p \otimes p)\\ &= U_A\big((\lambda - T) + \lambda\big) + U_0 + \lambda U_1.\end{aligned}$$

For K_1, K_3, K_4, K_5, the first two factors have the form $p_l(A^m p)_i = \lambda^m p_l p_i$. They are symmetric in l, i, so their antisymmetrizations vanish. The remaining term satisfies

$$d\, p_l(K_{lijk} - K_{iljk}) = K_2(p_j A_{ik} - \lambda p_i p_j p_k).$$

Its contraction with V is $K_2((2\lambda - T) - \lambda) = K_2(\lambda - T)$. Finally, $p \cdot A^j p = \lambda^j$ gives the displayed vector contribution, and the scalar contribution is $-\lambda F$. This proves (A.9) by polynomial addition.

The polynomial C is positive on the whole real line. Its leading coefficient is positive, and

$$4(1913784625)(3927806547) - (5321331390)^2 = 1751335356292427400 > 0.$$

Step 2: the sharp cubic bound. If $y_1 + \cdots + y_4 = 0$, then

$$\sum_{i=1}^{4} y_i^3 \le \frac{1}{\sqrt{3}} \left(\sum_{i=1}^{4} y_i^2 \right)^{3/2}. \tag{A.10}$$

This is (A.3) with $m = 4$, whose proof was given above.

Apply (A.10) to $y_i = x_i - \theta/4$. Then

$$\sum y_i = 0, \qquad \sum y_i^2 = 1 - \frac{5}{4}\theta^2.$$

Expanding $x_i = y_i + \theta/4$ gives

$$\begin{aligned}T &= \sum_i x_i^3 - \theta^3\\ &= \sum_i y_i^3 + \frac{3\theta}{4} \sum_i y_i^2 + \frac{\theta^3}{16} - \theta^3\\ &= \sum_i y_i^3 + \frac{3}{4}\theta - \frac{15}{8}\theta^3.\end{aligned}$$

Applying (A.10) now gives

$$T \le \frac{3}{4}\theta - \frac{15}{8}\theta^3 + \frac{1}{\sqrt{3}} \left(1 - \frac{5}{4}\theta^2\right)^{3/2}. \tag{A.11}$$

Step 3: a positive polynomial on a rational interval. Define the integer polynomials

$$\begin{aligned}L(\theta) &:= 8P(\theta) - C(\theta)(6\theta - 15\theta^3) - 80000\\ &= 28706769375\theta^5 - 69462602666\theta^4 - 15960533291\theta^3\\ &\quad + 12899417771 6\theta^2 - 12681497789 8\theta + 39484941048,\\ D(\theta) &:= 3L(\theta)^2 - C(\theta)^2(4 - 5\theta^2)^3.\end{aligned}$$

We next verify

$$L(\theta) \geq 4 \cdot 10^7, \qquad D(\theta) \geq 5 \cdot 10^{15} \quad \left(0 \leq \theta \leq \frac{9}{10}\right). \tag{A.12}$$

We include the arithmetic rule and all coefficient bounds, so this step is a finite polynomial inequality with no software hypothesis.

For a polynomial $R(x) = \sum_{j=0}^{n} r_j x^j$, define

$$b_i^{[a,b]}(R) := \sum_{k=0}^{i} \frac{\binom{i}{k}}{\binom{n}{k}} (b-a)^k \sum_{j=k}^{n} r_j \binom{j}{k} a^{j-k}, \qquad 0 \leq i \leq n. \tag{A.13}$$

To derive this rule, first note the power-to-Bernstein identity

$$v^k = \sum_{i=k}^{n} \frac{\binom{i}{k}}{\binom{n}{k}} \binom{n}{i} v^i (1-v)^{n-i}.$$

Indeed, $\binom{n}{i}\binom{i}{k} = \binom{n}{k}\binom{n-k}{i-k}$, so its right side equals $v^k(v + (1-v))^{n-k} = v^k$. The coefficient of v^k in the ordinary power expansion of $R(a + (b-a)v)$ is $(b-a)^k \sum_{j=k}^{n} r_j \binom{j}{k} a^{j-k}$. Substituting the identity above into this expansion gives

$$R(a + (b-a)v) = \sum_{i=0}^{n} b_i^{[a,b]}(R) \binom{n}{i} v^i (1-v)^{n-i}.$$

For $0 \leq v \leq 1$, the weights on the right are nonnegative and sum to one. Therefore a lower bound for every coefficient gives the same lower bound for R on $[a, b]$.

For L on $[0, 9/10]$, with $n = 5$, write $b_i = b_i^{[0,9/10]}(L)$. The coefficients are

$$b_0 = 39484941048, \qquad b_1 = \frac{416456125659}{25}, \qquad b_2 = \frac{1070019349929}{250},$$
$$b_3 = \frac{22340858803839}{10000}, \quad b_4 = \frac{32225886471177}{25000}, \quad b_5 = \frac{997373040003}{20000}.$$

Each is greater than $4 \cdot 10^7$.

For D, use $n = 10$ and the consecutive intervals

$$[a_j, b_j] = \left[0, \frac{9}{20}\right], \quad \left[\frac{9}{20}, \frac{81}{160}\right], \quad \left[\frac{81}{160}, \frac{171}{320}\right], \quad \left[\frac{171}{320}, \frac{9}{16}\right], \quad \left[\frac{9}{16}, \frac{27}{40}\right], \quad \left[\frac{27}{40}, \frac{9}{10}\right].$$

The entries in the following table are the exact integers

$$m_{ji} = \left\lfloor 10^{-15} b_i^{[a_j,b_j]}(D) \right\rfloor.$$

Consequently they satisfy

$$b_i^{[a_j,b_j]}(D) \geq 10^{15} m_{ji}, \qquad 1 \leq j \leq 6, \quad 0 \leq i \leq 10.$$

$j\backslash i$	0	1	2	3	4	5	6	7	8	9	10
1	3689811	2458235	1585464	990049	598393	349594	196884	106427	54890	26822	12361
2	12361	10553	8958	7556	6330	5263	4339	3544	2865	2291	1809
3	1809	1568	1351	1155	980	824	686	565	459	368	291
4	291	214	151	100	61	33	15	6	5	12	26
5	26	80	246	477	740	1008	1263	1490	1682	1835	1944
6	1944	2164	2216	2084	1801	1419	998	595	263	53	7

All these integers follow by substituting the displayed polynomials L and C into (A.13). If a resulting coefficient is N/d with $d > 0$, its listed integer m is verified by $10^{15}md \leq N < 10^{15}(m+1)d$. For example, the coefficient producing the smallest table entry is

$$b_8^{[171/320,\,9/16]}(D) = \frac{37529941696050313344544605054 9}{68719476736000} \in (5 \cdot 10^{15}, 6 \cdot 10^{15}).$$

Thus that entry is $m_{4,8} = 5$. The table gives a lower bound for every coefficient, not evaluations of D at finitely many points. Its smallest entry is 5. The intervals cover $[0, 9/10]$, so (A.12) follows.

Step 4: conclusion. On the smaller interval in (A.8), $4 - 5\theta^2 \geq 0$. Since $C > 0$ and $L > 0$, the strict positivity of D gives

$$L(\theta) > \frac{C(\theta)}{\sqrt{3}}(4 - 5\theta^2)^{3/2}.$$

Since $(4 - 5\theta^2)^{3/2} = 8(1 - 5\theta^2/4)^{3/2}$, division by eight and the definition of L give

$$P(\theta) - C(\theta)\left[\frac{3}{4}\theta - \frac{15}{8}\theta^3 + \frac{1}{\sqrt{3}}\left(1 - \frac{5}{4}\theta^2\right)^{3/2}\right] > \frac{80000}{8} = 10000.$$

Combining this with (A.9), $C > 0$, and (A.11) proves $S(A, p) > 10^{-6}$, hence (A.7).

Appendix B. The multiplier in ordinary tensor notation

The scalar polynomials displayed below have integer coefficients. They are numerators: each field f, U, M, K is obtained by dividing its entire displayed tensor numerator by the common denominator d once. Put

$$d = 10^{10}, \qquad s = |p|^2, \qquad t = p \cdot Ap, \qquad r = p \cdot A^2 p, \qquad z = p \cdot A^3 p, \qquad T = \operatorname{tr}(A^3).$$

The letter r in this appendix is an algebraic invariant, not a ball radius. The unindexed F below is the scalar numerator df; it is distinct from the vector F_l in (3.4). The fields used in Section 3 are defined by

$$\begin{aligned} df &= F, \\ dU_{jk} &= U_A\, A_{jk} + U_0 p_j p_k + U_1 p_j (Ap)_k, \\ dM &= M_0 p + M_1 Ap + M_2 A^2 p + M_3 A^3 p, \\ dK_{lijk} &= K_1 p_l (Ap)_i A_{jk} + K_2 p_l p_j A_{ik} + K_3 p_l (A^2 p)_i p_j p_k \\ &\quad + K_4 p_l (Ap)_i p_j p_k + K_5 p_l (A^2 p)_i A_{jk}. \end{aligned}$$

Here the scalar polynomials on the right are given below. No symmetrization is implicit in these formulas. In particular, the term $p_j(Ap)_k$ appears with exactly its displayed coefficient. The antisymmetrization needed in the proof remains $G_{lijk} = \delta_{li} U_{jk} + K_{lijk} - K_{iljk}$. More explicitly, exchanging only the slots l, i in the five terms for K and collecting terms gives

$$\begin{aligned} d(K_{lijk} - K_{iljk}) &= [p_l (Ap)_i - p_i (Ap)_l](K_1 A_{jk} + K_4 p_j p_k) \\ &\quad + [p_l (A^2 p)_i - p_i (A^2 p)_l](K_5 A_{jk} + K_3 p_j p_k) \\ &\quad + K_2 p_j (p_l A_{ik} - p_i A_{lk}). \end{aligned}$$

The scalar coefficients are unchanged by this slot exchange because they depend only on the invariants s, t, r, z, T. Each displayed term changes sign under $l \leftrightarrow i$. The definition of G uses this full difference, so no factor $1/2$ is inserted.

The scalar field and the rank-two field.

$$\begin{aligned} F &= 10000000000 - 3755863437s + 1873014544s^2 - 1120830321s^3 + 1997035026s^4 \\ &\quad + t(10584640366 + 2183994295s + 43186584s^2 + 2289921003s^3) \\ &\quad + r(482004749 + 4850194906s - 357668087s^2) \\ &\quad + t^2(2907579685 + 1007736382s) - 1294671023\, tr, \end{aligned}$$

$$\begin{aligned} U_A &= 8540301744 + 913852507s - 463760052s^2 - 2365562345s^3 + 1744509278s^4 \\ &\quad + t(1809347173 + 2377153028s + 1198331357s^2) + 1913784625t^2, \end{aligned}$$

$$\begin{aligned}U_0 &= 9432844503 - 5824514733s - 534960237s^2 - 1042568884s^3\\ &\quad + t(3708751228 - 2160684084s - 940422548s^2)\\ &\quad + r(-1912140724 + 504025693s) - 2220644478t^2,\\ U_1 &= -4362460942 + 4956816046s + 2579465917s^2 - 2709298930s^3\\ &\quad + t(1589302871 + 2330314887s) + 518966937(1-s)r.\end{aligned}$$

The vector field.

$$\begin{aligned}M_0 &= 4410933670 - 3296076367s - 1651925812s^2 + 3441895491s^3\\ &\quad + t(-1988049487 - 1403316402s + 2107980686s^2)\\ &\quad + r(1129025766 + 4681348439s - 377359858s^2)\\ &\quad + z(2039361060 + 2891322933s)\\ &\quad + T(2015047792 + 557588072s - 206880379s^2)\\ &\quad + t^2(4523731080 - 2651636105s) - 2064640466\,tr,\\ M_1 &= 131480326 + 9234756275s - 691429908s^2 - 1227118202s^3\\ &\quad + t(1058048125 + 4577417724s + 2314211645s^2)\\ &\quad + r(-55152205 + 2589071439s) - 1199150559t^2,\\ M_2 &= 88566419 + 5416034553s + 3575586612s^2 - 8096979677s^3\\ &\quad + t(1126525592 + 2452747605s),\\ M_3 &= 604588671 + 1301400010s + 1505009778s^2.\end{aligned}$$

The rank-four field.

$$\begin{aligned}K_1 &= 5579900926 - 2967104052s + 779158574s^2 + t(63488030 - 2112695908s),\\ K_2 &= -5968962410 + 2723150122s + 1170033188s^2 + t(484602132 - 548102300s)\\ &\quad + 259483469(1-s)r,\\ K_3 &= -3682696842 + 1332493297s,\\ K_4 &= 1514122987 + 157839553s + 1547669922t,\\ K_5 &= 4085463237 + 1495930839s + 1379882037s^2.\end{aligned}$$

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